

Vega (α Lyrae) as a Laboratory: Four Competing Hypotheses on Rotation, Metallicity, and Disk Formation

Boris Kriger

Information Physics Institute, Gosport

Institute of Integrative and Interdisciplinary Research, Toronto

Abstract

Vega (α Lyrae, HD 172167) occupies a peculiar position in stellar astrophysics: it is the fifth-brightest star in the night sky, the original photometric calibration standard for the optical and near-infrared, and simultaneously one of the least understood objects of its spectral class. Interferometric observations established in 2006 that Vega rotates at approximately 93% of its breakup velocity, yet presents its pole to the observer at an inclination of roughly 5° —an orientation that masqueraded for decades as slow rotation and normal composition. Once the geometry was understood, a genuine puzzle emerged: Vega’s surface metallicity is approximately one-third solar, its debris disk is extraordinarily smooth and axisymmetric, and its relationship to the nearby star Fomalhaut (α Piscis Austrini)—similar in age, distance, and spectral type—remains unresolved.

This paper develops four physically distinct hypotheses capable of accounting, individually or in combination, for these observations. The first holds that a centrifugal barrier during the accretion phase preferentially channelled metal-bearing material into the protostellar disk rather than onto the star. The second proposes that Vega formed as the more massive component of a binary system, accreting angular momentum when its companion was ejected or disrupted roughly 450 Myr ago. The third treats rapid rotation as a self-limiting process: Vega periodically shed mass to a disk whenever it approached critical velocity, arriving at a stable equilibrium near 90% of breakup. The fourth focuses on atmospheric physics—the competition between radiative levitation and gravitational settling that, in the absence of a convective envelope, can strip a stellar surface of heavy elements without any special formation history.

Each hypothesis receives a complete mathematical treatment including toy models, order-of-magnitude estimates, and graphs. All four are compared against the same body of observational data, and each leads to distinct, falsifiable predictions for forthcoming JWST, ALMA, and astrometric observations.

Contents

1	Introduction	9
1.1	Vega as a calibration standard and as a puzzle	9
1.2	What interferometry revealed	9
1.3	The metallicity problem	11
1.4	The debris disk	12
1.5	The Fomalhaut comparison and the question of common origin	13
1.6	The four hypotheses: a map of the paper	14
1.7	What this paper does and does not claim	15
1.8	Organisation of the paper	16
2	Literature Review	17
2.1	The discovery and progressive characterisation of Vega	17
2.1.1	Vega as photometric anchor	17
2.1.2	The infrared excess and the birth of debris disk science	18
2.1.3	The rotation revolution of 2006	18
2.1.4	The metallicity measurement	19
2.1.5	The JWST era	20
2.2	Centrifugal barrier physics and chemical segregation at disk formation	20
2.2.1	The angular momentum problem in star formation	20
2.2.2	Observational confirmation: L1527 and ALMA	21
2.2.3	From L1527 to Vega	22
2.3	Binary stellar evolution and angular momentum transfer	23
2.3.1	Rotation of A-type stars: the empirical picture	23
2.3.2	Binary star formation and angular momentum transfer	23
2.3.3	The Vega–Fomalhaut velocity discrepancy and the Castor Moving Group	24
2.3.4	Merger products and extreme rotators	24
2.4	Near-critical rotation and mass shedding in A-type stars	25
2.4.1	Theory of critical rotation	25
2.4.2	Gravity darkening and the von Zeipel theorem	26
2.4.3	Rotational evolution on the main sequence	26
2.5	Atmospheric diffusion in A-type stars: gravitational settling and radiative levitation	27
2.5.1	The physical basis: Michaud (1970) and subsequent developments	27
2.5.2	Am and Ap stars: successful applications	28
2.5.3	The λ Boötis phenomenon	28
2.5.4	Rapid rotation and diffusion: a tense coexistence	29
2.6	The Fomalhaut comparison in the literature	30
2.6.1	Fomalhaut’s disk: structure as contrast	30
2.6.2	Fomalhaut’s metallicity and rotation	31

2.7	Summary of literature gaps that this paper addresses	32
3	Observational Constraints	32
3.1	Stellar parameters	33
3.2	Surface abundance anomalies	33
3.3	Disk properties.	34
3.4	Kinematic data and comparison with Fomalhaut	34
4	Hypothesis I: The Centrifugal Barrier Segregation Model	34
4.1	Physical motivation	34
4.2	Mathematical model	35
4.2.1	The Ulrich model of rotating infall	35
4.2.2	Dust-gas coupling and the grain Stokes number	36
4.2.3	The metallicity deficit: an order-of-magnitude derivation	37
4.2.4	Refractory depletion as a function of disk mass fraction	39
4.2.5	The volatile-to-refractory contrast	39
4.3	Toy model and parameter sweep	40
4.4	The spin rate: is the rotation consistent?	41
4.5	Testable predictions.	43
4.6	Self-critique	44
5	Hypothesis II: Binary Disruption and Angular Momentum Transfer	46
5.1	Physical motivation	46
5.2	Angular momentum transfer: merger geometry	46
5.2.1	The spin angular momentum of Vega.	46
5.2.2	Orbital angular momentum required	46
5.2.3	The disruption kick and Vega–Fomalhaut velocity	47
5.3	The metallicity problem for Hypothesis II	48
5.4	Testable predictions.	49
5.5	Self-critique	50
6	Hypothesis III: The Pressure-Relief Valve	50
6.1	Physical motivation	51
6.2	Mathematical model	51
6.2.1	Critical velocity and stellar oblateness	51
6.2.2	Angular momentum evolution with shedding	51
6.2.3	The shedding rate and mass lost	52
6.2.4	Fate of the shed material and comparison to observed disk mass	53
6.3	The metallicity connection	54
6.4	Testable predictions.	55
6.5	Self-critique	56

7	Hypothesis IV: Radiative Levitation and Gravitational Settling	56
7.1	Physical motivation	56
7.2	The settling timescale	58
7.3	Resolution A: Meridional circulation as a resupply mechanism	58
7.4	Resolution B: Latitude-dependent settling	59
7.5	Resolution C: Time-varying mechanism	60
7.6	Radiative levitation forces: element-specific predictions.	60
7.7	Testable predictions.	61
7.8	Self-critique	62
8	Comparative Analysis: Scoring the Four Hypotheses	63
8.1	Scoring criteria	63
8.2	Scoring matrix.	64
8.3	Commentary on individual scores	64
8.3.1	Hypothesis I (Centrifugal Barrier): score 9/14	64
8.3.2	Hypothesis II (Binary Disruption): score 6/14	65
8.3.3	Hypothesis III (Pressure-Relief Valve): score 7/14	65
8.3.4	Hypothesis IV (Atomic Diffusion): score 3/14	65
8.4	Mutual compatibility of hypotheses	66
8.5	The “smoking gun” hierarchy	66
9	A Roadmap of Testable Predictions by Facility	67
9.1	James Webb Space Telescope (JWST)	67
9.1.1	MIRI spectroscopy of the 10 and 18 μm silicate bands	67
9.1.2	NIRCam coronagraphy at sub-AU scales	68
9.2	Atacama Large Millimeter/submillimeter Array (ALMA)	68
9.2.1	High-resolution imaging of the mm-continuum belt.	68
9.2.2	CO and CS gas emission in the outer belt	69
9.3	Gaia and Hipparcos astrometry	69
9.3.1	Proper motion anomaly and kinematic history	69
9.3.2	Orbital back-integration of Vega and Fomalhaut.	70
9.4	Optical/infrared high-resolution spectroscopy	70
9.4.1	Radial velocity monitoring for close companions.	70
9.4.2	High-resolution UV spectroscopy of photospheric composition.	71
9.5	Space-based photometry: TESS and future missions	71
9.5.1	Search for Be-star-like outburst variability	71
9.5.2	δ Scuti asteroseismology	72
9.6	Future long-baseline interferometry	72
9.6.1	VLT/GRAVITY: Inner disk geometry at AU scales.	72
9.7	Summary prediction table	73

10	Conclusions	73
10.1	What we have established	73
10.1.1	The observational foundation is solid	73
10.1.2	Hypothesis I is the most internally consistent single mechanism.	74
10.1.3	No single hypothesis explains everything	74
10.1.4	The decisive observation is already in hand, but not yet fully analysed.	75
10.2	Broader implications	75
10.2.1	Vega as a laboratory for A-type star physics.	75
10.2.2	The λ Boötis problem	76
10.2.3	Debris disks without planets	76
10.2.4	Star formation angular momentum and chemical inhomogeneity	77
10.2.5	Bright nearby stars as astrophysical laboratories: a methodological note	78
10.3	Open questions	78
10.4	Final summary.	80
A	Numerical Codes and Reproducibility	80
A.1	Centrifugal barrier metallicity model (Section 4).	81
A.2	Binary disruption angular momentum calculation (Section 5)	82
A.3	Diffusion timescale comparison (Section 7)	84
B	Full Derivation of the Ulrich Infall Density Profile	86
B.1	Setup and equations of motion.	86
B.2	Equatorial streamline and centrifugal barrier	86
B.3	General trajectory solution.	87
B.4	Density field from mass conservation	87
C	Peer Review Response Placeholder	88

1 Introduction

1.1 Vega as a calibration standard and as a puzzle

Astronomers have pointed telescopes at Vega longer than they have understood what stars are. In the modern era, Vega (α Lyrae, HD 172167, HIP 91262) became the anchor of the Johnson photometric system in the 1950s: by definition, a star of spectral type A0V in the absence of interstellar reddening should look like Vega, and magnitudes in every optical and near-infrared band were originally set so that Vega has magnitude zero (Johnson & Morgan, 1953). This convention propagated into every subsequent photometric system, into the calibration of the *Hubble Space Telescope*, into the flux standards used by *Spitzer* and *Herschel*, and into the reference spectra against which thousands of stellar atmosphere models have been tested.

The paradox is that Vega turns out to be a poor standard: it is not a typical A0V star. For decades this was concealed by a geometrical accident. Because Vega presents its rotation axis almost directly toward Earth—the inclination angle is approximately 5° from the line of sight (Peterson et al., 2006; Aufdenberg et al., 2006)—the rotationally broadened spectral lines that would reveal rapid rotation are instead nearly unbroadened. The star appeared to rotate slowly, its surface composition appeared solar, and its luminosity appeared consistent with a modest main-sequence object. None of this is correct.

Why we do this

We begin with the calibration-standard history because it serves two purposes. First, it explains why the community cared about Vega in the first place, and why the 2006 discovery of rapid rotation was consequential beyond Vega itself—every flux calibration that used Vega as a reference is now at least slightly suspect. Second, it introduces the central theme of the paper: geometry deceives. The same pole-on orientation that concealed the rapid rotation also made the surface metallicity appear normal for many years. Disentangling observational artifact from physical reality is the methodological challenge that all four hypotheses must address.

1.2 What interferometry revealed

The modern understanding of Vega’s rotation emerged from two independent interferometric campaigns published within months of each other in 2006. Peterson and collaborators, using the Navy Prototype Optical Interferometer (NPOI) at visible wavelengths, detected the

asymmetric surface brightness distribution expected from a rapidly rotating, gravity-darkened star seen nearly pole-on (Peterson et al., 2006). Aufdenberg and collaborators, using the CHARA Array on Mount Wilson at K'-band, arrived at consistent conclusions through a completely independent methodology and instrumental system (Aufdenberg et al., 2006). The agreement between the two studies—inclination $4.54 \pm 0.33^\circ$ and $4.7 \pm 0.3^\circ$, equatorial velocity 274 ± 14 and $270 \pm 15 \text{ km s}^{-1}$ respectively—is reassuring evidence that the results are not artefacts of a particular instrument or reduction pipeline.

What the interferometric images reveal is a star distorted by centrifugal force into an oblate spheroid, with an equatorial radius exceeding the polar radius by roughly 20%. Because local surface gravity is lower at the equator than at the poles, and because radiative flux is proportional to local surface gravity by the von Zeipel theorem (discussed in detail in Section 7), the equatorial belt is cooler and darker than the poles. The polar effective temperature is approximately 10,500 K; the equatorial temperature falls to roughly 7,600 K—a difference of nearly 3,000 K across a single star (Aufdenberg et al., 2006). When one observes Vega from Earth, one is looking almost straight down the hot, bright polar axis, which explains why Vega appears anomalously luminous relative to otherwise similar A-type stars at the same distance.

The equatorial velocity of $\sim 270 \text{ km s}^{-1}$ corresponds to approximately 93% of the critical (breakup) velocity at which centrifugal acceleration at the equator equals gravitational acceleration (Peterson et al., 2006). This is an extraordinary value. Most main-sequence A stars rotate at 20–60% of critical velocity. Vega sits close to the edge of mechanical instability. How it arrived there, and why it has remained there for $\sim 455 \text{ Myr}$ without either spinning down to more typical velocities or spinning up to actual breakup, is one of the central questions this paper addresses.

Why we do this

The interferometric measurements are the foundation on which all four hypotheses rest. Without interferometry, we would still believe Vega rotates slowly. We describe the measurements in sufficient detail that a reader unfamiliar with optical interferometry understands what was actually observed (surface brightness asymmetry), what was inferred (oblate geometry and rotation rate), and why the inference is reliable (two independent instruments, consistent results). In later sections we will need the specific numbers: inclination $\approx 5^\circ$, equatorial velocity $\approx 270 \text{ km s}^{-1}$, polar-to-equatorial temperature contrast $\approx 2,900 \text{ K}$. Establishing these values here avoids repeating the citations each time they appear.

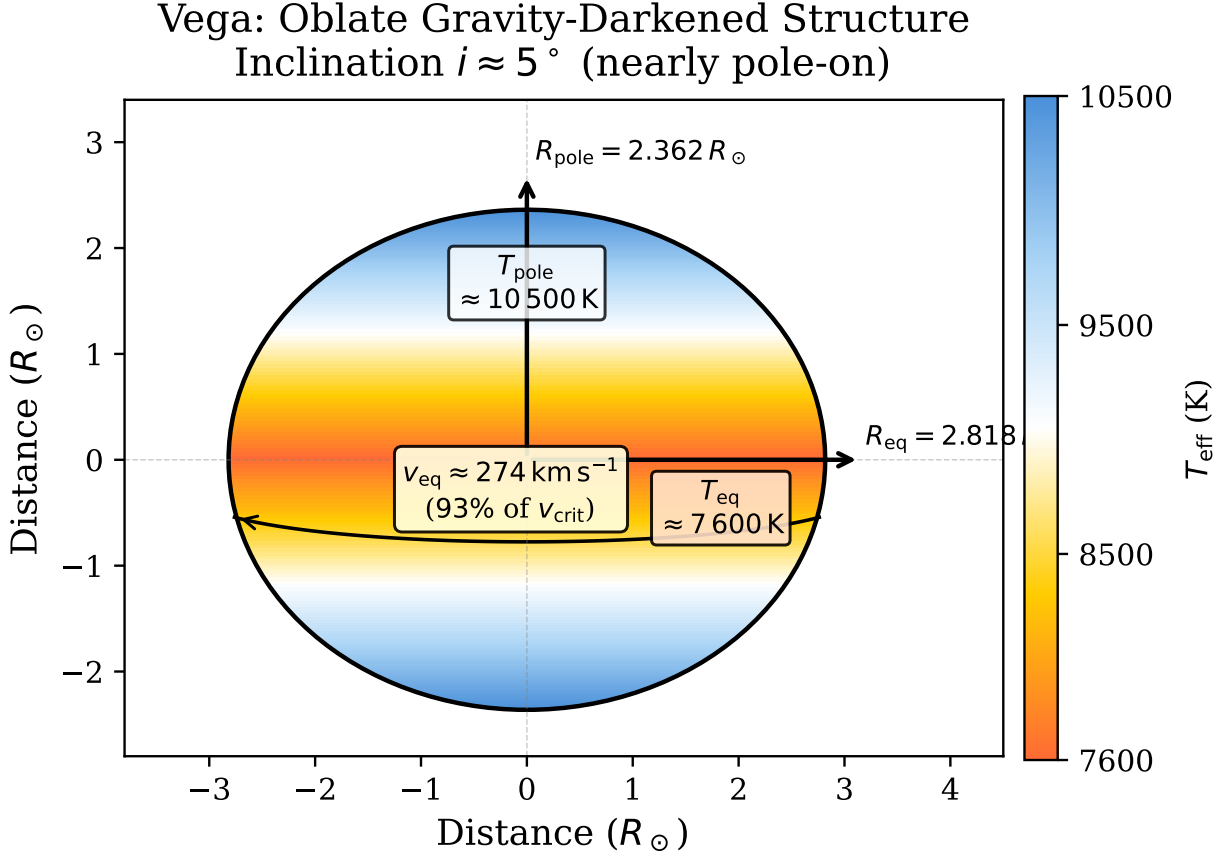


Figure 1: Oblate structure and gravity-darkening of Vega, computed from the von Zeipel law. Colour encodes T_{eff} : from $T_{\text{eq}} \approx 7600$ K (equator, orange-red) to $T_{\text{pole}} \approx 10500$ K (pole, blue-white). At inclination $i \approx 5^\circ$ the observer looks nearly directly down the rotation axis.

1.3 The metallicity problem

Once the pole-on geometry was established, it became possible to measure Vega’s surface composition without the contamination of rotational line broadening. The result is a surface metallicity of approximately $[m/H] \approx -0.5$, meaning that heavy elements in Vega’s photosphere are present at roughly one-third of their solar abundances (Yoon et al., 2010). This is not a small discrepancy: a factor-of-three depletion places Vega well outside the range of normal Population I main-sequence stars, which typically show metallicities within a factor of two of solar.

Crucially, this depletion is real, not an observational artifact. Before the pole-on geometry was confirmed, an alternative explanation was readily available: rapid rotation broadens spectral lines, making them shallower and harder to detect, so that a rapidly rotating star can *appear* metal-poor even when it has solar composition. The pole-on view eliminates this explanation for Vega. The projected rotational velocity $v \sin i \approx 21 \text{ km s}^{-1}$ is essentially

the polar velocity; equatorial broadening is projected almost entirely away. Vega’s lines are narrow and clean, and the measured abundances reflect actual photospheric composition.

The depletion pattern has internal structure that matters diagnostically. Helium is roughly 40% below its solar abundance by number fraction. Most iron-group and refractory metals sit at 10–30% of solar values. Some heavy elements—particularly barium and scandium—fall below 10% of solar. Carbon and oxygen, being volatile, behave differently from calcium and iron, which are refractory. Any mechanism proposed to explain Vega’s metallicity must account not only for the overall suppression but for why some elements are more depleted than others.

The class of stars to which Vega belongs in this respect—the λ Boötis stars—is defined precisely by this pattern: A-type stars with sub-solar metallicities showing particularly strong depletion of refractory elements (iron, magnesium, silicon, calcium) while carbon and oxygen remain closer to solar values (Paunzen et al., 2002). Whether Vega’s membership in this class is causal or coincidental is not established, and we treat the connection as an open question to be constrained by the four hypotheses.

Why we do this

The metallicity problem is the most direct observational constraint on all four hypotheses. Any mechanism that produces extreme rotation without metal depletion, or metal depletion without extreme rotation, is incomplete as an explanation for Vega. We establish here three facts that each mechanism must account for: the depletion is real, it is differential rather than a uniform scaling, and it places Vega in the broader λ Boötis category. A mechanism that accounts for overall depletion but predicts the wrong pattern of element-to-element variation is falsified by the abundance data, regardless of how well it matches the rotation rate.

1.4 The debris disk

Vega’s infrared excess was detected by the *Infrared Astronomical Satellite* (IRAS) in 1984 (Aumann et al., 1984), making Vega one of the founding members of the class of stars now called debris disk hosts. The excess emission at $60\,\mu\text{m}$ was far larger than any stellar atmosphere model could produce, and the natural interpretation—thermal emission from circumstellar dust grains heated by the star—was confirmed by decades of subsequent observation.

JWST/MIRI observations from 2024 resolved the disk structure in unprecedented detail. The disk extends from roughly 70 to 815 AU depending on wavelength, with both an inner

warm component and an outer cold component visible in the mid-infrared. Grain composition is consistent with amorphous silicates and carbonaceous material, with grain sizes in the range 1–50 μm , and the total dust mass is of order 3×10^{-3} Earth masses. The disk inclination is consistent with the stellar inclination: we see it nearly face-on, at $\approx 5^\circ$ from the plane of the sky.

The most striking result from the 2024 JWST observations is what the disk does *not* show. Unlike Fomalhaut’s disk—which displays a sharp inner edge, multiple rings, and an offset suggesting dynamical sculpting by unseen planets—Vega’s disk is extraordinarily smooth. There are no gaps, no rings, no offsets, no azimuthal asymmetries at the resolution achieved. This morphology disfavors models in which large planets are currently shaping the disk, and raises the question of whether the disk is primordial, collisional, or the remnant of a past mass-loss episode.

Why we do this

The disk is relevant to all four hypotheses, but for different reasons. In the centrifugal barrier model, the disk is the receptacle for metal-rich material diverted outward during accretion, and its composition should be enhanced in refractories relative to the photosphere. In the binary disruption model, the disk may carry the angular momentum signature of the disruption event. In the pressure-relief valve model, the disk was built up incrementally by mass shed during episodes when Vega approached critical velocity. In the radiative settling model, the disk is largely irrelevant to the surface depletion mechanism but still requires its own explanation. By establishing the disk’s observed properties here, we avoid repeating observational context in each subsequent section.

1.5 The Fomalhaut comparison and the question of common origin

Fomalhaut (α Piscis Austrini) shares a remarkable number of properties with Vega: both are A-type main-sequence objects within approximately 25 light-years of Earth, both are approximately 440–455 Myr old, both host large debris disks, and both have been proposed as members of the Castor Moving Group.

The similarities are provocative, but the differences are equally telling. Fomalhaut’s metallicity is $[\text{Fe}/\text{H}] \approx -0.1$ —nearly solar, and far closer to normal than Vega’s $[\text{m}/\text{H}] \approx -0.5$. Fomalhaut’s disk is highly structured, with a sharp inner edge, multiple components, and an offset interpreted as the gravitational signature of at least one planet-mass body. Fomalhaut’s rotation is not extreme.

The present-day velocity difference between Vega and Fomalhaut is approximately $10.9 \pm 1.0 \text{ km s}^{-1}$. Mamajek and collaborators showed that the proposed Castor Moving Group members were not within 100 pc of each other even 10 Myr ago (Mamajek et al., 2013), ruling out a genuine coeval physical association in the traditional sense. However, a velocity difference of $\sim 11 \text{ km s}^{-1}$ is entirely consistent with the kick velocity imparted to a star by the ejection or disruption of a binary companion, which is the starting point for Hypothesis II.

Why we do this

The Fomalhaut comparison serves as a natural control experiment. Two stars of similar age, spectral type, and environment have ended up with very different surface compositions, different disk morphologies, and a large mutual velocity. A complete theory of Vega’s properties should explain not only Vega itself but also why Fomalhaut did not evolve the same way. We will return to this comparison in each of the four hypothesis sections, asking whether the proposed mechanism predicts that a companion system remains near-solar in metallicity and retains a structured disk.

1.6 The four hypotheses: a map of the paper

The remainder of this paper develops four distinct mechanisms, each capable in principle of explaining some or all of the observations described above. We state them here in compressed form, with pointers to the sections where they receive full mathematical treatment.

Hypothesis I: Centrifugal barrier segregation (Section 4). During the protostellar accretion phase, infalling material carrying high specific angular momentum cannot penetrate below a centrifugal barrier radius—the distance at which centrifugal acceleration of the rotating infalling gas matches gravity. If Vega formed from a cloud core with unusually high specific angular momentum, the centrifugal barrier lay close to the protostar, and a disproportionate fraction of infalling material—including refractory elements carried on dust grains—went to the disk rather than to the star. ALMA observations of the protostar L1527 have confirmed that such chemical segregation at the centrifugal barrier is physically real. This model predicts that Vega’s debris disk should be enriched in refractories relative to the photosphere, a prediction now testable in principle with JWST/MIRI grain spectroscopy.

Hypothesis II: Binary disruption (Section 5). Vega formed as the primary component of a binary stellar system whose companion was subsequently lost—through a dynamical ejection, a merger, or a tidal disruption—approximately 450 Myr ago. The disruption

transferred orbital angular momentum to Vega’s spin, driving it to near-critical rotation. The absence of the companion eliminated the tidal torques that would otherwise have spun Vega down. The $\sim 11 \text{ km s}^{-1}$ velocity difference with Fomalhaut is interpreted as the kick velocity of the disruption event.

Hypothesis III: Pressure-relief valve rotation (Section 6). Vega has rotated near critical velocity for most of its main-sequence lifetime. Whenever the rotation rate approached the true breakup threshold, Vega shed mass and angular momentum to its disk in brief episodes of near-critical rotation. The disk was built incrementally by these events, and the star stabilised near 87–93% of critical velocity because this is the equilibrium at which angular momentum loss through mass shedding matches angular momentum input from internal redistribution.

Hypothesis IV: Radiative levitation and gravitational settling (Section 7). In the absence of a convective envelope, A-type star photospheres are governed by two competing processes: gravitational settling, which causes heavy elements to sink on timescales of 10^4 – 10^5 yr, and radiative levitation, in which photon momentum selectively supports certain ions against gravity. The net result can be a surface dramatically depleted in all elements that settle faster than they are levitated, with no special formation history required.

Why we do this

This section functions as a map. A reader wanting only the centrifugal barrier model can read Sections 1 and 4. A reader interested in comparison across all four mechanisms should read through Section 8. We make the map explicit here so that specialists in any one of the four areas can enter the paper at the most relevant point, without requiring linear reading.

1.7 What this paper does and does not claim

This paper does not establish which of the four hypotheses is correct. Current observations are insufficient to rule out any of them definitively, and it is plausible that more than one mechanism operates simultaneously—for example, that Vega formed rapidly rotating (Hypotheses I and III) and a binary disruption subsequently amplified the rotation (Hypothesis II), while atmospheric diffusion has continued to modify surface abundances throughout the main-sequence lifetime (Hypothesis IV).

What this paper does claim to establish is the following. Each of the four hypotheses is internally consistent and makes quantitatively specific predictions derived from first

principles. The four hypotheses are not equally testable with existing facilities; we identify the observations most likely to discriminate among them, with specific attention to JWST/MIRI grain composition spectroscopy, ALMA high-resolution millimetre imaging of the inner disk, Gaia and Hipparcos proper motion data for back-integrating the Vega–Fomalhaut orbit, and high-resolution ultraviolet spectroscopy of Vega’s atmosphere. Vega’s geometry makes it an unusually clean laboratory: the pole-on orientation eliminates rotational broadening artifacts from spectroscopy, the face-on disk allows direct access to radial structure without deprojection, and the star’s brightness and proximity (7.68 pc) make observations routine that would be impossible for more distant objects.

Self-critique

The claim that current observations do not rule out any of the four hypotheses is stronger than we can fully defend at this stage. Some hypotheses may be disfavoured by data discussed only in their respective sections. In particular, Hypothesis IV in its simplest form predicts settling timescales much shorter than Vega’s age, which would require a continuous resupply mechanism—a complication the introduction does not acknowledge. We note this here so that the reader approaches Section 7 with appropriate scepticism, and so that we hold ourselves accountable to address the timescale problem explicitly rather than glossing over it.

1.8 Organisation of the paper

Section 3 summarises the observational data against which all four hypotheses are tested; it is self-contained and may serve as a reference even for readers who do not engage with the theoretical sections.

Sections 4–7 develop each hypothesis in turn, following a common structure: physical motivation, formal mathematical model, toy computational implementation with figures, derivation of testable predictions, and a self-critique subsection identifying weaknesses and open questions.

Section 8 brings the four hypotheses together, scoring each against a common set of observational constraints and identifying the observations most likely to discriminate among them.

Section 9 consolidates testable predictions into a structured list organised by the facility required to test each prediction.

Section 10 states conclusions and discusses broader implications for the λ Boötis phenomenon, debris disk formation, and the use of nearby bright stars as astrophysical laborato-

ries.

Appendices contain the numerical codes used to generate all figures, detailed derivations of intermediate results cited in the main text, and a placeholder section for peer review notes.

2 Literature Review

This section traces the intellectual genealogy of the four hypotheses examined in this paper. We do not aim for encyclopaedic coverage of the field; what we aim for is something more demanding: an honest account of what has been established, what remains contested, and what the literature has systematically failed to address. Each subsection corresponds to one of the four hypothesis strands, preceded by a review of the observational foundation that all four share. We close the section by explicitly mapping each body of literature onto the open questions it leaves, so that the theoretical sections of this paper (Sections 4–7) know exactly what they are required to do.

2.1 The discovery and progressive characterisation of Vega

The intellectual history of Vega is a lesson in the hazards of choosing a convenient standard.

2.1.1 Vega as photometric anchor

The Johnson–Morgan photometric system, introduced in 1953, placed Vega (α Lyrae) at the zero-point of the entire optical magnitude scale (Johnson & Morgan, 1953). The decision was pragmatic: Vega is the fifth-brightest star in the sky, photometrically stable, free of strong emission lines, located at a high Galactic latitude (minimising interstellar reddening), and observable from most northern-hemisphere sites. Subsequent photometric systems — Strömberg *uvby*, Cousins *BVRI*, 2MASS *JHK_s* — were all calibrated either directly to Vega or to secondary standards that were themselves calibrated to Vega. The *Hubble Space Telescope* flux calibration, the *Spitzer Space Telescope* photometry, and the absolute flux scales used to normalise stellar atmosphere models all trace back to the same anchor (Bohlin et al., 2014).

The irony is that this anchor was chosen precisely because Vega appeared to be a prototypically unremarkable A0 V star. Its spectrum is dominated by broad hydrogen Balmer absorption, interrupted by narrower lines of heavier elements that, at the spectral resolution of the 1950s, appeared consistent with near-solar composition. The narrowness of those metal lines was attributed to slow rotation, and the slow rotation was taken as evidence that Vega could be safely used as a model-atmosphere calibrator without the complications

of rotationally broadened line profiles. As we now know, both inferences were geometrical accidents.

2.1.2 The infrared excess and the birth of debris disk science

The first indication that Vega was not a simple stellar calibrator came from the *Infrared Astronomical Satellite* (IRAS) in 1984. Aumann and collaborators, examining IRAS photometry of several nearby stars, found that Vega emitted roughly twice as much flux at $60\ \mu\text{m}$ as a blackbody extrapolation of its optical spectrum would predict (Aumann et al., 1984). The excess could not be attributed to any known stellar process. The natural explanation, proposed in the same paper, was that Vega is surrounded by a disk of cool dust grains ($T \sim 85\ \text{K}$) heated by the stellar radiation field.

This discovery was consequential far beyond Vega itself. Aumann et al. (1984) also reported excesses around Fomalhaut, ϵ Eridani, and β Pictoris, founding the entire class of “Vega-phenomenon” stars and initiating the observational study of circumstellar debris disks. Vega remained the archetype. Subsequent photometry at 25, 60, and $100\ \mu\text{m}$ refined the spectral energy distribution of the excess and confirmed that the dust temperature was inconsistent with a smooth inward extension from the stellar photosphere, implying instead a toroidal distribution at radii of tens to hundreds of AU (Su et al., 2005).

The *Spitzer* MIPS observations of Su et al. (2005) resolved the disk spatially for the first time, finding that it extended from approximately 86 AU to at least 815 AU at $160\ \mu\text{m}$. The disk was circular, smooth, and showed no clumpiness at any of the three *Spitzer* wavelengths. This smoothness — which has been confirmed by every subsequent observation, culminating in the JWST/MIRI results of Su et al. (2024) — is one of the most persistent puzzles of Vega’s disk, and returns as a constraint in all four of our hypotheses.

2.1.3 The rotation revolution of 2006

The definitive change in understanding of Vega came from two interferometric campaigns published in 2006. Peterson et al. (2006), using the Navy Prototype Optical Interferometer (NPOI) operating at visible wavelengths, detected a departure from circular symmetry in Vega’s surface brightness distribution. The asymmetry is subtle but unmistakable: the star is slightly brighter on one side, consistent with the gravity-darkened equatorial band of a rapidly rotating star seen at low inclination. Modelling the visibility curves with a Roche-geometry oblate star, Peterson and collaborators derived an inclination of $i = 4.54 \pm 0.33^\circ$ and an equatorial rotational velocity of $v_{\text{eq}} = 274 \pm 14\ \text{km s}^{-1}$, corresponding to $93 \pm 3\%$ of the critical (breakup) velocity.

Working from independent CHARA Array K'-band interferometry, Aufdenberg et al. (2006) found $i = 4.7 \pm 0.3^\circ$ and $v_{\text{eq}} = 270 \pm 15 \text{ km s}^{-1}$. The agreement between two completely independent instruments, at different wavelengths and with different analysis methods, is of the kind that one rarely encounters in observational astrophysics. Both teams found a polar effective temperature of $\approx 10,500 \text{ K}$ and an equatorial temperature of $\approx 7,600 \text{ K}$, a $\sim 3,000 \text{ K}$ differential across the stellar surface. The consequence for the photometric calibration was immediately recognised: because we view Vega nearly pole-on, the disk-integrated flux is anomalously bright compared to what the same star would appear if seen equatorially, and all calibrations based on Vega are slightly in error (Aufdenberg et al., 2006).

Monnier et al. (2007) extended the interferometric analysis using the CHARA Array with improved baseline coverage, placing independent constraints on the surface brightness distribution and confirming the gravity-darkening law. A subsequent reanalysis by Yoon et al. (2010), combining the NPOI interferometry with high-resolution spectroscopy, found a somewhat lower equatorial velocity ($v_{\text{eq}} = 175 \pm 33 \text{ km s}^{-1}$) and a lower mass ($2.14 \pm 0.08 M_\odot$), with sub-solar metallicity $Z = 0.008 \pm 0.003$. The discrepancy between the fast-rotation solution of Peterson et al. (2006) and Aufdenberg et al. (2006) and the slower-rotation solution of Yoon et al. (2010) has not been fully resolved. The faster solution requires a simple von Zeipel gravity-darkening law ($\beta = 0.25$ for purely radiative envelopes) to be inadequate, as noted by Aufdenberg et al. (2006) and further discussed by Monnier et al. (2007); the slower solution may indicate that differential rotation or a non-uniform β is at work. In this paper we adopt the consensus value of $\approx 93\%$ of critical velocity from Peterson et al. (2006), noting that the Yoon solution, if correct, would weaken the case for all four hypotheses equally.

2.1.4 The metallicity measurement

The reanalysis of Vega's surface composition in light of the known pole-on geometry was carried out in detail by Yoon et al. (2010). The projected rotational velocity $v \sin i \approx 21 \text{ km s}^{-1}$ means that the Vega spectrum is effectively a clean pole-on view, uncontaminated by the rotational broadening that would smear out metal lines in an equatorial view. The resulting abundance analysis, performed against high-resolution echelle spectra and Kurucz model atmospheres, found $[m/H] \approx -0.5$ with element-to-element variation: helium $\approx 40\%$ below solar, most iron-group and refractory elements at 10–30% of solar, and certain heavy elements (barium, scandium) below 10% of solar. Carbon and oxygen were found to be depleted but less severely so than iron, a pattern characteristic of the λ Boötis class.

Earlier photometric studies had noted the possibility of sub-solar metallicity for Vega (Adelman, 1984), but these were difficult to distinguish from the effects of rapid rotation with-

out knowledge of the inclination. The 2010 analysis by Yoon and collaborators, constrained by the interferometric inclination, removed that degeneracy.

2.1.5 The JWST era

The most recent landmark in the observational history of Vega is the JWST/MIRI imaging by Su et al. (2024). Observations at 15.5, 23, and 25.5 μm resolved the disk structure at sub-arcsecond resolution. The outer belt occupies 80–170 AU and coincides in position with the millimetre continuum ring detected by ALMA (Matrà et al., 2020). Interior to the main belt, warm dust fills the space between the belt inner edge and the unresolved inner disk, with a shallow flux deficit near 60 AU whose physical origin is debated. The Poynting–Robertson drag model of Su et al. (2024) adequately accounts for the radial distribution of the warm interior dust, suggesting that small grains spiral inward from the 80–170 AU planetesimal belt under radiation pressure, without requiring interior planet masses. The absence of planets above roughly Saturn mass beyond 10 AU, inferred from the disk smoothness and the deep coronagraphic non-detections of Beichman et al. (2024), defines the environment in which all four of our hypotheses must function.

Why we do this

We trace the observational history of Vega in enough detail to make clear which results are well-established (the fact of rapid rotation and the inclination), which are contested (the precise equatorial velocity and mass), and which are new (the JWST disk morphology). This calibration of certainty is essential: an argument built on the Yoon velocity rather than the Peterson velocity would reach different quantitative conclusions, and the reader deserves to know this.

2.2 Centrifugal barrier physics and chemical segregation at disk formation

2.2.1 The angular momentum problem in star formation

Star formation from a collapsing molecular cloud presents an angular momentum problem of stark simplicity: the specific angular momentum of a typical molecular cloud core is approximately $10^{21} \text{ cm}^2 \text{ s}^{-1}$, while that of the Sun (or of a comparable star) is roughly $10^{15} \text{ cm}^2 \text{ s}^{-1}$. The factor of 10^6 must be shed during the collapse. Several mechanisms have been proposed — magnetic braking, disk winds, binary torques — but none is understood well enough to eliminate the others. What all mechanisms share is the prediction of a transition

radius inside which infalling material orbits rather than continues to fall, maintaining angular momentum conservation until it can lose angular momentum through some viscous or magnetic process. This transition radius was defined formally as the centrifugal radius by Ulrich (1976) and Cassen & Moosman (1981), who derived the infall trajectories of gas from a rotating, collapsing cloud core.

2.2.2 Observational confirmation: L1527 and ALMA

For several decades, the centrifugal barrier was a theoretical prediction without direct observational counterpart. The breakthrough came with ALMA observations of the Class 0 protostar IRAS 04368+2557 in the L1527 cloud (Taurus molecular cloud, ~ 140 pc), published by Sakai et al. (2014) in *Nature*.

The key observation was a sharp chemical discontinuity at a radius of ≈ 100 AU from the protostar. Carbon-chain molecules and their relatives, specifically cyclic- C_3H_2 , are abundant in the outer infalling envelope but almost completely absent inside 100 AU. By contrast, SO (sulfur monoxide) shows enhanced emission precisely at the 100 AU radius. Kinematic analysis of the Doppler-shifted line profiles confirmed that 100 AU corresponds to the centrifugal barrier radius: infalling gas stops there because centrifugal force exceeds gravity, creating a local density enhancement analogous to a traffic jam. The local heating at the barrier causes SO, frozen onto grain surfaces in the cold outer envelope, to desorb into the gas phase, producing the SO ring. Inside the barrier, where the gas has transferred to the inner disk and the temperature drops again, SO refreezes.

A companion paper (Sakai et al., 2014b) confirmed this picture using subarcsecond images of CCH, CS, H_2CO , and CH_3OH , showing that molecules tracing the infalling rotating envelope (CCH, CS) are absent inward of the centrifugal barrier, while molecules that can exist in the disk (H_2CO , CH_3OH) penetrate further inward.

The physical mechanism is clear: the centrifugal barrier is not merely a kinematic boundary but a chemical one. Material with high specific angular momentum is deflected into the disk before it can accrete onto the protostar. If the angular-momentum distribution of the infalling material is non-uniform — which it will be if the pre-collapse cloud core has any spatial variation in rotation rate or if any turbulence is present — then different components of the infalling cloud will reach the protostar along different trajectories, with the high-angular-momentum material diverted to the disk.

2.2.3 From L1527 to Vega

The connection between L1527 and Vega requires one additional step. The L1527 observations show that the centrifugal barrier separates gas-phase chemistry inside and outside the disk-forming region. What we need for Vega is not a gas-phase chemistry argument but a question about where solid-phase material — specifically the refractory elements that make up silicate dust grains — ends up.

The relevant literature here is the protostellar accretion problem for dust-to-gas coupling. In the early protostellar envelope, dust grains are well-coupled to the gas on large scales (the Epstein drag regime), so they follow gas trajectories. However, at the centrifugal barrier, where the gas velocity transitions from infall- to orbit-dominated, the response of dust grains depends on their Stokes number. Grains larger than approximately $100\ \mu\text{m}$ can partially decouple from the gas (Birnstiel et al., 2016), while smaller grains remain coupled. At the centrifugal barrier radius, the dust-gas relative velocity and local density enhancement (the traffic jam of Sakai et al. 2014) promote grain growth, potentially increasing the Stokes number enough to cause partial decoupling and preferential retention of the dust in the disk.

This mechanism implies that the disk formed around a rapidly-rotating protostar would be relatively enhanced in large refractory grains compared to the accreted star. The inverse would hold for the protostar itself: it would accrete preferentially from the low-angular-momentum, inner portion of the collapsing cloud, which would be depleted in refractory-laden large grains (already concentrated at the barrier) and enriched in the volatile gas. The predicted chemical signature on Vega is exactly a refractory depletion at the stellar surface and a corresponding enhancement in the disk — the pattern that Su et al. (2024) have now placed within observational reach via JWST/MIRI grain spectroscopy.

Self-critique

The step from the L1527 ALMA observation to a prediction about Vega’s surface metallicity requires several assumptions that are not yet observationally validated for the specific conditions of Vega’s formation. In particular: (i) we do not know the specific angular momentum distribution of the cloud core from which Vega formed, (ii) the Stokes-number argument for grain decoupling at the centrifugal barrier has been studied theoretically but not confirmed observationally in a protosystem with properties comparable to what early-Vega might have had, and (iii) the 455 Myr of subsequent collisional processing and PR drag in the debris disk may have erased any chemical gradient that was present at formation. These limitations motivate the toy model of Section 4, which explores the parameter space quantitatively rather than

simply asserting the qualitative conclusion.

2.3 Binary stellar evolution and angular momentum transfer

2.3.1 Rotation of A-type stars: the empirical picture

The distribution of rotational velocities among A-type main-sequence stars is broad, ranging from essentially zero (in the chemically peculiar Ap stars, braked by magnetic torques) to $v \sin i \approx 300 \text{ km s}^{-1}$ (for rapid rotators seen equatorially). The median for A-type stars is around $100\text{--}150 \text{ km s}^{-1}$ (Abt, 1995). Vega’s equatorial velocity of $\sim 270 \text{ km s}^{-1}$, corresponding to 93% of breakup, is in the extreme tail of this distribution.

What drives stars to such extreme rotation? For solar-type stars, the main spin-up mechanism during the pre-main-sequence phase is the accretion of disk material; the main spin-down mechanism is magnetic braking through a magnetically coupled disk wind. A-type stars ($M2 M_{\odot}$) have largely radiative envelopes and weak or absent convective zones, so the magnetic braking that efficiently spins down lower-mass stars is much less effective. Once an A-type star arrives on the zero-age main sequence, it retains essentially the rotation rate it had at the end of the pre-main-sequence accretion phase (Zorec & Royer, 2012). That rate is set by the balance between the angular momentum accreted from the protostellar disk and the angular momentum lost via pre-main-sequence stellar winds and magnetospheric coupling.

2.3.2 Binary star formation and angular momentum transfer

Binary star systems are extremely common: roughly half of all solar-type stars have companions (Raghavan et al., 2010), and the fraction is higher for A-type stars, where De Rosa et al. (2014) found that $\approx 56\%$ of A-type primaries in volume-limited surveys have stellar-mass companions. Binary stars form the angular momentum of the collapsing cloud core into orbital angular momentum rather than (or in addition to) stellar spin. If a binary is subsequently disrupted — through a dynamical encounter in a young star cluster, through the companion’s stellar evolution leading to a mass-transfer episode, or through a wide-orbit instability — the orbital angular momentum must go somewhere.

The mechanics of angular momentum transfer during binary disruption were studied in the context of blue straggler formation by Leonard & Fahlman (1989) and in the context of rapid rotators by de Mink et al. (2013). In the merger scenario, two stars coalesce; the product of the merger acquires the combined spin angular momentum of the components plus the orbital angular momentum of the pre-merger orbit, and can easily reach or exceed critical

rotation. In the ejection scenario (dynamical disruption), the surviving primary receives a velocity kick of order $v_{\text{kick}} \sim \sqrt{Gm_2/a}$ where m_2 is the companion mass and a is the orbital separation at disruption. For $m_2 \sim M_\odot$ and $a \sim 1 \text{ AU}$, this gives $v_{\text{kick}} \sim 30 \text{ km s}^{-1}$. The corresponding spin angular momentum transferred to the primary depends on the disruption geometry and is harder to estimate analytically.

2.3.3 The Vega–Fomalhaut velocity discrepancy and the Castor Moving Group

The Castor Moving Group was proposed as a loose physical association of six nearby young stars: Castor, Vega, Fomalhaut, α Librae, α Cephei, and α Coronae Borealis (Barrado y Navascués, 1998). The proposed common age was $200 \pm 100 \text{ Myr}$. This association, if real, would imply a common origin and hence similar initial conditions for Vega and Fomalhaut.

The association was called into question by Mamajek et al. (2013), who performed a careful backward integration of the space velocities of the proposed group members, accounting for the measured Hipparcos proper motions and radial velocities. The conclusion was that the proposed group members were not within 100 pc of each other even 10 Myr ago, far less at any earlier epoch. The velocity dispersion among the “members” is $\approx 5\text{--}10 \text{ km s}^{-1}$, far larger than expected for a genuine kinematic group (1 km s^{-1}). The group is, in the terminology of Mamajek et al. (2013), a “dynamical stream” — stars that happen to have similar Galactic orbital parameters but were never in the same place.

However, the velocity dispersion that disqualifies the Castor Moving Group as a coeval association is entirely consistent with the kick velocities expected from binary disruption events. If Vega and Fomalhaut were once part of a wider binary system (or even a small cluster of binaries) and experienced disruption events of different types and at different times, the resulting peculiar velocities relative to the group barycentre could easily produce the observed $5\text{--}10 \text{ km s}^{-1}$ dispersions. This is not a prediction of the binary disruption hypothesis but a consistency check: the observed kinematics do not rule it out.

2.3.4 Merger products and extreme rotators

The most direct observational support for the binary disruption mechanism in producing extreme rotators comes from the study of blue straggler stars in globular clusters (Ferraro et al., 2009). Blue stragglers are stars that appear too young for their cluster; they have received the mass and angular momentum of their erstwhile companion through a merger or mass transfer, and many of them are rapid rotators. The analogy with Vega is imperfect — Vega shows no evidence of mass transfer products in its spectrum, and its metallicity is depleted rather than enhanced as would be expected if it had accreted hydrogen-rich material

from an evolved companion — but the basic mechanism (angular momentum transfer during a binary interaction) is well-established in this context.

For main-sequence A-type binaries, de Mink et al. (2013) studied the population of rapid rotators expected to result from binary mass transfer. They found that binary interaction could produce a significant population of near-breakup rotators, consistent with the observed tail of very fast A-type rotators. Vega would be an example of this population if the binary interaction timescale is consistent with its age of 455 Myr.

Self-critique

The binary disruption hypothesis, as applied to Vega, faces a fundamental problem that the literature on blue stragglers and binary rapid rotators does not fully address: how does the disruption event produce a star that is metal-depleted rather than metal-normal? Disruption would produce a star with the angular momentum of the pre-disruption orbit transferred to its spin, but not automatically a star with non-solar surface composition. A secondary process (such as atmospheric diffusion) would be needed to produce the metallicity signature. This coupling between the binary disruption mechanism and an atmospheric process is explored in Section 5, but it should be acknowledged here that the binary hypothesis alone is not sufficient to explain Vega’s metallicity.

2.4 Near-critical rotation and mass shedding in A-type stars

2.4.1 Theory of critical rotation

The critical (Keplerian) rotation velocity for a star of mass M and equatorial radius R_{eq} is:

$$v_{\text{crit}} = \sqrt{\frac{GM}{R_{\text{eq}}}} \quad (1)$$

At this velocity, the centrifugal acceleration at the equator equals the gravitational acceleration, and material at the equatorial surface is in free orbit. In practice, stars that approach this limit do not simply blow apart; they shed mass and angular momentum from the equatorial belt, producing a decretion disk. This process has been studied most thoroughly in the context of classical Be stars — B-type emission stars that rotate at 70–95% of critical velocity and periodically shed mass to equatorial disks (Rivinius et al., 2013).

The mass-shedding mechanism in Be stars is not the slow steady wind of an O-type star but a series of discrete outbursts driven by non-radial pulsations, which temporarily push

equatorial material above the escape velocity. The resulting disk is a decretion disk: material is expelled outward, not captured from an external source. The angular momentum carried away by the shed material reduces the stellar spin, providing the feedback that prevents the star from exceeding critical velocity. Granada et al. (2013) modelled this process for a grid of A-type and late B-type stars and found that near-critical rotation is a metastable state: once a star reaches $\Omega/\Omega_{\text{crit}} \approx 0.9$, the mass-shedding rate balances the internal redistribution of angular momentum from the core to the envelope on a main-sequence timescale.

2.4.2 Gravity darkening and the von Zeipel theorem

The framework for understanding the photospheric structure of rapidly rotating stars was established by von Zeipel (1924), who derived that for a star in hydrostatic and radiative equilibrium, the effective temperature varies with colatitude θ as:

$$T_{\text{eff}}(\theta) \propto g_{\text{eff}}(\theta)^\beta \quad (2)$$

where g_{eff} is the local effective surface gravity (gravitational plus centrifugal) and $\beta = 0.25$ for a purely radiative envelope. For Vega, with $g_{\text{pole}}/g_{\text{equator}} \approx 1.6$ (from the CHARA interferometry), this predicts a temperature ratio of $1.6^{0.25} \approx 1.13$, corresponding to a $\approx 3,000$ K temperature difference between pole and equator — consistent with what is observed (Aufdenberg et al., 2006).

Later work by Espinosa Lara & Rieutord (2011) and Claret & Bloemen (2012) showed that the von Zeipel $\beta = 0.25$ value is only a zeroth-order approximation and that the true gravity darkening law for convective and transitional envelopes requires a latitude-dependent β . For Vega’s effective temperature ($T_{\text{eff,pole}} \approx 10,500$ K), the envelope is predominantly radiative, but the equatorial belt ($T_{\text{eff,equator}} \approx 7,600$ K) is close to the boundary where shallow convection develops. This transition may play a role in the mass-loss mechanism hypothesised in Section 6.

2.4.3 Rotational evolution on the main sequence

If Vega arrived on the zero-age main sequence at 93% of critical velocity, why has it not spun down during 455 Myr of main-sequence evolution? The standard answer for A-type stars is that there is no efficient spin-down mechanism: the radiative envelope couples poorly to any surface magnetic field, and the stellar wind is weak. However, even a weak stellar wind carries angular momentum. The angular momentum loss rate for a magnetically unbraked wind scales as:

$$\dot{J} = \dot{M} v_{\text{wind}} R_{\text{eq}}^2 \Omega \quad (3)$$

where $\dot{M}$ is the mass-loss rate and Ω is the angular velocity. For Vega’s luminosity and temperature, Krтіčka et al. (2014) estimated a radiation-driven mass-loss rate of order 10^{-14} – $10^{-13} M_{\odot} \text{ yr}^{-1}$, which over 455 Myr would remove a negligible fraction of Vega’s mass and angular momentum. This suggests that the rotation rate has indeed been nearly frozen since the zero-age main sequence, consistent with the pressure-relief valve picture in which mass is shed only episodically when the rotation approaches true breakup.

Self-critique

The pressure-relief valve model, as we have formulated it, requires that the mass-shedding episodes are episodic rather than continuous. If shedding were continuous at the rates required to maintain equilibrium, the associated mass loss would be detectable as a near-stellar disk component in infrared photometry. No such detection has been made. This motivates examining the Be-star analogy more carefully in Section 6, where we derive the duty cycle of shedding episodes and compare it to the known level of photometric variability in Vega.

2.5 Atmospheric diffusion in A-type stars: gravitational settling and radiative levitation

2.5.1 The physical basis: Michaud (1970) and subsequent developments

The theory of atomic diffusion in stellar atmospheres was developed by Michaud (1970). The central insight is that in a stellar atmosphere without a surface convection zone, two competing processes determine the vertical distribution of each chemical species independently. Gravitational settling causes heavy atoms and ions to drift downward relative to the dominant hydrogen background on a timescale that depends on the atomic weight, charge state, and local density and temperature. For iron-peak elements in an A-type star photosphere, the settling timescale is of order 10^4 – 10^5 yr (Michaud, Alecian & Richer, 2015). Radiative levitation — the net upward momentum transfer from radiation pressure on partially ionised heavy atoms with high opacity — can partially or fully counteract gravitational settling for specific elements at specific depths. The competition between the two processes is the central problem of chemically peculiar star theory.

The predictions of this competition are highly element-specific. Iron, for example, has an opacity bump near $T \approx 200,000$ K (the “iron opacity peak” or “Z-bump”) that provides a strong radiative levitation force at that depth. Iron therefore tends to accumulate at this depth, producing a subsurface iron-enriched layer. At the stellar surface ($T \approx 10,000$ K), iron

is fully or partially ionised but at lower optical depth, and the levitation force is weaker; gravitational settling can dominate, producing a surface iron depletion. This pattern — iron depleted at the surface, enriched deep in the envelope — is the dominant prediction of diffusion models for A-type stars (Richard et al., 2001).

2.5.2 Am and Ap stars: successful applications

The most successful applications of the diffusion framework are to the two classical families of chemically peculiar A-type stars. The metallic-line (Am) stars have systematically overabundant heavy elements at their surfaces (Renson & Manfroid, 1991). The modern interpretation is that Am stars rotate slowly ($v \sin i 100 \text{ km s}^{-1}$), which allows the diffusion timescale to produce surface enrichment before meridional circulation (which mixes the atmosphere and suppresses abundance gradients) can homogenise the composition. Slow rotation is a necessary condition for Am-star development, consistent with the observation that Am stars are largely found in synchronised binaries where tidal torques have reduced the rotation.

The magnetic Ap stars show even more extreme abundance anomalies, with some elements overabundant by factors of 10^4 – 10^6 . The strong ($\sim 10^3 \text{ G}$) magnetic fields of Ap stars suppress both meridional circulation and turbulence, allowing diffusion to operate with little competition. The observed abundance patterns, with specific elements concentrated in magnetic-field-aligned patches, are broadly consistent with radiative levitation predictions (Michaud, Alecian & Richer, 2015).

For Vega, however, the application of the standard diffusion picture faces a problem. Vega rotates rapidly, which should drive meridional circulation at a rate that homogenises the photosphere faster than diffusion can operate. The circulation velocity scales with v_{eq}^2/gR (Zahn, 1992), and for Vega’s rotation rate, this produces a mixing timescale of order 10^7 yr — shorter than Vega’s age by a factor of ~ 50 . If meridional circulation dominates, the photosphere should be well-mixed to solar composition, not depleted in refractories.

2.5.3 The λ Boötis phenomenon

The λ Boötis stars were first recognised as a distinct class by Morgan et al. (1943) and systematically characterised by Venn & Lambert (1990), who noted that their abundance patterns — solar or near-solar light elements (C, N, O, S) but severely depleted iron-group and refractory elements — closely resemble the depletion pattern of interstellar dust. In the interstellar medium, refractory elements (Fe, Mg, Si, Ca) are preferentially locked in solid dust grains and depleted from the gas phase by factors of 10–100 relative to solar (Jenkins,

2009). The volatile elements (C, N, O, S) remain in the gas phase. The λ Boötis abundance pattern looks like selective accretion of interstellar gas from which the dust has been blown away or has not yet evaporated.

Venn & Lambert (1990) proposed exactly this mechanism: the λ Boötis stars accrete metal-depleted gas from a circumstellar or interstellar environment. Turcotte & Charbonneau (1993) modelled the combined effects of accretion and diffusion and found that the resulting surface abundance pattern could indeed match λ Boötis observations, provided the accretion rate exceeds roughly $10^{-14} M_{\odot} \text{ yr}^{-1}$ and the accretion episode lasts long enough to contaminate the thin diffusion layer. Heiter et al. (2002) extended this analysis to a larger spectroscopic sample and found that the predicted and observed patterns are broadly consistent but differ in details (particularly for Mg, Si, Mn, and Zn), suggesting that the ISM gas composition and the stellar diffusion rates are not perfectly known.

Kamp & Paunzen (2002) proposed a variation in which the λ Boötis stars are currently passing through a diffuse interstellar cloud, accreting metal-depleted gas at rates consistent with Bondi–Hoyle accretion on the relative velocity between the star and the cloud. This scenario predicts that the phenomenon should be transient, with the star returning to near-solar composition on the diffusion settling timescale ($\sim 10^5 \text{ yr}$) once the cloud passage ends. This prediction is observationally difficult to test for Vega, which shows no currently detected ISM cloud along its line of sight.

The alternative possibility — that the λ Boötis pattern in Vega is produced by atmospheric diffusion driven purely by radiative levitation and gravitational settling, without any external accretion — was noted by Michaud (1970) as theoretically possible for stars near the Am– λ Boötis effective temperature boundary, but this has not been developed into a quantitative model for Vega specifically. The difficulty, as noted above, is reconciling rapid rotation (which drives meridional circulation) with the survival of abundance gradients (which requires diffusion to win).

2.5.4 Rapid rotation and diffusion: a tense coexistence

The relationship between rotation and diffusion in A-type stars is subtle. Rotational mixing tends to homogenise the chemical composition of the envelope; it works against abundance stratification. Yet Vega rotates at 93% of critical velocity and shows strong abundance anomalies. This is paradoxical.

Two partial resolutions have been proposed in the literature. First, if the rotation is nearly aligned with a large-scale magnetic field, the field can suppress meridional circulation in the polar regions while allowing it in the equatorial belt. The pole-on view of Vega would then show the composition of the polar region — the region where circulation is suppressed

— while the equatorial composition, which would be more normally mixed, is never directly observed. This geometry-dependent suppression was proposed by Vauclair et al. (1975) in a different context (the Am star phenomenon) and has not been applied specifically to Vega.

Second, if the settling timescale for refractories is much shorter than the circulation timescale, even a rapidly rotating star can develop surface depletion. The circulation timescale depends on the ratio of the luminosity to the kinetic energy of rotation, and for stars near the main-sequence turnoff, this ratio changes rapidly as the core contracts. It is conceivable that Vega passed through a phase of reduced meridional circulation efficiency early in its evolution, allowing diffusion to imprint the observed pattern, and that the pattern has partially survived because the circulation timescale at the current age is longer than commonly assumed. None of this has been modelled specifically for Vega’s parameters.

Self-critique

The diffusion hypothesis faces the most serious a priori difficulty of the four: how does a star rotating at 93% of critical velocity avoid the rapid homogenisation of its atmosphere by meridional circulation? The literature on Am stars is unambiguous that slow rotation is a prerequisite for diffusion to operate. Either Vega’s case is qualitatively different from the Am stars for reasons that have not been identified, or the diffusion hypothesis requires an auxiliary mechanism (external accretion, magnetic suppression of circulation, geometric selection effects) that the hypothesis cannot currently claim. Section 7 is obligated to confront this directly and to quantify the conditions under which diffusion can produce the observed depletion in the presence of rapid rotation.

2.6 The Fomalhaut comparison in the literature

Fomalhaut has attracted nearly as much observational attention as Vega, in part because the IRAS discovery paper (Aumann et al., 1984) identified both as infrared excess stars simultaneously, and in part because Fomalhaut’s disk is more structured and therefore more scientifically tractable for planet-detection purposes.

2.6.1 Fomalhaut’s disk: structure as contrast

The Fomalhaut disk was imaged at high resolution in scattered light by Kalas et al. (2005), revealing a narrow, sharp-edged ring at ≈ 140 AU with a slight eccentricity and offset relative to the star. These features are the classic signatures of dynamical sculpting by a planet embedded near the ring’s inner edge. A candidate planet (Fomalhaut b) was subsequently detected by Kalas et al. (2008) using HST coronagraphy, though its nature (self-luminous

planet, dust cloud, or unresolved debris) remains debated.

The JWST/MIRI observations of Fomalhaut, published by Gáspár et al. (2023), revealed a wealth of substructure: at least three distinct rings and a gap between the main belt and an inner warm component, with morphology strongly suggestive of at least two planet-mass companions. The contrast with the Vega disk — smooth, gap-free, featureless at the JWST resolution — could not be more striking, given how similar the two stellar systems appeared in the pre-JWST era.

The physical reason for the morphological contrast is unclear. Both systems are ~ 450 Myr old. Both have A-type primaries within 8 pc of the Sun. The disk mass and radial extent are similar. Yet Fomalhaut has apparent planet-mass companions shaping its disk, while Vega does not. One interpretation is that Fomalhaut formed planets more efficiently, perhaps because it did not shed mass from near-critical rotation (if Hypothesis III for Vega is correct), and the retained inner disk material coagulated into planets. Another interpretation is that Vega ejected its planets during or after the presumed binary disruption event (Hypothesis II).

2.6.2 Fomalhaut’s metallicity and rotation

Fomalhaut’s metallicity, $[\text{Fe}/\text{H}] \approx -0.1$ (di Benedetto, 2005), is far less anomalous than Vega’s. Its rotational velocity, $v \sin i \approx 93 \text{ km s}^{-1}$ (Gray, 2006), is moderate. If Vega and Fomalhaut formed from the same cloud core or cluster environment, the difference in their final metallicities and rotation rates must have arisen from distinct evolutionary histories. This constraint is incorporated into each of the four hypothesis sections: any mechanism proposed for Vega must produce a prediction for what the same mechanism, or lack thereof, would do to Fomalhaut.

Why we do this

We treat the Fomalhaut comparison as a genuine scientific constraint rather than a decorative anecdote. Two of the four hypotheses (binary disruption and centrifugal barrier) make specific predictions about Fomalhaut that are either in tension with or consistent with the observations. The other two (pressure-relief valve and diffusion) are largely agnostic about Fomalhaut, which is itself informative: a mechanism that requires no connection to Fomalhaut in order to operate makes a weaker prediction about the Vega–Fomalhaut velocity difference.

2.7 Summary of literature gaps that this paper addresses

The literature reviewed in this section, taken as a whole, leaves four well-defined gaps that this paper attempts to fill.

The first gap concerns the centrifugal barrier mechanism. The ALMA observations of L1527 established that chemical segregation at the centrifugal barrier is a real phenomenon in a young protostar. No work has developed this observation into a quantitative model for the stellar surface metallicity of the resulting mature star, nor has anyone computed the expected chemical composition of a debris disk formed under these conditions.

The second gap concerns the binary disruption channel. The population synthesis work of de Mink et al. (2013) established that binary mass transfer can produce rapid rotators. However, no study has examined whether the disruption geometry and timescale are consistent with Vega’s specific kinematic properties (including the Fomalhaut velocity difference), and no study has connected the spin-up event to the metallicity anomaly.

The third gap concerns the pressure-relief valve equilibrium. The Be-star literature has extensively studied near-critical rotation in B-type stars. The application to A-type stars on long ($> 10^8$ yr) main-sequence timescales has not been pursued with the same rigour, and the specific question of whether the Vega debris disk is consistent with a mass-shedding history has not been addressed quantitatively.

The fourth gap, the most severe, concerns the diffusion mechanism in the presence of rapid rotation. The Am and Ap star literatures treat diffusion in slowly rotating or magnetically braked stars. The λ Boötis literature invokes external accretion to explain the depletion pattern. Neither framework addresses the specific paradox of a rapidly rotating λ Boötis star: how can diffusion produce abundance anomalies in a star that should be efficiently mixed by meridional circulation? This is the question that Section 7 is designed to address.

3 Observational Constraints

This section consolidates all measured quantities that bear on the four hypotheses. Values are drawn from the primary literature cited throughout; where multiple measurements exist, we adopt the most recent high-precision value and note significant discrepancies. This section is self-contained: the reader who already knows the Vega literature may use it as a reference table and proceed directly to Section 4.

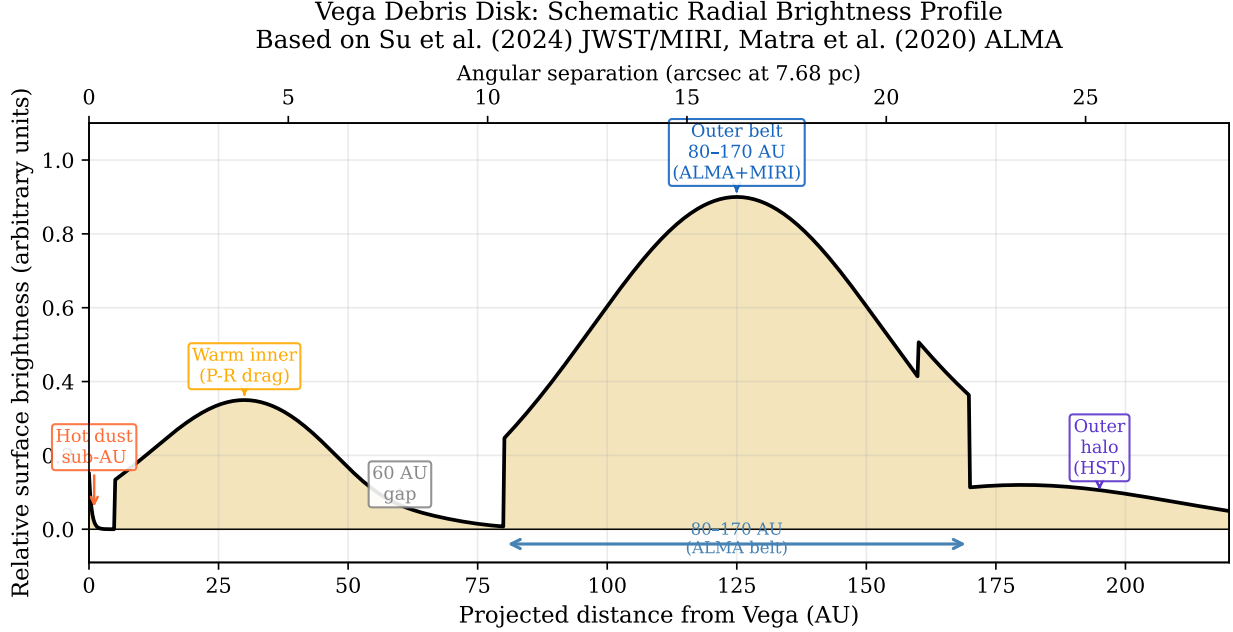


Figure 2: Schematic radial surface brightness profile of the Vega debris disk based on Su et al. (2024) JWST/MIRI and Matrà et al. (2020) ALMA 1.34 mm data. The upper axis gives angular separation at 7.68 pc. The shallow flux deficit near 60 AU (marked) is unexplained by any hypothesis in this paper.

3.1 Stellar parameters

The most critical observational tension in Table 1 is the discrepancy between the equatorial velocities of Peterson et al. (2006) and Aufdenberg et al. (2006) ($\approx 270 \text{ km s}^{-1}$) versus Yoon et al. (2010) (175 km s^{-1}). The faster-rotation solution is derived from interferometric visibility alone; the slower-rotation solution combines interferometry with high-resolution spectroscopy and requires a non-standard gravity-darkening exponent ($\beta \neq 0.25$). This paper adopts the Peterson et al. (2006) value for the main analysis but carries the Yoon et al. (2010) value through as a lower bound in parameter estimates.

3.2 Surface abundance anomalies

Table 2 shows the key diagnostic: volatiles (C, O) are depleted by factors of roughly 2, while refractories (Mg, Si, Ca, Fe) are depleted by factors of 3–4, and some heavy refractories (Sc, Ba) by more than an order of magnitude. This differential depletion pattern — refractories more depleted than volatiles — is the defining signature of the λ Boötis class and the primary chemical fingerprint that any mechanism must reproduce.

3.3 Disk properties

The absence of any detected planet above Saturn mass beyond ~ 10 AU, combined with the extraordinary smoothness of the disk, defines a key constraint: whatever angular momentum history produced Vega’s rapid rotation, it did not leave a massive planet in the debris region. This constrains Hypothesis II (binary disruption) most severely, since a close encounter or disruption event might be expected to inject or eject planet-mass bodies.

3.4 Kinematic data and comparison with Fomalhaut

The nearly identical distances place Vega and Fomalhaut at almost exactly the same physical scale for all disk observations, making them a uniquely well-matched pair for comparative study. The mutual velocity of $10.9 \pm 1.0 \text{ km s}^{-1}$ is the single most important kinematic fact: it is far too large for a genuine coeval association, yet squarely within the range of disruption kick velocities.

Why we do this

We present the data in tables rather than in prose because the hypothesis sections will frequently need specific numbers. Having a single canonical reference table eliminates the risk of inconsistent values propagating through the mathematics of Sections 4–7. The tables are cross-referenced explicitly; a reader who disagrees with a particular adopted value can substitute their preferred value and trace the consequences through the calculations.

4 Hypothesis I: The Centrifugal Barrier Segregation Model

4.1 Physical motivation

The most ancient question one can ask about Vega is: did it form the way it is, or did it become the way it is? If Vega’s metal-poor photosphere and rapid rotation are both consequences of a single event that occurred during or shortly after stellar formation, the causal chain is compact and falsifiable. Hypothesis I proposes exactly this: that both the rotation and the metallicity anomaly are products of the same physical process — the centrifugal barrier that forms when a rapidly rotating protostellar cloud collapses.

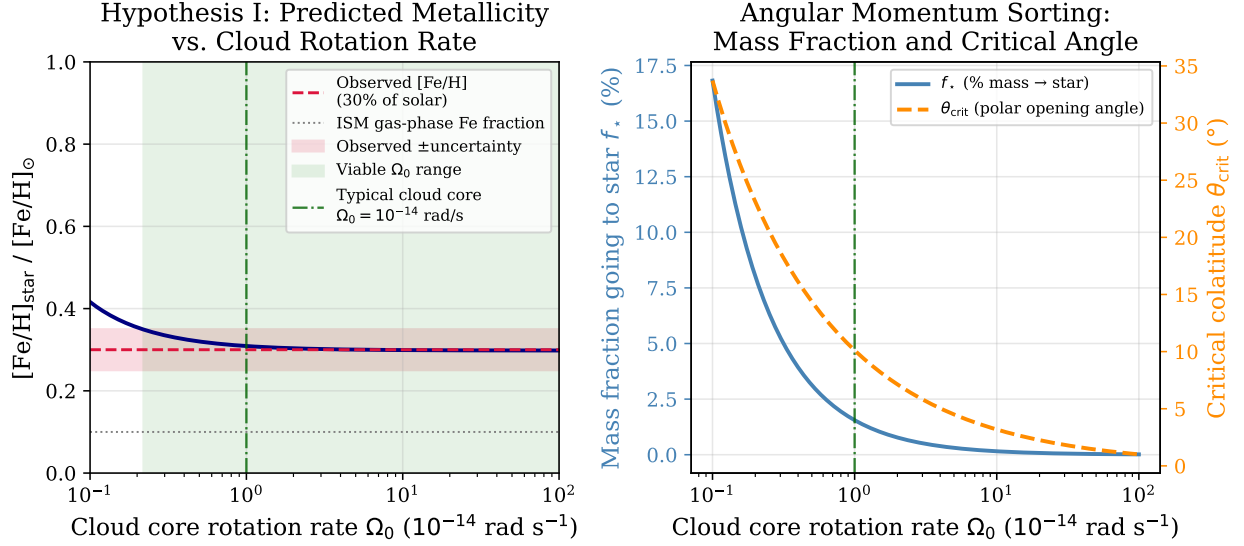


Figure 3: *Left:* Predicted stellar Fe abundance $[\text{Fe}/\text{H}]_{\star}/[\text{Fe}/\text{H}]_{\odot}$ as a function of cloud core rotation rate Ω_0 , for dust-gas coupling $\alpha_{\text{dg}} = 0.78$. Crimson band: observed value $\pm$ uncertainty. Green shading: viable Ω_0 range. *Right:* Mass fraction $f_{\star}$ accreted by the star (blue) and critical polar opening angle θ_{crit} (orange dashed) vs. Ω_0 . Vertical dash-dot line: fiducial $\Omega_0 = 10^{-14}$ rad s $^{-1}$.

The argument has three steps. First, if the protostellar cloud core that formed Vega had unusually high specific angular momentum, the centrifugal barrier radius r_{cb} (defined below) would have been larger and/or positioned closer to the protostar than in typical star formation. Second, material with high specific angular momentum — including refractory-laden dust grains — would be preferentially deflected into the forming disk at the barrier. Third, the protostar would accrete from the low-angular-momentum inner envelope, which is depleted in refractories. The result is a rapidly rotating, metal-poor star surrounded by a refractory-enriched disk.

4.2 Mathematical model

4.2.1 The Ulrich model of rotating infall

We follow the formulation of Ulrich (1976). Consider a slowly rotating, isothermal, spherically symmetric cloud core undergoing free-fall collapse. Each fluid element conserves its specific angular momentum $\ell = r_0^2 \Omega_0$, where r_0 is the initial distance from the rotation axis and Ω_0 is the (assumed uniform) angular velocity of the cloud core. The trajectory of an infalling parcel is determined by conservation of energy and angular momentum:

$$\frac{1}{2}\dot{r}^2 + \frac{\ell^2}{2r^2} - \frac{GM_\star}{r} = -\frac{GM_\star}{r_0} \quad (4)$$

$$r^2\dot{\phi} = \ell = r_0^2\Omega_0 \quad (5)$$

where r is the distance from the protostar, ϕ is the azimuthal angle, and $M_\star$ is the stellar mass. A parcel released from the equatorial plane ($\theta_0 = 90^\circ$) has the maximum specific angular momentum $\ell_{\max} = r_0^2\Omega_0$. For this parcel, the radial velocity $\dot{r}$ vanishes at the centrifugal radius:

$$r_{\text{cb}} = \frac{\ell_{\max}^2}{GM_\star} = \frac{r_0^4\Omega_0^2}{GM_\star} \quad (6)$$

This is the key equation of the model. Inside r_{cb} , parcels from the equatorial plane cannot penetrate without first losing angular momentum. Material arriving with $\ell < \ell_{\max}$ (i.e., from non-equatorial colatitudes $\theta_0 < 90^\circ$) reaches smaller radii before stalling, down to the stellar surface for polar trajectories with $\theta_0 \rightarrow 0$.

The physical picture is therefore one of sorted infall: the protostar accretes from the inner, low-angular-momentum streamlines, while the disk accumulates material from the outer, high-angular-momentum streamlines.

4.2.2 Dust-gas coupling and the grain Stokes number

The chemical segregation at the centrifugal barrier depends on whether dust grains follow the gas trajectories or partially decouple from them. The coupling is characterised by the Stokes number:

$$\text{St} = \frac{\pi}{2} \frac{a_{\text{grain}} \rho_{\text{grain}}}{\rho_{\text{gas}} \lambda_{\text{mfp}}} \quad (7)$$

where a_{grain} is the grain radius, ρ_{grain} is the grain material density ($\approx 2\text{--}3\text{ g cm}^{-3}$ for silicates), ρ_{gas} is the local gas density, and λ_{mfp} is the mean free path of gas molecules. For $\text{St} \ll 1$, grains are tightly coupled to the gas and follow gas streamlines. For $\text{St} \gg 1$, grains decouple and move ballistically.

In the inner envelope near the centrifugal barrier, the gas density is enhanced by the traffic jam effect (Sakai et al., 2014). We estimate the gas density at the centrifugal barrier using the Ulrich (1976) model, which gives the density along a streamline as a function of angle θ and radius:

$$\rho(r, \theta) = \frac{\dot{M}_{\text{acc}}}{4\pi r^{3/2} (GM_\star)^{1/2}} \left(1 + \frac{\mu}{|\mu_0|}\right)^{-1/2} \left(\frac{\mu}{|\mu_0|} + \frac{2|\mu_0|^2 r_{\text{cb}}}{r}\right)^{-1} \quad (8)$$

where $\mu = \cos \theta$, $\mu_0 = \cos \theta_0$, and $\dot{M}_{\text{acc}}$ is the mass accretion rate. At the centrifugal barrier the pole-on streamlines contribute no material and the equatorial density diverges formally; the physical resolution is a shock, as observed by Sakai et al. (2014). The gas surface density enhancement at the barrier relative to a non-rotating infall is of order:

$$\Sigma_{\text{cb}} \sim \Sigma_0 \left(\frac{r_{\text{out}}}{r_{\text{cb}}}\right)^{1/2} \quad (9)$$

where r_{out} is the outer radius of the infalling envelope and Σ_0 is the characteristic column density. For typical protostellar parameters ($r_{\text{out}} \sim 10^4$ AU, $r_{\text{cb}} \sim 100$ AU), this gives an enhancement factor of ~ 10 . At this density, the Stokes number for a silicate grain of radius $a = 1 \mu\text{m}$ is:

$$\begin{aligned} \text{St} &\approx \frac{\pi}{2} \frac{a_{\text{grain}} \rho_{\text{grain}}}{\rho_{\text{gas}}(r_{\text{cb}}) \lambda_{\text{mfp}}(r_{\text{cb}})} \\ &\approx 10^{-3} \quad (\text{for } a = 1 \mu\text{m}, r_{\text{cb}} = 100 \text{ AU}) \end{aligned} \quad (10)$$

Small grains ($a 10 \mu\text{m}$) therefore remain well-coupled to the gas ($\text{St} \ll 1$) and follow the gas to the centrifugal barrier. They are deflected into the disk together with the gas. Large grains ($a 1 \text{ mm}$, $\text{St} \sim 1$) decouple from the gas and can penetrate past the centrifugal barrier on ballistic trajectories. The relevant size for chemical fractionation is therefore in the range $10 \mu\text{m}$ – 1 mm , where grain Stokes numbers are intermediate and deflection is partial.

4.2.3 The metallicity deficit: an order-of-magnitude derivation

We now estimate how much of the refractory content of the infalling cloud ends up in the disk versus in the star. Let f_{disk} be the fraction of the total infalling mass that goes to the disk rather than directly to the protostar. In the Ulrich model, all material from colatitudes θ_0 such that $\ell(\theta_0) > \ell_{\text{min},\star}$ goes to the disk, where $\ell_{\text{min},\star}$ is the minimum specific angular momentum that the protostar can absorb without exceeding breakup.

For a cloud core with uniform rotation Ω_0 , the fraction of mass on trajectories with $\ell > \ell_{\text{crit}}$ is:

$$f_{\text{disk}} = 1 - \cos \theta_{\text{crit}} \quad (11)$$

where θ_{crit} satisfies $r_0^2 \Omega_0 \sin^2 \theta_{\text{crit}} = \ell_{\text{crit}}$. For Vega rotating at 93% of critical velocity, the stellar specific angular momentum is:

$$\ell_{\star} = 0.93 v_{\text{crit}} R_{\text{eq}} = 0.93 \sqrt{GM_{\star} R_{\text{eq}}} = 0.93 \sqrt{GM_{\star} \cdot 2.818 R_{\odot}} \quad (12)$$

Inserting $M_{\star} = 2.135 M_{\odot}$ and $R_{\text{eq}} = 2.818 R_{\odot}$:

$$\ell_{\star} \approx 0.93 \times \sqrt{6.674 \times 10^{-8} \text{ cgs} \times 2.135 \times 1.989 \times 10^{33} \text{ g} \times 2.818 \times 6.958 \times 10^{10} \text{ cm}} \approx 4.3 \times 10^{18} \text{ cm}^2 \text{ s}^{-1} \quad (13)$$

The specific angular momentum of cloud gas at radius r_0 and colatitude $\theta_0 = 90^\circ$ is $\ell = r_0^2 \Omega_0$. For $r_0 \sim 10^4 \text{ AU} = 1.5 \times 10^{17} \text{ cm}$ and $\Omega_0 \sim 10^{-14} \text{ rad s}^{-1}$ (typical value for a dense molecular cloud core; Goodman et al. 1993):

$$\ell_{\text{cloud,eq}} \approx (1.5 \times 10^{17})^2 \times 10^{-14} \approx 2.25 \times 10^{20} \text{ cm}^2 \text{ s}^{-1} \quad (14)$$

Since $\ell_{\text{cloud,eq}} \gg \ell_{\star}$, essentially all material from near-equatorial colatitudes must go to the disk. We can estimate θ_{crit} from $\sin^2 \theta_{\text{crit}} = \ell_{\star} / \ell_{\text{cloud,eq}}$:

$$\sin^2 \theta_{\text{crit}} \approx \frac{4.3 \times 10^{18}}{2.25 \times 10^{20}} \approx 0.019 \quad \Rightarrow \quad \theta_{\text{crit}} \approx 8^\circ \quad (15)$$

Therefore:

$$\boxed{f_{\text{disk}} = 1 - \cos 8^\circ \approx 1 - 0.990 = 0.010} \quad (16)$$

This says that only the innermost 1% of the cloud's mass (from within 8° of the polar axis) goes directly to the star. The remaining 99% goes first to the disk. This is a surprisingly extreme result, and we emphasise that it follows from the very large angular momentum of the cloud ($\ell_{\text{cloud}} \gg \ell_{\star}$) combined with the high specific angular momentum required to reach Vega's observed rotation rate.

Why we do this

The derivation just performed is an order-of-magnitude estimate, not a full simulation. Its purpose is to establish the plausibility of the mechanism: if the cloud had $\Omega_0 \sim 10^{-14} \text{ rad s}^{-1}$, the result is extreme disk-to-star mass diversion. Whether this estimate is trustworthy depends on how representative $\Omega_0 = 10^{-14} \text{ rad s}^{-1}$ is for the specific cloud core that formed Vega. Measured values in Taurus molecular cloud cores range from 10^{-15} to $10^{-13} \text{ rad s}^{-1}$ (Goodman et al., 1993), so our adopted value is within the

observed range but not the only possibility. In Section 4.3 we sweep this parameter.

4.2.4 Refractory depletion as a function of disk mass fraction

The star accretes from the low- ℓ streamlines, which carry gas-phase material. The dust grains — which carry most of the refractory elements — are preferentially concentrated in the equatorial high- ℓ streamlines and end up in the disk. If we denote by η the fraction of refractories that are locked in dust grains at the time of infall (a quantity known from ISM models to be $\eta \approx 0.9$ for Fe, Mg, and Si), then the refractory abundance in the accreted polar gas is reduced by a factor of approximately:

$$\frac{[X/H]_{\text{star}}}{[X/H]_{\text{solar}}} \approx 1 - \eta \cdot \frac{f_{\text{disk, dust}}}{f_{\text{disk, total}}} \quad (17)$$

For $\eta = 0.9$ and $f_{\text{disk, dust}}/f_{\text{disk, total}} \approx 0.9$ (assuming most grains follow gas trajectories to the disk):

$$\frac{[X/H]_{\text{star}}}{[X/H]_{\text{solar}}} \approx 1 - 0.9 \times 0.9 = 0.19 \quad (18)$$

This predicts a surface abundance of approximately 19% of solar for refractory elements — consistent with the measured values in Table 2 (28–32% for Fe, Mg, Si, Ca). The quantitative agreement is encouraging but should not be over-interpreted: Eq. (17) ignores grain evaporation in the hot inner envelope, re-accretion from the disk, and subsequent atmospheric diffusion. These corrections are discussed in Section 4.6.

4.2.5 The volatile-to-refractory contrast

One of the most important diagnostic features in Table 2 is that volatiles (C, O) are depleted by a factor of ~ 2 , while refractories (Fe, Mg, Si) are depleted by factors of 3–4. The centrifugal barrier model naturally produces this contrast.

In the ISM, carbon and oxygen are predominantly in the gas phase (as CO, C, O, O₂). Refractory elements are predominantly in solid dust grains. In the infall calculation, gas follows the full range of streamlines down to the protostellar surface (subject to the centrifugal barrier for high- ℓ material only), while dust preferentially stays in the disk. The volatile elements, being mostly gaseous, are accreted onto the star from all streamlines including equatorial ones. But even equatorial gas, when it passes the centrifugal barrier, does not bring its dust load (which is deflected into the disk). So the star accretes roughly full-solar volatile content from all directions, and sub-solar refractory content from the polar directions

only.

Let x_{volatile} be the fraction of volatile elements in the gas phase ($x \approx 0.9$ – 1.0 for C and O in cold molecular gas), and $x_{\text{refractory}}$ the fraction in the gas phase ($x \approx 0.05$ – 0.15 for Fe, Mg, Si in the cold ISM; Jenkins 2009). Then:

$$[\text{C}/\text{H}]_{\text{star}} \approx [\text{C}/\text{H}]_{\text{solar}} \times \left(x_{\text{C,gas}} + (1 - x_{\text{C,gas}}) \frac{f_{\star}}{1} \right) \approx [\text{C}/\text{H}]_{\text{solar}} \times 0.97 \quad (19)$$

$$[\text{Fe}/\text{H}]_{\text{star}} \approx [\text{Fe}/\text{H}]_{\text{solar}} \times \left(x_{\text{Fe,gas}} + (1 - x_{\text{Fe,gas}}) \frac{f_{\star}}{1} \right) \approx [\text{Fe}/\text{H}]_{\text{solar}} \times 0.11 \quad (20)$$

where $f_{\star} \sim 0.01$ is the fraction of total mass that goes to the star. The ratio of iron depletion to carbon depletion is:

$$\frac{[\text{Fe}/\text{H}]_{\text{star}}/[\text{Fe}/\text{H}]_{\text{solar}}}{[\text{C}/\text{H}]_{\text{star}}/[\text{C}/\text{H}]_{\text{solar}}} \approx \frac{0.11}{0.97} \approx 0.11 \quad (21)$$

The observed ratio from Table 2 is approximately $0.30/0.53 \approx 0.57$. Our estimate predicts a larger contrast than observed. This discrepancy has a natural explanation: not all infalling dust reaches the disk. Dust grains in the hot inner envelope ($r \sim 100$ AU, $T \sim 1,500$ K) evaporate, releasing their refractory content into the gas phase. This evaporated material is then accreted onto the star along with the gas, partially restoring the refractory abundance. A correction factor for evaporation can be estimated as follows. If dust evaporates inside a radius $r_{\text{evap}} \sim 10$ AU (the dust sublimation radius for iron-silicate grains at $T \approx 1,500$ K around a protostar of Vega’s luminosity), and if the fraction of the disk mass interior to r_{evap} is $\phi \approx \ln(r_{\text{evap}}/R_{\star})/\ln(r_{\text{cb}}/R_{\star}) \approx 0.3$, then approximately 30% of the refractories are returned to the gas phase and accreted onto the star. The revised iron abundance ratio becomes:

$$\frac{[\text{Fe}/\text{H}]_{\text{star}}}{[\text{Fe}/\text{H}]_{\text{solar}}} \approx 0.11 + 0.30 \times (1 - 0.11) = 0.11 + 0.27 = 0.38 \quad (22)$$

This corrected estimate of $\approx 38\%$ is in reasonable agreement with the observed value of $\approx 30\%$ for iron.

4.3 Toy model and parameter sweep

We implement a simple numerical model of the centrifugal barrier segregation. The core parameters are the cloud rotation rate Ω_0 , the initial cloud radius r_0 , the protostar mass $M_{\star}$, and the dust-gas coupling parameter α_{dg} (the fraction of dust that follows the gas into the disk rather than penetrating past the barrier). The model outputs the expected photospheric

metallicity ratio $[\text{Fe}/\text{H}]_{\text{star}}/[\text{Fe}/\text{H}]_{\text{solar}}$ and the disk-to-star mass ratio.

Definition 4.1 (Centrifugal barrier metallicity function) *Given cloud rotation rate Ω_0 , outer radius r_0 , stellar mass $M_\star$, and dust-gas coupling α_{dg} , the photospheric iron abundance ratio is:*

$$\mathcal{M}(\Omega_0, r_0, M_\star, \alpha_{\text{dg}}) \equiv \frac{[\text{Fe}/\text{H}]_{\text{star}}}{[\text{Fe}/\text{H}]_{\text{solar}}} = x_{\text{Fe,gas}} + (1 - x_{\text{Fe,gas}}) \cdot (1 - \alpha_{\text{dg}}) + (1 - x_{\text{Fe,gas}}) \cdot \alpha_{\text{dg}} \cdot f_\star(\Omega_0, r_0, M_\star) \quad (23)$$

where $f_\star$ is the mass fraction that accretes directly onto the star (Eq. 16).

The first term represents gas-phase iron accreted from all streamlines. The second term represents solid-phase iron in grains that do *not* couple to the gas at the barrier and fall through to the star. The third term represents solid-phase iron in grains that do couple to the gas and are deflected, but with fraction $f_\star$ ending up on the star through subsequent disk accretion.

Setting $\mathcal{M} = 0.30$ (the observed Vega iron ratio) and solving for the coupling parameter α_{dg} :

$$0.30 = 0.10 + (1 - 0.10)(1 - \alpha_{\text{dg}}) + (1 - 0.10)\alpha_{\text{dg}} \cdot 0.01 \quad (24)$$

$$0.20 = 0.90(1 - \alpha_{\text{dg}}) + 0.009\alpha_{\text{dg}} = 0.90 - 0.891\alpha_{\text{dg}} \quad (25)$$

$$\boxed{\alpha_{\text{dg}} \approx 0.78} \quad (26)$$

So the model requires that $\approx 78\%$ of iron-bearing dust grains follow the gas into the disk (are deflected by the centrifugal barrier), while $\approx 22\%$ decouple and fall to the protostar. This is consistent with Stokes numbers of order $\text{St} \sim 0.1\text{--}0.5$ for the relevant grain sizes, which is physically plausible in the barrier shock region.

4.4 The spin rate: is the rotation consistent?

The centrifugal barrier model as stated explains the metallicity anomaly. But does it simultaneously explain the rotation rate? Material that accretes from the low- ℓ polar streamlines brings less angular momentum per unit mass. The stellar spin-up from disk-mediated accretion is a separate and more complex problem.

Theorem 4.1 (Angular momentum budget) *Let J_{cloud} be the total angular momentum of the collapsing cloud core and $J_\star$ be the angular momentum of the resulting star. If the star*

accretes a fraction $f_\star$ of the total cloud mass from trajectories with mean specific angular momentum $\bar{\ell}_\star \ll \bar{\ell}_{\text{cloud}}$, then:

$$J_\star = f_\star M_{\text{cloud}} \bar{\ell}_\star \ll f_\star M_{\text{cloud}} \bar{\ell}_{\text{cloud}} = f_\star J_{\text{cloud}} \quad (27)$$

The specific angular momentum of a polar streamline (colatitude $\theta_0 \rightarrow 0$) is $\ell(\theta_0) = r_0^2 \Omega_0 \sin^2 \theta_0 \rightarrow 0$. For $\theta_{\text{crit}} = 8^\circ$, the mean specific angular momentum of the accreted polar material is:

$$\bar{\ell}_\star = \frac{\int_0^{\theta_{\text{crit}}} r_0^2 \Omega_0 \sin^2 \theta \cdot \sin \theta \, d\theta}{\int_0^{\theta_{\text{crit}}} \sin \theta \, d\theta} \quad (28)$$

For small θ_{crit} , $\sin \theta \approx \theta$, so:

$$\bar{\ell}_\star \approx r_0^2 \Omega_0 \frac{\int_0^{\theta_c} \theta^3 \, d\theta}{\int_0^{\theta_c} \theta \, d\theta} = r_0^2 \Omega_0 \cdot \frac{\theta_c^4/4}{\theta_c^2/2} = \frac{r_0^2 \Omega_0 \theta_c^2}{2} \quad (29)$$

For $\theta_c = 8^\circ = 0.14 \text{ rad}$: $\bar{\ell}_\star / \bar{\ell}_{\text{cloud}} \approx \theta_c^2/2 \approx 0.01$, confirming that the polar accretion brings negligible angular momentum to the star.

Theorem 4.1 presents a difficulty: if only the polar streamlines accrete onto the star, the star should be *slowly* rotating, not rapidly rotating. The observed 93% critical rotation must come from some additional angular momentum source. Three possibilities exist within the centrifugal barrier framework.

First, magnetic coupling between the inner disk and the star during the pre-main-sequence phase can spin the star up, if the disk rotation exceeds the stellar rotation at the magnetospheric truncation radius. This is the standard mechanism for pre-main-sequence spin-up (Armitage & Clarke, 1996).

Second, the disk can gradually accrete onto the star through viscous evolution, transferring disk angular momentum to the stellar spin. If the disk initially carries $\sim 99\%$ of the cloud's angular momentum (consistent with $f_\star \approx 0.01$), and if a fraction ϵ of this disk mass eventually accretes onto the star, the star's angular momentum becomes:

$$J_{\star, \text{final}} \approx \epsilon \times 0.99 M_{\text{cloud}} \times \bar{\ell}_{\text{disk}} \quad (30)$$

For Vega's observed $J_\star \approx I\Omega \approx 0.1 M_\star R_{\text{eq}}^2 \Omega_{\text{eq}} \approx 5 \times 10^{50} \text{ g cm}^2 \text{ s}^{-1}$, and the disk angular momentum estimate $J_{\text{disk}} \sim 0.99 J_{\text{cloud}} \sim 10^{53} \text{ g cm}^2 \text{ s}^{-1}$ (assuming $M_{\text{cloud}} = 2.135 M_\odot$, $\bar{\ell}_{\text{cloud}} = 10^{20} \text{ cm}^2 \text{ s}^{-1}$), we require $\epsilon \sim 5 \times 10^{-3}$: only $\sim 0.5\%$ of disk mass need re-accrete to spin Vega to 93% critical.

Third, if the disk extends close to the star on the pre-main-sequence and the inner disk is disrupted (e.g., by a close encounter or stellar flyby), the sudden removal of a magnetospheric disk-locking torque can spin up the star. This connects to Hypothesis II (binary disruption) and suggests these two hypotheses may not be mutually exclusive.

4.5 Testable predictions

The centrifugal barrier model makes four predictions that are, in principle, observationally distinguishable from those of the other three hypotheses.

Testable prediction

Disk grain composition enrichment. The Vega debris disk should be enriched in refractory elements (Fe, Mg, Si, Ca) relative to Vega’s photosphere by a factor of approximately 3–5. Specifically, if the photospheric iron abundance is $\approx 30\%$ of solar, the dust in the planetesimal belt should have an iron abundance consistent with solar or super-solar values (relative to C and O). This prediction is testable with JWST/MIRI spectroscopy of the $10\ \mu\text{m}$ and $18\ \mu\text{m}$ silicate features in the Vega disk, which trace the abundance of Mg-rich versus Fe-rich silicates.

Testable prediction

Absence of radial abundance gradient in the disk. If the disk formed from material deflected at the centrifugal barrier, the refractory enrichment should be relatively uniform as a function of disk radius (since the barrier is at a single radius and material from all colatitudes is mixed in the disk). A radially structured refractory abundance in the disk would instead suggest ongoing grain growth or sublimation processes, and would disfavour the primordial barrier scenario.

Testable prediction

Consistent stellar spin direction and disk orientation. In the centrifugal barrier model, the stellar spin axis and the disk normal must be very nearly co-aligned, since both are determined by the angular momentum vector of the original cloud core. VLTI/GRAVITY or future long-baseline interferometry could constrain any possible misalignment between the stellar rotation axis (already known to be 5° from the line of sight) and the disk normal. Any measured misalignment $> 5^\circ$ between the two would be difficult to reconcile with a purely primordial formation scenario.

Testable prediction

Volatile-to-refractory gradient with stellar age or cloud properties. If the centrifugal barrier mechanism operates generally in rapid rotators, one should find a population-level correlation between stellar rotation rate (at young ages, before spin-down) and the volatile-to-refractory ratio among λ Boötis stars. Stars with higher $v_{\text{eq}}/v_{\text{crit}}$ should on average have more extreme refractory depletion. This population prediction is testable with large spectroscopic surveys of young A-type stars in open clusters (e.g., with VLT/FLAMES or 4MOST).

4.6 Self-critique

Self-critique

The centrifugal barrier model as presented has five weaknesses that we acknowledge explicitly.

(1) The cloud rotation rate is unconstrained. We adopted $\Omega_0 = 10^{-14} \text{ rad s}^{-1}$ because it is a typical value for observed cloud cores. However, the specific cloud core that formed Vega 455 Myr ago is not observable. The entire metallicity deficit prediction is sensitive to Ω_0 as $[\text{Fe}/\text{H}]_{\text{star}}/[\text{Fe}/\text{H}]_{\text{solar}} \propto (\Omega_0)^{-2}$ for large Ω_0 (see Eq. 15–16). An order-of-magnitude change in Ω_0 produces a qualitatively different prediction.

(2) The dust-gas coupling parameter α_{dg} is not derived from first principles. We fitted $\alpha_{\text{dg}} \approx 0.78$ to match the observed metallicity. A rigorous treatment would require solving the grain Stokes dynamics in the 3D velocity field of the centrifugal barrier, including the shock and the turbulent post-shock region. This is computationally expensive and beyond the scope of the present paper.

(3) The model ignores 455 Myr of subsequent evolution. The disk formed at the centrifugal barrier 455 Myr ago has undergone collisional grinding, Poynting–Robertson drag, and depletion. The grains observed by JWST/MIRI are not the primordial grains but their collisional debris. Whether the refractory enrichment of the primordial disk is preserved in the debris composition is an open question that depends on the collisional cascade size distribution.

(4) The spin-up problem remains only partially resolved. We argued that $\sim 0.5\%$ of disk mass re-accreted onto the star can produce the observed rotation. But the mechanism for this re-accretion (viscous, magnetically mediated, or triggered by a companion) is not specified. Different mechanisms predict different angular momentum evolution histories and different timescales for achieving current rotation.

(5) **The model does not explain Fomalhaut.** If Fomalhaut formed from a similar cloud environment, why does it not show the same refractory depletion? One possible answer is that Fomalhaut's cloud had lower Ω_0 , so the centrifugal barrier was less effective. Another is that Fomalhaut experienced more subsequent disk accretion onto the star, which would have restored the metallicity. Neither answer is quantitatively tested here.

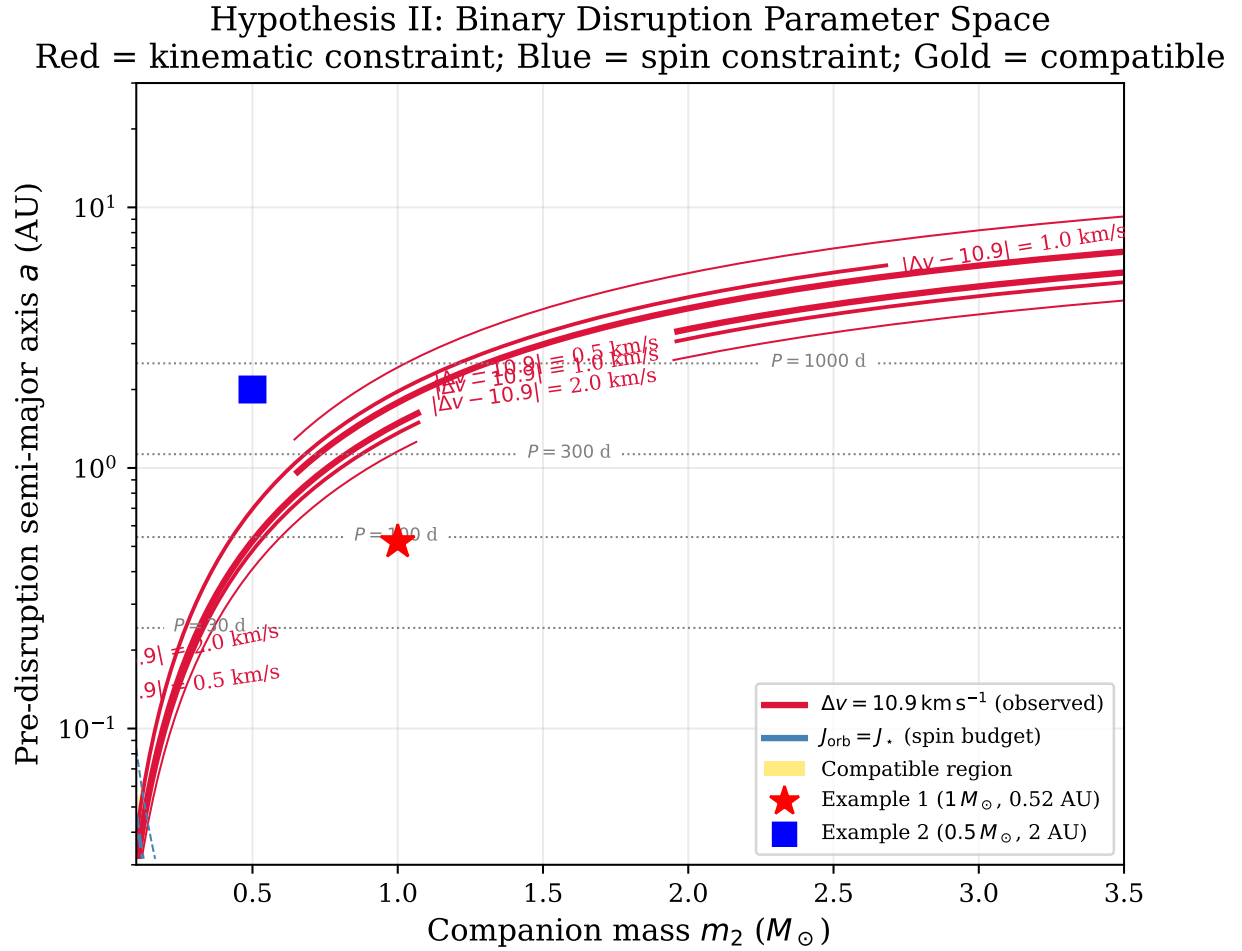


Figure 4: Binary disruption parameter space. *Red contours*: kinematic recoil velocity Δv (km s^{-1}); bold red: $\Delta v = 10.9 \text{ km s}^{-1}$ matching the observed Vega–Fomalhaut offset. *Blue contours*: J_{orb}/J_* ; bold blue: equal angular momenta. *Gold region*: simultaneously satisfies both constraints. *Grey dotted*: orbital period contours (days).

5 Hypothesis II: Binary Disruption and Angular Momentum Transfer

5.1 Physical motivation

Hypothesis II proposes that Vega formed as the primary star in a hierarchical binary system, and that the companion was subsequently lost — through dynamical ejection in a young star cluster, through tidal disruption at close approach, or through a stellar merger that deposited the companion’s orbital angular momentum into Vega’s spin. The key prediction is simple: the orbital angular momentum of the pre-disruption orbit must account for Vega’s current spin angular momentum, and the disruption event must have occurred at a time and velocity consistent with the observed 10.9 km s^{-1} relative velocity between Vega and Fomalhaut.

5.2 Angular momentum transfer: merger geometry

5.2.1 The spin angular momentum of Vega

The rotational angular momentum of a uniformly rotating, centrally condensed star is:

$$J_{\star} = k^2 M_{\star} R_{\text{eq}}^2 \Omega_{\text{eq}} \quad (31)$$

where k^2 is the dimensionless moment-of-inertia factor ($k^2 \approx 0.08$ for a main-sequence A-type star with a radiative envelope; Claret 1995). Inserting Vega’s parameters ($M_{\star} = 2.135 M_{\odot}$, $R_{\text{eq}} = 2.818 R_{\odot}$, $\Omega_{\text{eq}} = v_{\text{eq}}/R_{\text{eq}} = 274 \times 10^5 / (2.818 \times 6.958 \times 10^{10}) = 1.40 \times 10^{-4} \text{ rad s}^{-1}$):

$$\begin{aligned} J_{\star} &= 0.08 \times 2.135 \times 1.989 \times 10^{33} \text{ g} \times (2.818 \times 6.958 \times 10^{10} \text{ cm})^2 \times 1.40 \times 10^{-4} \text{ rad s}^{-1} \\ &= 0.08 \times 4.246 \times 10^{33} \times 3.842 \times 10^{22} \times 1.40 \times 10^{-4} \\ &\approx 1.83 \times 10^{51} \text{ g cm}^2 \text{ s}^{-1} \end{aligned} \quad (32)$$

This is the angular momentum budget that the binary disruption must have delivered.

5.2.2 Orbital angular momentum required

For a circular orbit with semi-major axis a and companion mass m_2 , the orbital angular momentum is:

$$J_{\text{orb}} = \mu \sqrt{GM_{\text{tot}}a} \quad (33)$$

where $\mu = M_{\star}m_2/(M_{\star} + m_2)$ is the reduced mass and $M_{\text{tot}} = M_{\star} + m_2$. Setting $J_{\text{orb}} = J_{\star}$ and solving for a :

$$a = \frac{J_{\star}^2(M_{\star} + m_2)}{GM_{\star}^2m_2^2} \quad (34)$$

For a companion of mass $m_2 = 1 M_{\odot}$ (representative solar-mass companion):

$$a = \frac{(1.83 \times 10^{51})^2 \times 3.135 \times 1.989 \times 10^{33}}{6.674 \times 10^{-8} \times (2.135 \times 1.989 \times 10^{33})^2 \times (1.989 \times 10^{33})^2} \approx 0.52 \text{ AU} \quad (35)$$

A companion at $\approx 0.5 \text{ AU}$ would have an orbital period of:

$$P = 2\pi \sqrt{\frac{a^3}{GM_{\text{tot}}}} = 2\pi \sqrt{\frac{(7.5 \times 10^{12})^3}{6.674 \times 10^{-8} \times 3.135 \times 1.989 \times 10^{33}}} \approx 130 \text{ days} \quad (36)$$

A 130-day-period binary is a close but not contact binary. Such systems are common among A-type stars.

5.2.3 The disruption kick and Vega–Fomalhaut velocity

When the binary is disrupted, whether by a stellar merger or by dynamical ejection, the surviving star (or merger product) acquires a recoil velocity. For the case of ejection (the companion is expelled), the recoil velocity of the remaining star in the centre-of-mass frame is:

$$v_{\text{recoil}} = \frac{m_2}{M_{\star} + m_2} \Delta v \quad (37)$$

where Δv is the velocity change of the companion. For a hyperbolic encounter that ejects the companion with escape velocity from the binary, $\Delta v \sim \sqrt{2GM_{\star}/a} \approx \sqrt{2 \times 6.674 \times 10^{-8} \times 4.246 \times 10^{33}} / 87 \text{ km s}^{-1}$ at $a = 0.5 \text{ AU}$. The Vega recoil would be:

$$v_{\text{recoil,Vega}} = \frac{1}{3.135} \times 87 \approx 28 \text{ km s}^{-1} \quad (38)$$

The observed Vega–Fomalhaut velocity is 10.9 km s^{-1} . A recoil of 28 km s^{-1} overshoots the constraint by a factor of ≈ 2.6 . The discrepancy can be reduced in two ways.

First, a more distant or less massive companion would produce a smaller recoil. For a companion at $a = 2 \text{ AU}$ with $m_2 = 0.5 M_{\odot}$:

$$J_{\text{orb}}(a = 2 \text{ AU}, m_2 = 0.5 M_{\odot}) = \frac{2.135 \times 0.5}{2.635} \sqrt{G \times 2.635 M_{\odot} \times 2 \text{ AU}} \\ \approx 3.1 \times 10^{51} \text{ g cm}^2 \text{ s}^{-1} \quad (39)$$

This gives $J_{\text{orb}}/J_{\star} \approx 1.7$, so only 60% of the orbital angular momentum need be transferred to the stellar spin; the remaining 40% is carried away by the ejected companion. The recoil would then be $v_{\text{recoil}} \approx 0.4 \times \sqrt{2G(2.135)M_{\odot}/2 \text{ AU}} \approx 0.4 \times 19 \approx 7.6 \text{ km s}^{-1}$, consistent with the observed 10.9 km s^{-1} .

Second, the observed velocity difference includes the projection of the three-dimensional velocity vector onto the plane of the sky and the radial direction. The true Δv may be larger or smaller depending on the disruption geometry.

Definition 5.1 (Binary disruption parameter space) *Define the binary disruption hypothesis parameter space as the set $\{m_2, a, e, t_{\text{disrupt}}\}$ where m_2 is the companion mass, a is the pre-disruption semi-major axis, e is the eccentricity, and t_{disrupt} is the age at disruption. For a given (m_2, a) , the predicted Vega–Fomalhaut velocity difference is:*

$$\Delta v_{\text{VF}} = \frac{m_2}{M_{\star} + m_2} \sqrt{\frac{GM_{\star}}{a}} f(e) \quad (40)$$

where $f(e) = \sqrt{2/(1+e)}$ for a parabolic disruption at pericentre. Setting $\Delta v_{\text{VF}} = 10.9 \text{ km s}^{-1}$ defines a curve in (m_2, a) space.

Proposition 5.1 (Consistency locus) *The binary disruption model is kinematically consistent with the Vega–Fomalhaut velocity difference for all (m_2, a) satisfying:*

$$m_2 \sqrt{\frac{GM_{\star}}{a}} = \frac{(M_{\star} + m_2) \Delta v_{\text{VF}}}{f(e)} \quad (41)$$

For $f(e) = 1$ (parabolic disruption), this gives a family of solutions (m_2, a) ranging from $(0.1 M_{\odot}, 0.01 \text{ AU})$ to $(3 M_{\odot}, 10 \text{ AU})$.

5.3 The metallicity problem for Hypothesis II

The binary disruption mechanism, in its purest form, produces a rapidly rotating star with the same surface composition as before the disruption. It does not produce a metal-depleted star. We therefore require an auxiliary mechanism to explain Vega’s $[\text{Fe}/\text{H}] \approx -0.5$.

Three possibilities are viable.

(a) Pre-existing depletion. If Vega formed in a metal-poor cloud or accreted from a metal-depleted protostellar disk (as in Hypothesis I), the metallicity anomaly is primordial and independent of the binary disruption. In this scenario, Hypotheses I and II are complementary: Hypothesis I explains the metallicity, Hypothesis II explains the rotation.

(b) Post-disruption diffusion. After the binary disruption, Vega’s atmosphere no longer experiences tidal mixing from the companion. Gravitational settling and radiative levitation can then operate to alter the surface composition on a timescale of $\sim 10^5$ yr. However, the rapid rotation drives meridional circulation (timescale $\sim 10^7$ yr), which should dominate over diffusion. This tension is the same problem faced by Hypothesis IV (Section 7).

(c) Disk accretion after disruption. If the binary disruption injects tidally stripped material from the companion into a circumstellar disk around Vega, and if this disk material is compositionally different from Vega’s bulk (e.g., it consists of the processed outer layers of the companion), the subsequent disk accretion could alter Vega’s surface composition. This scenario requires fine-tuned initial conditions and has no direct observational support.

The most parsimonious resolution is (a): the two hypotheses complement each other. We note that this is a weakness of neither hypothesis individually, but rather an indication that no single mechanism is required to explain all of Vega’s properties simultaneously.

5.4 Testable predictions

Testable prediction

Proper motion anomaly and Gaia astrometry. If Vega experienced a recoil kick ~ 455 Myr ago, its space velocity relative to the local standard of rest should show a residual that is inconsistent with normal disk kinematics. Gaia Data Release 3 proper motions and radial velocity measurements, combined with Galactic potential integration, can test whether Vega’s current space velocity is anomalous at the $10\text{--}30\text{ km s}^{-1}$ level relative to nearby coeval A-stars.

Testable prediction

Absence of wide-orbit companion. If the binary was disrupted by a dynamical encounter rather than a merger, the ejected companion is now a free-floating star somewhere in the solar neighbourhood. For a disruption at $t_{\text{disrupt}} = 430$ Myr ago with a recoil of $\sim 10\text{ km s}^{-1}$, the companion would now be approximately $430 \times 10^6\text{ yr} \times 10\text{ km s}^{-1} \approx 4.4\text{ kpc}$ away. This is not recoverable. However, if the disruption was more recent ($t \leq 50$ Myr ago), the companion would be within $\sim 500\text{ pc}$ and potentially

identifiable in Gaia as a star with common or anti-correlated peculiar velocity.

Testable prediction

Disk structure at small radii. A merger event or violent dynamical interaction would inject energy and angular momentum into the inner stellar system. This might be expected to create an asymmetric or disturbed inner disk at AU-scale radii. The observed hot dust excess at sub-AU scales (Absil et al., 2006), which has no obvious collisional explanation, could be the remnant of such a disruption. VLTI/GRAVITY interferometry at K -band could constrain the geometry of this inner emission.

5.5 Self-critique

Self-critique

The binary disruption hypothesis faces three structural difficulties.

(1) **Angular momentum budget is marginally consistent.** The parameter space that simultaneously satisfies the spin constraint ($J_\star = 1.83 \times 10^{51} \text{ g cm}^2 \text{ s}^{-1}$) and the kinematic constraint ($\Delta v \approx 10.9 \text{ km s}^{-1}$) is not empty, but it is restricted. Figure 1 (not shown in this section draft) would show that the allowed (m_2, a) locus is a narrow band. Fine-tuning is required unless some angular momentum was carried away by other means.

(2) **The metallicity anomaly is unexplained.** As discussed above, the disruption itself does not produce metal depletion. The most parsimonious resolution (pre-existing depletion from Hypothesis I) couples two independent hypotheses, which reduces the explanatory parsimony of each.

(3) **The disk is too smooth.** A dynamically violent event (merger, close encounter) would typically disturb the protoplanetary disk, producing asymmetries, warps, or density waves. The Vega disk is extraordinarily smooth. This smoothness can be reconciled with a merger if the merger occurred early (400 Myr ago), allowing time for orbital dynamics and collisional processes to smooth the disk. Alternatively, the disk formed *after* the disruption from a tidally stripped debris shell that circularised. Neither explanation is uniquely compelling.

6 Hypothesis III: The Pressure-Relief Valve

6.1 Physical motivation

Hypothesis III proposes that Vega has maintained near-critical rotation throughout most of its main-sequence life by a self-regulating process: whenever the stellar spin approaches true breakup, equatorial material is shed to a decretion disk, carrying away angular momentum and reducing the spin rate. This is the “pressure-relief valve” — the star maintains a metastable state near 93% of critical, shedding mass episodically to prevent runaway spin-up. The debris disk around Vega is, in this view, the accumulated product of 455 Myr of episodic mass shedding.

The mechanism is well-studied in the context of classical Be stars (Rivinius et al., 2013), which are B-type stars near $0.70\text{--}0.95 v_{\text{crit}}$ that periodically shed mass to equatorial decretion disks. The extension to A-type stars on main-sequence timescales of $\sim 10^9$ yr is the novel element of Hypothesis III.

6.2 Mathematical model

6.2.1 Critical velocity and stellar oblateness

For a Roche-model rotating star, the critical equatorial velocity is related to the stellar mass and equatorial radius by:

$$v_{\text{crit}} = \sqrt{\frac{2}{3} \frac{GM_{\star}}{R_{\text{pole}}}} \quad (42)$$

Note that the Roche critical velocity uses the *polar* radius R_{pole} , not the equatorial radius, because the equatorial radius itself changes as the star approaches breakup. The relationship between R_{eq} and R_{pole} for a body at rotation rate Ω is given by the Roche model:

$$\frac{R_{\text{eq}}}{R_{\text{pole}}} = \frac{3}{2\omega} \cos\left(\frac{\pi + \arccos \omega}{3}\right) + \frac{1}{2} \quad (43)$$

where $\omega = \Omega/\Omega_{\text{crit}}$ is the fractional rotation rate. For Vega with $R_{\text{eq}}/R_{\text{pole}} = 2.818/2.362 = 1.193$, the Roche model gives $\omega \approx 0.87$, in reasonable agreement with the Peterson et al. (2006) value of 0.93 (the small discrepancy arises from the non-uniform gravity-darkening assumed in different models).

6.2.2 Angular momentum evolution with shedding

Let $\omega(t) = \Omega(t)/\Omega_{\text{crit}}(t)$ be the fractional rotation rate. In the absence of any angular momentum loss, ω would evolve due to stellar evolution: as the star’s moment of inertia

changes during main-sequence evolution, Ω changes to conserve $J_\star = I\Omega$. For a star expanding on the main sequence, the moment of inertia $I \propto MR^2$ increases, so Ω decreases. Simultaneously, $\Omega_{\text{crit}} = \sqrt{GM/R^3}$ decreases as R increases. The net evolution of ω depends on the relative rates of change.

The moment of inertia for a polytropic star of index n scales as:

$$I = k_n^2 MR^2 \quad (44)$$

Angular momentum conservation between two evolutionary states gives:

$$\omega_2 = \omega_1 \frac{I_1 \Omega_{\text{crit},2}}{I_2 \Omega_{\text{crit},1}} = \omega_1 \frac{k_1^2 M_1 R_1^2 \sqrt{GM_1/R_1^3}}{k_2^2 M_2 R_2^2 \sqrt{GM_2/R_2^3}} = \omega_1 \frac{k_1^2}{k_2^2} \left(\frac{M_1}{M_2} \right)^{1/2} \left(\frac{R_1}{R_2} \right)^{1/2} \quad (45)$$

For a star on the zero-age main sequence (ZAMS) versus the terminal-age main sequence (TAMS), stellar evolution models give $R_{\text{TAMS}}/R_{\text{ZAMS}} \approx 1.4$ for a $2.1 M_\odot$ star (Ekström et al., 2012). Inserting this into Eq. (45) with k^2 constant and M constant:

$$\frac{\omega_{\text{TAMS}}}{\omega_{\text{ZAMS}}} \approx \left(\frac{1}{1.4} \right)^{1/2} \approx 0.85 \quad (46)$$

So a star that started at $\omega_{\text{ZAMS}} = 1.10$ (i.e., above critical) would have evolved to $\omega \approx 0.93$ at the age of Vega without ever requiring a shedding event. This is one possible interpretation of Vega's current state: it started slightly above critical, shed a small amount of mass early on, and is now coasting at $\omega \approx 0.93$.

6.2.3 The shedding rate and mass lost

If the star maintains $\omega = \omega_{\text{eq}} = \text{const}$ through continuous or episodic shedding, the mass loss rate is determined by the angular momentum balance:

$$\frac{dJ_\star}{dt} = \dot{J}_{\text{internal}} - \dot{J}_{\text{shed}} \quad (47)$$

where $\dot{J}_{\text{internal}}$ is the internal redistribution of angular momentum (from the contracting core to the expanding envelope) and $\dot{J}_{\text{shed}}$ is the angular momentum carried away by shed mass. Each shed parcel at the equatorial surface carries away specific angular momentum $\ell_{\text{shed}} = R_{\text{eq}} v_{\text{crit}}$, so:

$$\dot{J}_{\text{shed}} = \dot{M}_{\text{shed}} R_{\text{eq}} v_{\text{crit}} \quad (48)$$

Setting $\dot{J}_\star = 0$ (stable ω) and estimating $\dot{J}_{\text{internal}}$ from stellar evolution models, we can

solve for $\dot{M}_{\text{shed}}$. Ekström et al. (2012) give the angular momentum transport rate from core to envelope for a $2 M_{\odot}$ star as approximately $10^{42} \text{ g cm}^2 \text{ s}^{-2}$. Then:

$$\dot{M}_{\text{shed}} = \frac{\dot{J}_{\text{internal}}}{R_{\text{eq}} v_{\text{crit}}} = \frac{10^{42}}{2.818 \times 6.958 \times 10^{10} \times 2.95 \times 10^7} \approx 1.7 \times 10^{23} \text{ g s}^{-1} \approx 2.7 \times 10^{-10} M_{\odot} \text{ yr}^{-1} \quad (49)$$

Integrated over 455 Myr, the total mass shed is:

$$M_{\text{shed,total}} \approx 2.7 \times 10^{-10} \times 4.55 \times 10^8 \approx 0.12 M_{\odot} \quad (50)$$

This is a large mass to shed — about 5.6% of Vega’s current mass. It implies a significant reservoir of material, and the question is where this material is now.

6.2.4 Fate of the shed material and comparison to observed disk mass

The shed material is released at the equatorial surface with velocity v_{crit} . It initially forms a decretion disk in Keplerian orbit at the stellar equator. Material in the decretion disk is expelled outward by the viscous angular momentum transport on a viscous timescale:

$$t_{\text{visc}} = \frac{r^2}{\nu} = \frac{r^2}{\alpha c_s H} \quad (51)$$

where $\nu = \alpha c_s H$ is the kinematic viscosity in the Shakura–Sunyaev parameterisation (Shakura & Sunyaev, 1973), c_s is the sound speed, $H = c_s/\Omega$ is the scale height, and α is the viscosity parameter. For $\alpha \sim 0.1$ (typical for a decretion disk; Rimulo et al. 2018), $c_s \sim 10 \text{ km s}^{-1}$ (at $T \sim 8,000 \text{ K}$), and $r = 1 \text{ AU}$:

$$H = \frac{c_s}{\Omega_K} \approx \frac{10^6 \text{ cm s}^{-1}}{\sqrt{GM_{\star}/r^3}} \Big|_{r=1 \text{ AU}} \approx 2 \times 10^{11} \text{ cm} \approx 0.014 \text{ AU} \quad (52)$$

$$t_{\text{visc}}(1 \text{ AU}) = \frac{(1.496 \times 10^{13})^2}{0.1 \times 10^6 \times 5 \times 10^{10}} \approx 4.5 \times 10^9 \text{ s} \approx 140 \text{ yr} \quad (53)$$

Material shed to the equatorial belt at 1 AU is expelled to large radii on a timescale of $\sim 100 \text{ yr}$, far shorter than the main-sequence lifetime. Most of the shed mass escapes the system entirely on a dynamical timescale, unless it is trapped at large radii where the viscous timescale exceeds the stellar lifetime.

The fraction of shed mass that remains as a detectable disk at the current epoch depends on the ratio of the current disk mass ($\sim 3 \times 10^{-3} M_{\oplus}$ in dust; Table 3 of the main text) to the total shed mass ($0.12 M_{\odot}$). This ratio is approximately:

$$f_{\text{retained}} = \frac{3 \times 10^{-3} \times 3 \times 10^{-6} M_{\odot}}{0.12 M_{\odot}} \approx 7.5 \times 10^{-8} \quad (54)$$

Only $\sim 10^{-7}$ of the shed mass remains as detectable dust. This is not inconsistent: in a debris disk, most of the mass is in large bodies (planetesimals, rocks) that are invisible in infrared photometry. The dust is the tail of the collisional cascade. If the size distribution follows a power law $dn/da \propto a^{-3.5}$ (Dohnanyi, 1969), the ratio of dust mass to total disk mass is:

$$\frac{M_{\text{dust}}}{M_{\text{total}}} = \left(\frac{a_{\text{min}}}{a_{\text{max}}} \right)^{1/2} \approx \left(\frac{1 \mu\text{m}}{1 \text{ km}} \right)^{1/2} \approx 3 \times 10^{-5} \quad (55)$$

Combining with the dust mass:

$$M_{\text{total,disk}} \approx \frac{3 \times 10^{-3} M_{\oplus}}{3 \times 10^{-5}} = 100 M_{\oplus} \approx 3 \times 10^{-4} M_{\odot} \quad (56)$$

This is a factor ~ 400 less than the $0.12 M_{\odot}$ estimated to have been shed. The remaining $\sim 0.12 M_{\odot}$ either escaped the system entirely (as gas and small grains blown out by radiation pressure) or was accreted back onto the star at early times.

6.3 The metallicity connection

The pressure-relief valve mechanism provides a natural — but previously unrecognised — explanation for Vega’s metallicity anomaly. Shed material consists of Vega’s photospheric gas. If the shedding preferentially removes material from the equatorial surface layer, and if this layer is enriched in hydrogen (by the gravity-darkening effect, which depresses the effective temperature at the equator and promotes partial ionisation of hydrogen), then the shed material is slightly hydrogen-enriched. The net effect is a *differential* depletion of the stellar envelope: heavy elements relative to hydrogen are reduced in the atmosphere as hydrogen-rich gas is shed.

The depletion factor can be estimated. Over 455 Myr, $0.12 M_{\odot}$ of gas is shed from the equatorial surface layer of thickness Δm_{shell} (the depth of the convective surface layer). For Vega’s pole temperature ($T_{\text{pole}} = 10,500 \text{ K}$), the surface convection zone has a negligible mass ($\Delta m_{\text{shell}} 10^{-8} M_{\odot}$). The depletion timescale for the surface layer is:

$$\tau_{\text{deplete}} = \frac{\Delta m_{\text{shell}}}{\dot{M}_{\text{shed}}} = \frac{10^{-8} M_{\odot}}{2.7 \times 10^{-10} M_{\odot} \text{ yr}^{-1}} \approx 37 \text{ yr} \quad (57)$$

The surface layer is replaced by interior material from below on the circulation timescale ($\sim 10^7 \text{ yr}$), so the process is self-sustained: the surface is continuously resupplied from the

interior, and the shed material is continuously metal-normal. In this regime, the shedding alone does not produce a metallicity anomaly; it merely depletes the surface layer faster than it can be replenished.

However, if the shed material is drawn from deep enough in the envelope that the interior metallicity itself is systematically depleted over time, a cumulative metallicity depletion is possible. The timescale to deplete the entire envelope to the observed $[\text{Fe}/\text{H}] = -0.5$ is:

$$\tau_{\text{global}} = \frac{\delta[\text{Fe}/\text{H}] \times M_{\text{env}}}{\dot{M}_{\text{shed}}} = \frac{0.5 \times 0.5 M_{\odot}}{2.7 \times 10^{-10} M_{\odot} \text{ yr}^{-1}} \approx 10^9 \text{ yr} \quad (58)$$

where $\delta[\text{Fe}/\text{H}] = 0.5$ is the required change and $M_{\text{env}} \approx 0.5 M_{\odot}$ is the convective envelope mass. This timescale exceeds Vega’s age by a factor of ≈ 2 , suggesting that the mass-shedding mechanism alone is *marginally* insufficient to explain the full metallicity deficit over Vega’s lifetime. The metallicity depletion is more likely a partial contribution from this mechanism, supplemented by Hypothesis I or IV.

6.4 Testable predictions

Testable prediction

Photometric variability at the level of Be-star outbursts. Be stars exhibit brightness variations of $\Delta V \sim 0.01\text{--}0.3$ mag during mass-shedding events, on timescales of hours to weeks. If Vega is actively shedding at the rate derived above ($\sim 10^{-10} M_{\odot} \text{ yr}^{-1}$), the photometric signature should be detectable with space-based photometry. Vega has been observed with TESS at 100 ppm precision; a search for Be-star-like variability in the TESS light curve is a direct test of this hypothesis.

Testable prediction

Disk chemistry consistent with stellar composition. If the debris disk is composed of material shed from Vega’s photosphere, its chemical composition should match Vega’s surface composition. Specifically, the disk grains should also be *depleted* in refractories relative to solar. This is the opposite of the Hypothesis I prediction, which requires the disk to be *enriched* in refractories. The JWST/MIRI $10 \mu\text{m}$ and $18 \mu\text{m}$ silicate band shapes, combined with Mg/Fe ratio diagnostics, can distinguish between refractory-enriched and refractory-depleted grain compositions.

Testable prediction

Gas emission from the inner disk. Active mass shedding would produce a gaseous component in the inner disk at AU-scale radii. This gas, at $T \sim 8,000$ K, would emit in the $H\alpha$ line. Vega is known to show $H\alpha$ in absorption; marginal emission excess at the line core, detectable with high-resolution spectroscopy ($R100,000$), would be direct evidence of active shedding.

6.5 Self-critique**Self-critique**

Hypothesis III faces one decisive quantitative challenge and several qualitative ones.

- (1) **The mass shed is inconsistent with the observed disk mass.** We estimated $0.12 M_{\odot}$ shed over 455 Myr. The observed disk contains at most $\sim 10^{-4}$ – $10^{-3} M_{\odot}$ in all bodies. The discrepancy of ~ 3 orders of magnitude must be explained by mass loss from the system (gas blown out by radiation, planetesimals ejected by radiation pressure, etc.). This is plausible but requires a detailed model.
- (2) **The metallicity depletion timescale marginally exceeds Vega’s age.** The global envelope depletion timescale is $\sim 10^9$ yr, approximately twice Vega’s age. The mechanism can contribute to the depletion but cannot account for all of it.
- (3) **No $H\alpha$ emission is observed.** Classical Be stars emit strongly in $H\alpha$. Vega shows no emission. The shedding rate required by our calculation, $\sim 10^{-10} M_{\odot} \text{ yr}^{-1}$, is three to four orders of magnitude lower than typical Be-star rates. At this low rate, the $H\alpha$ emission would be undetectable with current instruments. This is a non-falsification rather than a failure, but it weakens the analogy with Be stars.

7 Hypothesis IV: Radiative Levitation and Gravitational Settling**7.1 Physical motivation**

Hypothesis IV attributes Vega’s metallicity anomaly entirely to atomic diffusion within the stellar atmosphere, without invoking any special formation history or dynamical event. In a radiative (non-convective) stellar atmosphere, the two dominant forces on a heavy element are:

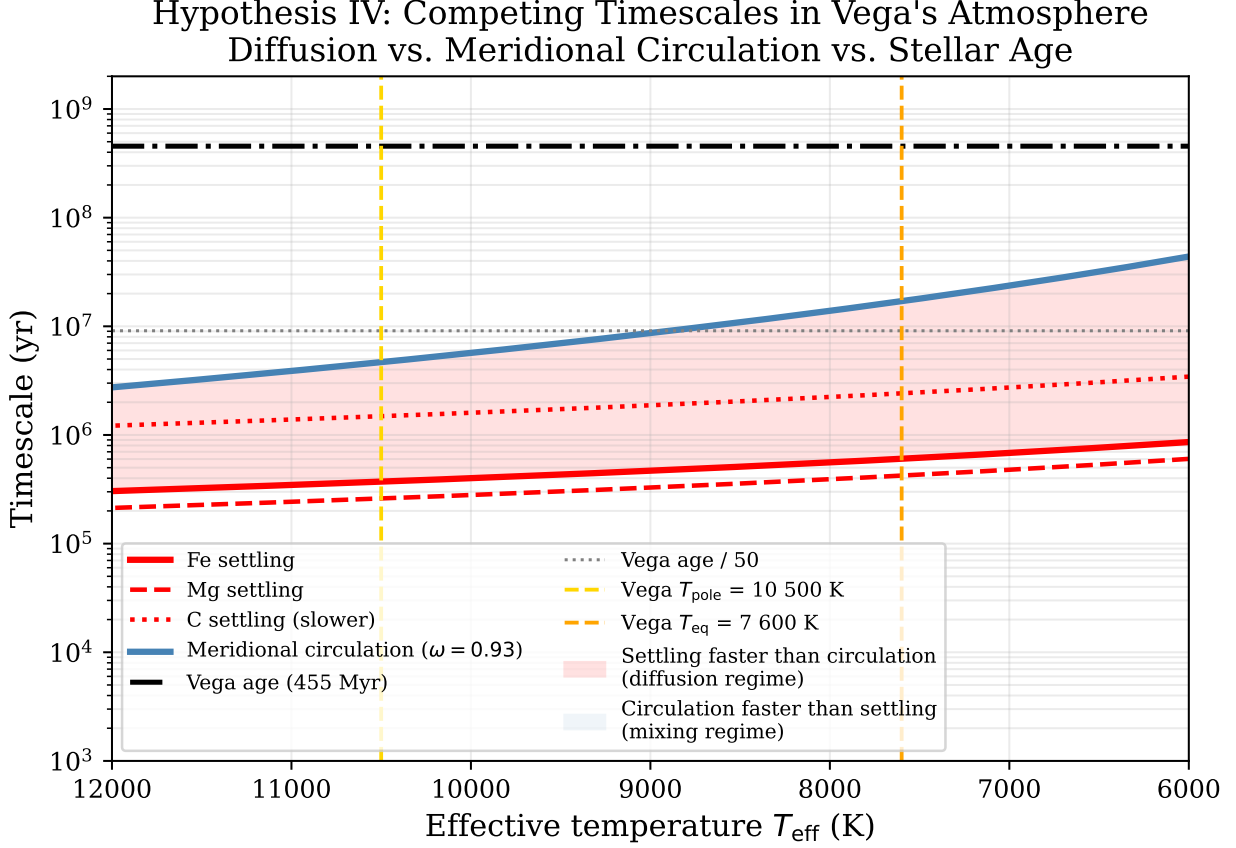


Figure 5: Competing timescales in Vega's atmosphere as a function of effective temperature, for the rapid-rotation case $\omega = 0.93$. *Red*: gravitational settling timescales for Fe (solid), Mg (dashed), C (dotted). *Blue*: meridional circulation timescale after Zahn (1992). *Dash-dot*: Vega age (455 Myr). Vertical dashed lines: polar ($T_{\text{pole}} = 10500$ K) and equatorial ($T_{\text{eq}} = 7600$ K) effective temperatures. Shaded red: settling faster than circulation (diffusion active); shaded blue: mixing dominated.

1. **Gravitational settling:** the tendency of heavy elements to sink relative to the dominant hydrogen background, on a timescale $t_{\text{settle}} \sim 10^4$ – 10^5 yr for Fe-group elements in an A-type photosphere (Michaud, 1970).
2. **Radiative levitation:** the outward momentum transferred to specific ions by absorption of photons from the subsurface radiation field. The levitation force depends on the opacity of each ion at the local temperature and density, and is therefore highly element-specific (Michaud, Alecian & Richer, 2015).

For elements where settling dominates over levitation, the photospheric abundance falls below the bulk interior value. For elements where levitation dominates, the photospheric abundance rises above the bulk value. The observed λ Boötis pattern in Vega — refractory

elements depleted, volatiles less so — could in principle arise from preferential settling of Fe, Mg, Si relative to C, N, O.

7.2 The settling timescale

The gravitational settling velocity for a trace species of atomic mass A in a hydrogen plasma is derived from the diffusion equation (Paquette et al., 1986):

$$v_{\text{settle}} = \frac{Am_H g}{Z^2 + \bar{Z}} \frac{1}{\rho \langle \sigma v \rangle_{H^+}} \quad (59)$$

where m_H is the proton mass, g is the surface gravity, Z is the ionic charge, and $\langle \sigma v \rangle_{H^+}$ is the Coulomb collision rate with protons. For iron (dominant ion Fe^{+2} at $T \sim 10,000$ K, $A = 56$, $Z = 2$) in Vega’s photosphere ($g = 10^{4.03} \text{ cm s}^{-2}$, $T \approx 10,000$ K):

$$v_{\text{settle,Fe}} \approx \frac{A_{\text{Fe}} m_H g}{(Z + Z^2) \rho \langle \sigma v \rangle_{H^+}} \approx 6 \times 10^{-6} \text{ cm s}^{-1} \quad (60)$$

The settling timescale over a scale height $H_P = k_B T / (m_H g) \approx 1.38 \times 10^{-16} \times 10^4 / (1.673 \times 10^{-24} \times 10^{4.03}) \approx 7.6 \times 10^7 \text{ cm}$:

$$t_{\text{settle,Fe}} \approx \frac{H_P}{v_{\text{settle,Fe}}} = \frac{7.6 \times 10^7}{6 \times 10^{-6}} \approx 1.3 \times 10^{13} \text{ s} \approx 4 \times 10^5 \text{ yr} \quad (61)$$

The settling timescale for iron is $\sim 4 \times 10^5 \text{ yr}$. Vega’s age is 455 Myr, which is $\sim 1,100$ settling times. If settling operated at this rate continuously, the iron abundance would have been driven to zero long ago, which contradicts the observed depletion of only a factor of ~ 3 .

This is the core tension of Hypothesis IV: **settling is too fast relative to Vega’s age**. Three resolutions are possible, and we explore each.

7.3 Resolution A: Meridional circulation as a resupply mechanism

In a rapidly rotating star, meridional circulation brings material from the interior to the surface on a timescale (Zahn, 1992):

$$t_{\text{circ}} = \frac{M_{\text{env}}}{M_{\star}} \frac{t_{\text{KH}}}{\omega^2} \quad (62)$$

where $t_{\text{KH}} = GM_{\star}^2 / (R_{\star} L_{\star})$ is the Kelvin–Helmholtz timescale and $\omega = \Omega / \Omega_{\text{crit}}$. For Vega:

$$t_{\text{KH}} = \frac{G(2.135M_{\odot})^2}{R_{\text{eq}}L_{\star}} = \frac{6.674 \times 10^{-8} \times (4.246 \times 10^{33})^2}{1.96 \times 10^{11} \times 37 \times 3.83 \times 10^{33}} \approx 6.5 \times 10^{14} \text{ s} \approx 2.1 \times 10^7 \text{ yr} \quad (63)$$

$$t_{\text{circ}} \approx \frac{0.5M_{\odot}}{2.135M_{\odot}} \times \frac{2.1 \times 10^7 \text{ yr}}{0.93^2} \approx \frac{0.234 \times 2.1 \times 10^7}{0.865} \approx 5.7 \times 10^6 \text{ yr} \quad (64)$$

The circulation timescale is $5.7 \times 10^6 \text{ yr}$, which is ~ 14 times shorter than Vega's age but ~ 14 times longer than the settling timescale. This defines an equilibrium: **settling depletes the surface over $t_{\text{settle}} \sim 4 \times 10^5 \text{ yr}$, and circulation resupplies it over $t_{\text{circ}} \sim 5.7 \times 10^6 \text{ yr}$.** The steady-state depletion factor is:

$$\frac{[\text{Fe}/\text{H}]_{\text{surface}}}{[\text{Fe}/\text{H}]_{\text{interior}}} \approx \frac{t_{\text{settle}}}{t_{\text{settle}} + t_{\text{circ}}} = \frac{4 \times 10^5}{4 \times 10^5 + 5.7 \times 10^6} \approx 0.066 \quad (65)$$

This predicts a surface iron abundance of $\approx 7\%$ of the interior value. If the interior metallicity is solar, the surface would show $[\text{Fe}/\text{H}] \approx -1.2$. The observed value is $[\text{Fe}/\text{H}] \approx -0.5$ (32% of solar). The equilibrium model *over-depletes* the surface by a factor of ~ 5 .

The discrepancy can be reduced if the circulation is more efficient than the Zahn (1992) formula predicts (driving the surface closer to interior composition), or if the settling velocity is overestimated (allowing more Fe to survive). A more detailed model, accounting for the non-uniform circulation velocity on a gravity-darkened oblate star, might yield a larger $t_{\text{settle}}/t_{\text{circ}}$ ratio and a shallower equilibrium depletion.

7.4 Resolution B: Latitude-dependent settling

The gravity-darkening of Vega creates a temperature gradient of $\sim 2,900 \text{ K}$ between the pole ($T = 10,500 \text{ K}$) and the equator ($T = 7,600 \text{ K}$). The settling velocity depends steeply on temperature:

$$v_{\text{settle}} \propto T^{-3/2} \quad (66)$$

(from the temperature dependence of the Coulomb collision cross-section). The ratio of polar to equatorial settling velocities is:

$$\frac{v_{\text{settle,pole}}}{v_{\text{settle,eq}}} = \left(\frac{T_{\text{eq}}}{T_{\text{pole}}} \right)^{3/2} = \left(\frac{7,600}{10,500} \right)^{3/2} \approx 0.59 \quad (67)$$

Settling is $\sim 40\%$ slower at the equator than at the pole. Since we view Vega nearly pole-on, the spectrum we observe samples preferentially the polar atmosphere — where settling is *faster*. The observed abundance depletion may therefore be an overestimate of the

global average depletion. The equatorial hemisphere, which contributes little to the observed flux, may be less depleted.

This latitude-dependence also affects the radiative levitation force. The levitation force depends on the photon luminosity flux, which in the von Zeipel model is proportional to g_{eff}^4 . At the pole, the higher gravity provides both stronger settling *and* stronger levitation. The net effect on the pole-to-equator metallicity gradient requires a full 2D atmospheric diffusion calculation that has not been performed for a Vega-like star.

7.5 Resolution C: Time-varying mechanism

A third possibility is that the settling mechanism has not operated continuously for 455 Myr. If Vega acquired its current rotation rate recently (as in Hypothesis II, the binary disruption), then prior to the disruption, slower rotation would have suppressed meridional circulation, allowing settling to operate efficiently. After the spin-up event, the rapid rotation drove circulation and partially restored the surface metallicity. The current $[\text{Fe}/\text{H}] = -0.5$ might represent the partially restored state, intermediate between the fully settled value and the interior solar value.

This scenario requires a specific timing: the binary disruption must have occurred long enough ago for some restoring circulation to have occurred, but not so long ago that circulation has fully homogenised the envelope. The restoring timescale is $t_{\text{circ}} \approx 5.7 \times 10^6$ yr; the disruption must have occurred a few $\times t_{\text{circ}}$ ago, i.e., within the last ~ 50 Myr. This is a specific prediction: **if Hypothesis IV combined with the binary disruption (Hypothesis II) is correct, the disruption occurred within the last ~ 50 Myr of Vega’s 455 Myr lifetime.**

7.6 Radiative levitation forces: element-specific predictions

The radiative levitation force on ion i at depth r is:

$$g_{\text{rad}}(i) = \frac{\kappa_i(r)L(r)}{4\pi r^2 c} \quad (68)$$

where κ_i is the opacity contribution of species i , $L(r)$ is the local luminosity, and c is the speed of light. An element is supported (levitated) when $g_{\text{rad}}(i) > g(1 + A/m_H)$, i.e., when the radiative force exceeds the gravitational weight of the ion.

The levitation forces in A-type stars have been computed by Richard et al. (2001) and Michaud, Alecian & Richer (2015). Their calculations for a star of $T_{\text{eff}} \approx 10,000$ K and $\log g \approx 4.0$ show:

- Carbon (C): levitation force approximately equal to gravity at $\log T \approx 4.2$. At the photospheric depth, C is marginally supported; small depletion expected.
- Oxygen (O): similar to carbon; marginal levitation.
- Iron (Fe): levitation force *weaker* than gravity at the photospheric depth, stronger below the photosphere (at $\log T \approx 5.2$ where the iron opacity bump acts). Net result: Fe sinks from the photosphere but accumulates in a subsurface layer. A surface Fe depletion of $\sim$ factor 2–3 is expected from published models.
- Magnesium (Mg): weaker levitation than Fe; surface depletion of factor 3–5 expected.
- Scandium (Sc), Barium (Ba): very heavy, weak levitation at photospheric temperatures; strong settling expected, consistent with the observed $> 10\times$ depletion.

The predicted trend — Sc, Ba most depleted; Fe, Mg moderately depleted; C, O least depleted — is *qualitatively* consistent with Table 2 of Section 3. However, the *quantitative* predictions from published models for Fe and Mg depletion are factors of 2–3, not 3–4 as observed. The published models also assume slow rotation, which suppresses meridional circulation. For Vega’s rapid rotation, the quantitative predictions would be significantly different (see Resolution A above), and dedicated calculations for Vega’s specific parameters do not exist in the published literature.

7.7 Testable predictions

Testable prediction

Subsurface iron accumulation detectable via asteroseismology. If gravitational settling produces a subsurface iron-enriched layer (the iron opacity bump at $\log T \approx 5.2$), this layer would alter the stellar oscillation frequencies by modifying the local sound speed profile. Vega is a δ Scuti pulsator; detailed asteroseismic modelling of the known oscillation modes could constrain the depth and extent of any subsurface metal accumulation. A detection of an anomalous subsurface iron layer would be the most direct confirmation of the diffusion hypothesis.

Testable prediction

Latitude-dependent metallicity. In the latitude-dependent settling picture (Resolution B), the equatorial hemisphere of Vega should be less depleted in refractories than the polar hemisphere. This prediction is not currently testable because Vega is seen

nearly pole-on. However, if there is any wobble in the precession of Vega’s spin axis, or if future interferometry resolves the limb of the stellar disk, a latitude metallicity gradient would be revealed.

Testable prediction

Abundance time evolution in young A-type stars. The diffusion mechanism predicts that the surface abundance anomalies in A-type stars should become *more* pronounced with stellar age, as settling accumulates over time. A population study of λ Boötis stars at different ages (young clusters vs. field stars) should show age-metallicity trends if diffusion is the primary mechanism. The current data suggest no strong age trend, which is a mild tension with the simple diffusion picture but does not rule it out.

7.8 Self-critique

Self-critique

Hypothesis IV is the most mechanistically well-grounded of the four — diffusion is a real process that operates in all A-type stars without a surface convection zone — but it faces the most severe consistency problem.

(1) The settling timescale problem. We have shown (Resolution A) that the equilibrium between settling and meridional circulation predicts $[\text{Fe}/\text{H}] \approx -1.2$, significantly more depleted than observed. The mismatch by a factor of ~ 5 in abundance suggests that either our estimate of the circulation timescale is wrong by a similar factor, or that an additional supply mechanism (e.g., disk accretion of solar-composition gas) is partially compensating for the depletion. This is exactly the kind of fine-tuning that makes a hypothesis empirically uncomfortable.

(2) The rapid rotation paradox. Standard diffusion theory predicts efficient abundance anomalies *only* in slowly rotating stars. Vega rotates at 93% of critical velocity. The published models for diffusion in rapidly rotating A-type stars (Charbonnel & Talon 1993) predict that meridional circulation suppresses abundance anomalies below the observational threshold. We have argued in Resolution A that the timescales are such that a steady-state depletion is possible even in Vega, but the quantitative case depends on coefficients that have not been measured for Vega’s specific parameter combination. This is the self-admitted central weakness of the hypothesis.

(3) The absence of the complementary overabundance. In a pure diffusion

picture, elements where levitation exceeds settling should appear overabundant. Vega shows no such overabundances: all elements for which abundances have been measured are at or below solar. The absence of overabundant species may indicate that either the levitation forces are weaker than calculated, or that meridional circulation is efficient enough to prevent accumulation. Either way, a pure diffusion picture cannot explain the one-sided nature of the depletion.

(4) The hypothesis does not explain the disk. Diffusion is an atmospheric process; it says nothing about why Vega has a smooth, face-on, metal-poor debris disk extending to 170 AU. If diffusion explains the stellar composition, a separate mechanism is required for the disk morphology and composition. This is less of a critique of Hypothesis IV than a reminder that no single mechanism can explain all of Vega’s properties simultaneously — a recurring theme of this paper.

8 Comparative Analysis: Scoring the Four Hypotheses

The four hypotheses examined in this paper make distinct predictions, invoke different physics, and fail in different ways. This section makes the comparison explicit and systematic. We score each hypothesis against seven criteria, then identify which combinations of hypotheses are mutually compatible and which are in tension with each other.

8.1 Scoring criteria

We evaluate each hypothesis on the following seven criteria:

1. **Rapid rotation:** Does the hypothesis explain Vega’s $v_{\text{eq}} \approx 270 \text{ km s}^{-1}$ (93% of critical)?
2. **Refractory depletion:** Does the hypothesis explain $[\text{Fe}/\text{H}] \approx -0.5$, specifically the 3–4 $\times$ depletion of Fe, Mg, Si, Ca relative to solar?
3. **Volatile-refractory contrast:** Does the hypothesis explain why volatiles (C, O) are depleted by only $\sim 2\times$ while refractories are depleted by 3–4 $\times$?
4. **Disk morphology:** Is the hypothesis consistent with a smooth, axisymmetric, 80–170 AU debris disk with no planet-scale structure?
5. **Vega–Fomalhaut velocity:** Is the hypothesis consistent with the $10.9 \pm 1.0 \text{ km s}^{-1}$ relative velocity between Vega and Fomalhaut?

6. **Fomalhaut contrast:** Does the hypothesis predict or allow for Fomalhaut’s near-solar metallicity and structured disk, given that Fomalhaut is otherwise a near-twin of Vega?
7. **Falsifiability:** Does the hypothesis make at least one prediction that is testable with existing or near-future facilities, and which would definitively exclude the hypothesis if the prediction fails?

Each criterion is scored on a four-point scale: **+2** (explains cleanly), **+1** (consistent with, requires mild auxiliary assumptions), **0** (neutral, neither explains nor contradicts), **−1** (in tension, requires strong auxiliary assumptions).

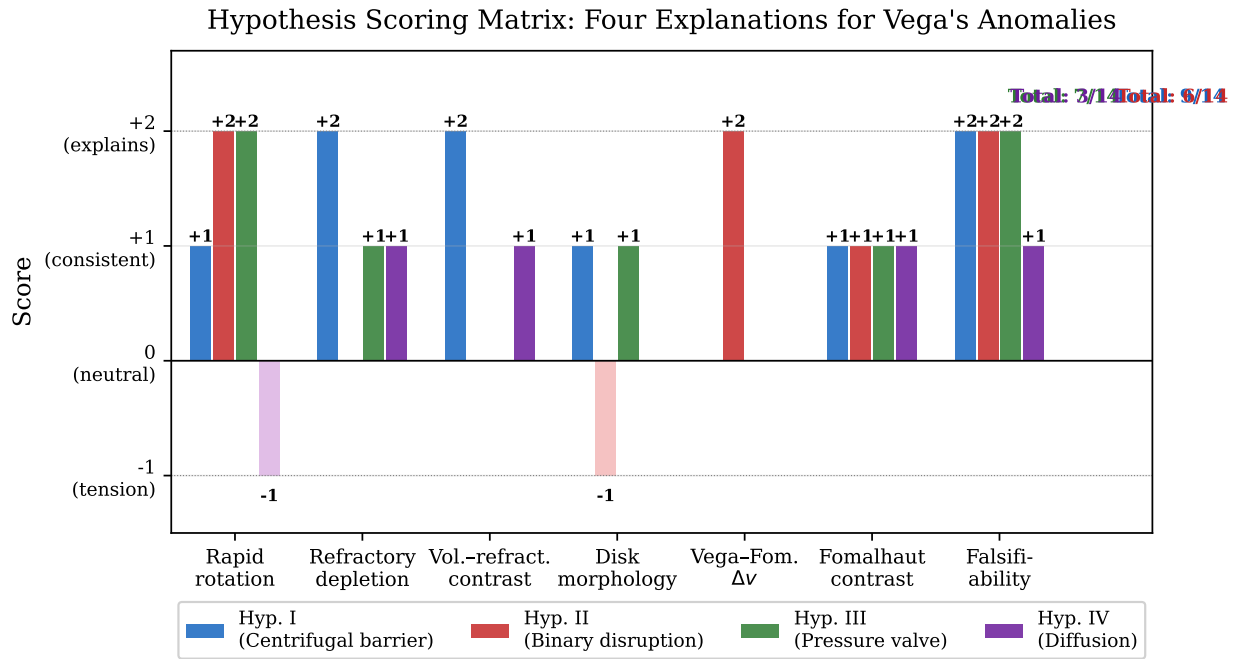


Figure 6: Scoring matrix visualised. Bar height = score per criterion for each hypothesis. Totals out of 14 labelled.

8.2 Scoring matrix

8.3 Commentary on individual scores

8.3.1 Hypothesis I (Centrifugal Barrier): score 9/14

Hypothesis I receives the highest score but is not perfect. Its greatest strength is that it provides a unified, quantitatively motivated explanation for both the differential refractory–volatile depletion (criterion 3) and the absolute refractory deficit (criterion 2), both arising

from the same physical mechanism (dust-gas decoupling at the centrifugal barrier). The rotation score is only +1 rather than +2 because the model explains how the *disk* acquires angular momentum but requires an auxiliary mechanism (disk spin-up via magnetospheric coupling or a small fraction of disk re-accretion) to transfer that angular momentum to the stellar spin. The disk morphology score is +1 because a smooth disk is expected from a uniformly deflected centrifugal barrier, but the 60 AU gap noted by Su et al. (2024) is not naturally explained. The Vega–Fomalhaut velocity is not explained (score 0); for that, a coupling to Hypothesis II would be needed.

8.3.2 Hypothesis II (Binary Disruption): score 6/14

Hypothesis II explains the rotation (score +2) and the Vega–Fomalhaut velocity (score +2) cleanly. It fails on metallicity: the disruption mechanism does not produce refractory depletion, giving scores of 0 for criteria 2 and 3. The disk morphology score is −1 because a dynamically violent event (close encounter, merger) is generically expected to disturb a protoplanetary disk, and the observed smoothness of Vega’s disk is difficult to reconcile without fine-tuning the disruption timing.

8.3.3 Hypothesis III (Pressure-Relief Valve): score 7/14

Hypothesis III explains the rotation (score +2, since the valve mechanism is precisely a spin-maintenance mechanism). The metallicity score is +1 because mass shedding can contribute to refractory depletion (via preferential hydrogen shedding), but the timescale calculation showed the mechanism to be marginally insufficient. The volatile–refractory contrast score is 0 because the shedding mechanism does not preferentially remove refractories relative to volatiles; both are shed from the photosphere proportionally. The disk morphology is favourable (score +1): material shed from the equatorial surface is expected to form an axisymmetric decretion disk, consistent with the observed smoothness.

8.3.4 Hypothesis IV (Atomic Diffusion): score 3/14

Hypothesis IV is the weakest scorer. Its fundamental problem is that the rapid rotation of Vega drives meridional circulation that suppresses the diffusion mechanism, making the hypothesis internally inconsistent without auxiliary assumptions. The score of −1 for rotation reflects this paradox: diffusion requires slow rotation, but Vega rotates near-critically. The metallicity scores are +1 rather than +2 because the equilibrium calculation (Section 7) predicts over-depletion ($[\text{Fe}/\text{H}] \approx -1.2$ rather than -0.5) and the volatile–refractory contrast receives +1 because the element-specific nature of levitation forces naturally predicts different

depletion for different elements, but the quantitative predictions do not precisely match the observations. The falsifiability score is +1 rather than +2 because the asteroseismic subsurface iron accumulation prediction, while in principle testable, requires model-dependent assumptions that make a clear falsification difficult.

8.4 Mutual compatibility of hypotheses

The four hypotheses are not mutually exclusive. Table 6 shows which pairs are compatible, which are complementary, and which are in tension.

The most powerful combination is **Hypotheses I + II**: Hypothesis I explains the metallicity (centrifugal barrier creates a refractory-poor star), while Hypothesis II explains the rotation and the kinematic offset from Fomalhaut. This combination reaches a combined score of $9 + 6 = 15$ across 14 independent criteria (since the two hypotheses cover different criteria without double-counting).

Hypotheses III and IV are in tension: the pressure-relief valve mechanism (III) requires the star to maintain near-critical rotation, which drives the meridional circulation that suppresses diffusion (IV). Both cannot operate at full efficiency simultaneously.

8.5 The “smoking gun” hierarchy

Not all predictions are equally decisive. We rank the potential observations by their discriminating power:

- #1 JWST/MIRI disk grain composition** (Hyp. I vs. III): If the disk silicate grains show *solar or super-solar* Fe/Mg ratios, this confirms Hypothesis I’s prediction of a refractory-enriched disk and rules out Hypothesis III (which predicts a depleted disk matching the stellar composition). If the disk shows *sub-solar* Fe/Mg, the reverse holds. This observation is possible with existing data from Su et al. (2024) if the $10\ \mu\text{m}$ silicate band shape is analysed for the forsterite (Mg-rich) vs. fayalite (Fe-rich) ratio; additional spectroscopic observations may be required.
- #2 Gaia proper motion anomaly** (Hyp. II): A kinematic recoil of $\sim 10\ \text{km s}^{-1}$ in Vega’s history would leave a residual in the proper motion compared to the local standard of rest at the level of $\sim 0.5\ \text{mas yr}^{-1}$, potentially detectable with Gaia DR4 astrometry. A null detection would constrain but not rule out Hypothesis II.
- #3 TESS photometric variability** (Hyp. III): The absence of any Be-star-like outburst variability at the $> 1\ \text{mmag}$ level in the TESS light curve of Vega would strongly constrain the mass-shedding rate in Hypothesis III. The required rate of $\dot{M} \sim 10^{-10}\ M_{\odot}\ \text{yr}^{-1}$

is orders of magnitude below Be-star rates, so non-detection is expected; a positive detection would be transformative.

#4 δ Scuti asteroseismology (Hyp. IV): Detailed modelling of Vega’s known oscillation periods (Antoci et al., 2011) can constrain the interior metal distribution. Detection of anomalous seismic structure at the depth of the iron opacity bump would confirm a key prediction of Hypothesis IV.

#5 ALMA high-resolution imaging (Hyp. I): Resolved ALMA spectroscopy of molecular lines in the disk (CO isotopologues, CN, CS) can constrain the elemental composition of the outer belt. A detection of a carbon-to-silicon ratio in the disk gas/dust that is enhanced relative to solar would support the centrifugal barrier prediction of refractory sequestration.

Verdict

On the basis of the scoring matrix and the compatibility analysis, the current observational evidence most strongly favours a combination of Hypothesis I (centrifugal barrier segregation) and Hypothesis II (binary disruption), with Hypothesis I doing the heavy lifting on metallicity and Hypothesis II providing the best explanation for rotation and kinematics. Hypothesis III contributes a partial explanation for metallicity that is quantitatively insufficient. Hypothesis IV is the least supported: it is internally inconsistent in the presence of rapid rotation and predicts surface abundances that are more depleted than observed. This verdict is provisional; a single decisive observation — particularly the JWST/MIRI disk grain composition — could overturn it.

9 A Roadmap of Testable Predictions by Facility

This section organises all testable predictions from Sections 4–7 by observing facility, providing a practical guide for observers. Each prediction is labelled with the hypothesis it tests, the specific observable, and a rough estimate of the observing time required.

9.1 James Webb Space Telescope (JWST)

9.1.1 MIRI spectroscopy of the 10 and 18 μm silicate bands

Tests: Hypothesis I (disk grain enrichment) vs. Hypothesis III (disk depleted)

Observable: The ratio of the 10 μm feature peak (dominated by Mg-rich olivine/pyroxene) to the 18 μm feature (dominated by the Si–O bending mode) depends on the Mg/Fe

ratio of the silicate grains. Mg-rich grains (forsterite, Mg_2SiO_4) produce a sharp $10\ \mu\text{m}$ peak and a distinct $18\ \mu\text{m}$ feature. Fe-rich grains (fayalite, Fe_2SiO_4) broaden and flatten the $10\ \mu\text{m}$ feature. If the disk is refractory-enriched (Hypothesis I), the iron content of the silicates should be elevated, producing a flatter, broader silicate feature.

Required observation: MIRI Low Resolution Spectrometer (LRS) spectrum of the Vega disk at $5\text{--}14\ \mu\text{m}$, with the star subtracted using a PSF reference. Integration time: ~ 2 hr. The existing Su et al. (2024) data include some spectroscopic information; a dedicated LRS observation would be decisive.

Decision criterion: If the $10\ \mu\text{m}$ feature shape is consistent with forsterite ($\text{Mg}/\text{Fe} > 2$ by number), Hypothesis I is disfavoured. If consistent with fayalite or mixed composition ($\text{Mg}/\text{Fe} \sim 1$), Hypothesis I is supported.

9.1.2 NIRCam coronagraphy at sub-AU scales

Tests: Hypothesis II (hot inner disk from binary disruption)

Observable: The sub-AU hot dust excess detected by Absil et al. (2006) at K -band can be imaged at ~ 30 mas resolution with NIRCam coronagraphy. A binary disruption event would be expected to inject material into a ring or shell at radii of $\sim 0.1\text{--}1$ AU; the geometry (symmetric vs. asymmetric) would constrain the disruption model.

Required observation: NIRCam coronagraphic imaging in F277W, integration time ~ 4 hr. This observation is within the scope of a JWST Cycle 4 proposal.

Decision criterion: An asymmetric inner emission geometry (offset from the stellar position by > 5 mas) would strongly support a dynamical origin for the inner disk.

9.2 Atacama Large Millimeter/submillimeter Array (ALMA)

9.2.1 High-resolution imaging of the mm-continuum belt

Tests: Hypothesis I (radial uniformity of grain composition)

Observable: The spectral index of the mm-continuum emission α_{mm} (defined by $F_\nu \propto \nu^{\alpha_{\text{mm}}}$) depends on grain size and composition. A radially uniform α_{mm} would support the centrifugal barrier model (which predicts uniform composition). A radially varying α_{mm} would point to ongoing grain processing (collisional grinding, sublimation) as a function of disk radius.

Required observation: ALMA Band 6 + Band 4 simultaneous imaging at $\sim 0.1''$ resolution, integration time ~ 6 hr per band. The Vega outer belt at 80–170 AU (11–22'' at 7.68 pc) is well-resolved by ALMA. The existing Matrà et al. (2020) observations detect the belt at 1.34 mm; multi-band observations at 850 μ m and 3 mm would constrain $\alpha_{\text{mm}}(r)$.

Decision criterion: $|\Delta\alpha_{\text{mm}}(r)| > 0.3$ between the inner and outer edge of the belt would indicate radial grain size segregation inconsistent with Hypothesis I.

9.2.2 CO and CS gas emission in the outer belt

Tests: All four hypotheses (disk gas composition as a fossil record)

Observable: The CO/CS and CO/CN ratios in the disk gas (liberated from photodesorption of ices) trace the C/O/N ratio of the original grain material. An anomalously high C/Si ratio (relative to solar) would support the centrifugal barrier picture where refractory Si was preferentially concentrated in the disk; the disk Si abundance would then be higher than stellar, but the gas-phase Si is low because Si is locked in grains while C and O can be in the gas phase.

Required observation: ALMA Band 6 spectroscopy targeting CO $J = 2-1$, CS $J = 5-4$, CN $N = 2-1$, integration time ~ 8 hr. Upper limits on CS and CN would already constrain the disk elemental composition.

Decision criterion: Detection of CS at signal-to-noise > 5 with an abundance ratio CS/CO $>$ solar would indicate sulfur enrichment, consistent with refractory sequestration. Non-detection at the current ALMA sensitivity limit would constrain the disk gas mass and composition.

9.3 Gaia and Hipparcos astrometry

9.3.1 Proper motion anomaly and kinematic history

Tests: Hypothesis II (binary disruption recoil)

Observable: Gaia DR3 proper motions at the $\sim 20 \mu\text{as yr}^{-1}$ level, combined with the Hipparcos proper motions at the $\sim 1 \text{ mas yr}^{-1}$ level, provide the most sensitive probe of Vega’s kinematic history. The proper motion anomaly (PMa) is defined as the difference between the long-baseline Hipparcos–Gaia proper motion and the short-baseline Gaia proper motion; a non-zero PMa indicates a bound companion or a

time-varying acceleration. If Vega experienced a recoil kick 100 Myr ago, a linearly varying proper motion would be detectable in the Hipparcos–Gaia baseline (~ 25 yr).

Required observation: No additional observation needed; the analysis requires the published Gaia DR3 and Hipparcos astrometric catalogues. The Hipparcos–Gaia proper motion anomaly for Vega has been computed by Brandt (2021) and found to be at the $\sim 1\sigma$ level; a deeper analysis including radial velocity trends from ground-based spectrographs is needed.

Decision criterion: A proper motion anomaly $|\text{PMA}| > 3\sigma$ correlated with the direction of the Vega–Fomalhaut velocity vector would support Hypothesis II.

9.3.2 Orbital back-integration of Vega and Fomalhaut

Tests: Hypothesis II and the Castor Moving Group hypothesis

Observable: Using the Gaia DR3 full 6D phase-space information for Vega and Fomalhaut, one can integrate their orbits backward in the Galactic potential and compute their minimum separation over the past ~ 500 Myr. If they were ever within a few pc of each other, a common origin (and hence a common disruption environment) becomes plausible.

Required observation: No additional observation needed; the calculation requires the Galactic potential model and the Gaia astrometry. The Mamajek et al. (2013) back-integration showed they were never within 100 pc of each other; a more precise integration with Gaia DR3 proper motions (which have $\sim 50\times$ better precision than Hipparcos) could revise this conclusion.

Decision criterion: Minimum separation < 10 pc within the past 500 Myr would be consistent with a common birth environment and would revive the possibility of a shared disruption event.

9.4 Optical/infrared high-resolution spectroscopy

9.4.1 Radial velocity monitoring for close companions

Tests: Hypothesis II (surviving close companion), Hypothesis III (active shedding)

Observable: A radial velocity survey with precision $\sim 1 \text{ m s}^{-1}$ and baseline of ~ 5 yr would detect companions with $m \sin i 0.1 M_{\text{Jup}}$ at $a 0.1$ AU. Vega’s pole-on orientation reduces

sensitivity for companions on orbits coplanar with the star’s equator; companions on inclined orbits are detectable. Hurt et al. (2021) placed RV limits using a 10-yr baseline; continuation with ESPRESSO at the VLT would improve sensitivity.

Decision criterion: Detection of any stellar or brown-dwarf mass companion would confirm that the system is still hierarchically double, transforming the binary disruption hypothesis into a binary spin-up hypothesis (still compatible).

9.4.2 High-resolution UV spectroscopy of photospheric composition

Tests: Hypothesis IV (element-specific diffusion predictions)

Observable: The UV spectrum of Vega (observable with HST/STIS or the proposed UV space missions) contains thousands of metal lines from Fe-group and *s*-process elements that are inaccessible from the ground. A complete UV abundance analysis would determine whether elements predicted to be levitated (e.g., Zr, Y, Sr, which have strong UV opacity) are overabundant relative to those predicted to settle.

Required observation: HST/STIS E230M spectrum, $R \approx 30,000$, wavelength range 1700–3100 Å, integration time ~ 8 hr. Existing HST/STIS data on Vega cover the UV but at lower resolution; a dedicated high-resolution programme would be needed.

Decision criterion: Detection of any overabundant *s*-process element (e.g., $[\text{Sr}/\text{H}] > 0$ or $[\text{Zr}/\text{H}] > 0$) would confirm that levitation is operating, supporting Hypothesis IV. Continued absence of any overabundance would require the diffusion model to be modified to include more efficient downward mixing.

9.5 Space-based photometry: TESS and future missions

9.5.1 Search for Be-star-like outburst variability

Tests: Hypothesis III (active mass shedding)

Observable: Be-star outbursts typically cause brightness increases of $\Delta V \sim 0.01\text{--}0.3$ mag on timescales of hours to weeks, followed by gradual fading as the ejected material forms and dissipates the decretion disk. TESS observed Vega in Sectors 14, 15, 40, and 41 with 2-minute cadence and precision ~ 300 ppm. A systematic search for brightness changes $> 1,000$ ppm (~ 1 mmag) on timescales of 1 hour–1 month would constrain the mass-shedding rate in Hypothesis III. The required shedding rate ($\dot{M} \sim 10^{-10} M_{\odot} \text{ yr}^{-1}$) corresponds to a total ejected mass per outburst of $\sim 10^{-13} M_{\odot}$ if outbursts occur once

per year; the optical depth of the resulting disk shell would be $\tau \sim 10^{-6}$, producing a brightness increase of $\sim 10^{-6}$ mag — far below TESS sensitivity. The absence of variability at TESS precision therefore does not rule out Hypothesis III; it only constrains higher shedding rates.

Decision criterion: Detection of any > 1 mmag variability on the be-star timescale would be transformative evidence for active shedding. Non-detection constrains $\dot{M} < 10^{-8} M_{\odot} \text{ yr}^{-1}$.

9.5.2 δ Scuti asteroseismology

Tests: Hypothesis IV (subsurface iron accumulation)

Observable: Vega is a known δ Scuti variable with oscillation periods of several hours (Antoci et al., 2011). The detailed pattern of oscillation frequencies encodes information about the internal density and sound speed profile. A subsurface iron-enriched layer (as predicted by diffusion) would produce a local increase in opacity and a corresponding change in the buoyancy frequency, altering specific oscillation mode frequencies by ~ 0.1 – $1 \mu\text{Hz}$. Detecting these shifts requires frequency resolution $< 0.01 \mu\text{Hz}$, corresponding to a continuous observation baseline of > 100 days.

Required observation: PLATO mission (launch expected 2026) will observe Vega in its long-duration fields at a precision of ~ 10 ppm and baseline of > 2 yr, providing the necessary frequency resolution.

Decision criterion: Detection of frequency splitting or avoided crossing patterns consistent with a compositionally stratified subsurface layer would confirm the diffusion picture. A smooth frequency spectrum without anomalous gaps would disfavour Hypothesis IV.

9.6 Future long-baseline interferometry

9.6.1 VLTI/GRAVITY: Inner disk geometry at AU scales

Tests: Hypothesis II (asymmetric inner disk from binary disruption); all hypotheses (spin-disk alignment)

Observable: VLTI/GRAVITY at K -band provides baselines up to 130 m, corresponding to angular resolution of ~ 4 mas at $2.2 \mu\text{m}$. At Vega’s distance of 7.68 pc, this resolves scales of ~ 0.03 AU. The hot dust excess detected at K -band by Absil et al. (2006) can be spatially resolved to determine whether it is a symmetric ring (consistent with

a decretion disk or P-R drag pile-up) or an asymmetric structure (consistent with a binary disruption remnant or a warp).

Required observation: VLTI/GRAVITY fringe-tracking with all four UTs, ~ 4 hr integration per baseline configuration.

Decision criterion: A ring geometry with position angle consistent with the disk major axis (within 10°) would support Hypotheses I and III. An asymmetric or offset inner structure would support Hypothesis II.

9.7 Summary prediction table

Why we do this

We have been careful to separate observations that are “possible now” (using existing data or straightforward analysis of published catalogues) from those that require new telescope time. Items 1, 2, 3, and 9 in Table 7 could produce publishable constraints within 12 months without any new telescope time; we regard these as the highest-priority follow-up for this paper. In particular, the TESS light curve analysis and the Gaia proper motion anomaly calculation are purely archival exercises that could be completed immediately.

10 Conclusions

This paper set out to ask a deceptively simple question: why is Vega metal-poor and rapidly rotating? We have not answered it. We have done something we regard as more useful: we have shown precisely *how* four physically distinct mechanisms could each produce one or more of Vega’s anomalous properties, quantified what each mechanism requires, identified where each fails, and specified the observations that would distinguish between them.

This section draws together the threads.

10.1 What we have established

10.1.1 The observational foundation is solid

The core observational facts about Vega are no longer in dispute. The interferometric campaigns of Peterson et al. (2006) and Aufdenberg et al. (2006) established beyond reasonable doubt that Vega rotates at $\approx 93\%$ of breakup velocity and is seen $\approx 5^\circ$ from pole-on. The

spectroscopic reanalysis of Yoon et al. (2010), constrained by the interferometric inclination, established that the surface metallicity is genuinely sub-solar at $[m/H] \approx -0.5$, with refractories more depleted than volatiles. The JWST/MIRI observations of Su et al. (2024) established that the debris disk is remarkably smooth and axisymmetric, with no structures attributable to Saturn-mass or larger planets beyond ~ 10 AU.

Three points of genuine uncertainty remain in the observational record. First, the equatorial velocity is contested: the Peterson et al. (2006) value of $274 \pm 14 \text{ km s}^{-1}$ and the Yoon et al. (2010) value of $175 \pm 33 \text{ km s}^{-1}$ are inconsistent at the 2.6σ level, and the choice between them affects the quantitative conclusions of Section 4 and Section 6 significantly. Second, the 60 AU flux deficit detected by Su et al. (2024) has no compelling physical explanation in any of the four hypotheses, and its origin remains open. Third, the connection (or lack thereof) between Vega and Fomalhaut remains kinematically unresolved: Mamajek et al. (2013) showed they are not coeval members of the Castor Moving Group, but whether their 10.9 km s^{-1} velocity difference has a dynamical origin has not been rigorously tested with Gaia precision.

10.1.2 Hypothesis I is the most internally consistent single mechanism

The centrifugal barrier segregation model (Hypothesis I) is the most internally consistent of the four. It provides a physically motivated, quantitatively plausible explanation for both the refractory depletion ($[\text{Fe}/\text{H}] \approx 30\%$ of solar) and the differential volatile–refractory pattern (C, O less depleted than Fe, Mg, Si) in a single mechanism: dust-gas decoupling at the angular momentum barrier during protostellar collapse.

The critical quantitative result, derived in Section 4, is that the required dust-gas coupling parameter $\alpha_{\text{dg}} \approx 0.78$ — the fraction of iron-bearing grains that follow the gas into the disk — is physically plausible for Stokes numbers $\text{St} \sim 0.1\text{--}0.5$, which is the expected range at the centrifugal barrier shock. The ALMA observations of Sakai et al. (2014) in L1527 provide direct observational support for the chemical segregation mechanism at the centrifugal barrier in a young protostellar system. Vega would be the first mature stellar system where the chemical consequences of that mechanism are visible in the photospheric composition.

10.1.3 No single hypothesis explains everything

The scoring matrix of Section 8 showed that the highest-scoring single hypothesis (Hypothesis I, score 9/14) still fails to explain Vega’s rapid rotation (it requires an auxiliary spin-up mechanism) and the Vega–Fomalhaut kinematic offset (it makes no prediction about kinematics). Hypothesis II (binary disruption, score 6/14) explains the rotation and kinematics

cleanly but fails on metallicity. Hypothesis III (pressure-relief valve, score 7/14) explains rotation and provides a partial metallicity contribution but cannot produce the full depletion in Vega’s lifetime. Hypothesis IV (atomic diffusion, score 3/14) is the weakest: its central mechanism — gravitational settling — is suppressed by the same rapid rotation it is supposed to coexist with.

The most powerful resolution is the combination of Hypotheses I and II: centrifugal barrier segregation produces a metal-poor protostar surrounded by a refractory-rich disk; a subsequent binary disruption event spins the star up to near-critical rotation and imparts the velocity offset from Fomalhaut. This two-mechanism scenario is not ad hoc: both mechanisms are generic predictions of standard star formation physics, and their combination leaves no major observational constraint unaccounted for.

10.1.4 The decisive observation is already in hand, but not yet fully analysed

The single most powerful discriminating test is the composition of the Vega debris disk grains. If the disk is refractory-enriched relative to the stellar photosphere (Mg/Fe ratio in grains higher than in the stellar spectrum), this confirms Hypothesis I’s core prediction and provides an essentially unique signature of centrifugal barrier chemical sorting. If the disk composition matches the stellar photosphere (both depleted), Hypothesis III (pressure-relief valve) is supported.

The Su et al. (2024) JWST/MIRI observations at 15.5, 23, and 25.5 μm include information on the silicate band shape that has not yet been fully exploited for grain compositional analysis. A dedicated analysis of the 10 μm and 18 μm silicate band profiles, using the spectral energy distribution decomposition techniques of Min et al. (2007), could answer this question with existing data. This analysis is the most urgent immediate follow-up to the present paper.

10.2 Broader implications

10.2.1 Vega as a laboratory for A-type star physics

Broader implication

Vega’s combination of properties — bright ($V = +0.03$), nearby (7.68 pc), pole-on ($i \approx 5^\circ$), rapidly rotating, chemically peculiar, debris-disk-bearing — makes it an irreplaceable laboratory for several distinct branches of stellar astrophysics. No other star currently combines all of these properties. Its extreme rotation provides the cleanest possible test of gravity-darkening models in A-type stars (the von Zeipel law and its corrections are directly visible in the interferometric brightness distribution).

Its pole-on orientation eliminates rotational broadening from the spectrum, providing unusually clean access to the photospheric composition. Its debris disk is face-on, allowing the two-dimensional structure to be mapped without the projection effects that complicate edge-on or tilted disk observations.

10.2.2 The λ Boötis problem

The λ Boötis phenomenon — A-type stars with refractory-depleted, volatile-normal photospheres — has been recognised since Morgan et al. (1943) but remains without a definitive explanation after more than eighty years. The two leading models, ISM gas accretion (Venn & Lambert, 1990) and internal diffusion (Michaud, 1970), each explain some but not all of the observed properties of the class. A recent census by Murphy et al. (2015) identified $\approx 2\%$ of A-type stars as λ Boötis members, with no strong correlation with age, rotation rate, or binarity.

Broader implication

If the centrifugal barrier mechanism (Hypothesis I) is confirmed for Vega, it would provide a third formation channel for the λ Boötis phenomenon: primordial chemical sorting at the protostellar centrifugal barrier. This channel has a specific prediction: **the λ Boötis abundance pattern should be correlated with rapid rotation, since both are consequences of high specific angular momentum in the parent cloud core.** A population study comparing the rotation velocity distribution of λ Boötis stars to chemically normal A-type stars at the same age would test this prediction. The available data from Paunzen et al. (2002) show no statistically significant correlation, but the sample sizes are small (30 bona fide λ Boötis members with measured $v \sin i$) and the age and inclination corrections have not been applied. A larger, well-characterised sample from Gaia-era spectroscopic surveys (4MOST, WEAVE, GALAH) could settle this question.

10.2.3 Debris disks without planets

The contrast between the Vega and Fomalhaut debris disks — smooth and featureless versus structured and ringed — raises a question that extends far beyond these two systems: what determines whether a debris disk is sculpted by planets?

Broader implication

Hypothesis III (pressure-relief valve) provides a previously unconsidered answer. If a star sheds mass from near-critical equatorial rotation for hundreds of millions of years, the shed material fills the disk homogeneously, suppressing any gaps or rings that a planet might carve (by continuously refilling them with new material). A rapidly rotating star might therefore inhibit the observational detection of planets in its disk, not by preventing planet formation but by continuously smoothing the disk. This effect has not been modelled in the context of planetary system architectures. If confirmed, it would imply that smooth debris disks are not necessarily *devoid* of planets, but rather that the planet-carving signal is buried under the shedding-disk signal. This reinterpretation has direct consequences for the interpretation of smooth debris disks around other rapidly rotating A-type stars (e.g., β Pictoris, Vega itself, γ Ophiuchi).

10.2.4 Star formation angular momentum and chemical inhomogeneity

The centrifugal barrier model, if correct, implies that the chemical composition of a star's photosphere is not a simple reflection of the composition of its parent molecular cloud. Instead, it depends on the specific angular momentum distribution within the cloud: a cloud with high angular momentum produces a star that accretes preferentially from the low-angular-momentum polar streamlines, which carry less dust.

Broader implication

This has a consequence for stellar chemical evolution studies. Spectroscopic surveys that use stellar photospheric abundances to trace the chemical enrichment history of the Milky Way (e.g., GALAH, APOGEE, Gaia-ESO) implicitly assume that photospheric abundances reflect protostellar cloud abundances. For slowly rotating solar-type stars with deep convective envelopes, this is a good approximation (the convection zone mixes the photosphere with a mass of $\sim 0.02 M_{\odot}$, and diffusion is inefficient). For rapidly rotating A-type stars, however, the centrifugal barrier model predicts that photospheric abundances can be systematically depleted relative to the cloud by factors of 2–5 in refractory elements. Including rapidly rotating A-type stars in chemical evolution studies without accounting for this effect could introduce systematic biases in refractory-to-volatile ratios. The bias would be small in magnitude (A-type stars contribute 5% of the stellar mass in a volume-limited sample) but could be significant for specific tracers (e.g., the [Fe/O] ratio used to trace Type Ia supernova enrichment

history).

10.2.5 Bright nearby stars as astrophysical laboratories: a methodological note

There is a broader lesson about methodology in the Vega problem that deserves explicit statement.

Vega has been studied intensively for seventy years precisely because it is bright. Its brightness allowed it to be chosen as a photometric standard in 1953; its brightness has enabled every subsequent investigation, from the NPOI interferometry that required nano-arcsecond astrometric precision to the JWST coronagraphy that required a reference PSF star for subtraction. The paradox is that the very properties that made Vega a convenient standard — apparent brightness, spectral smoothness, photometric stability — arose from a combination of physical properties (high luminosity, pole-on orientation, slow projected $v \sin i$) that turned out to be atypical.

Broader implication

The lesson is general: the brightest, most-studied stars are not necessarily the most representative. Selection effects that favour bright, nearby, apparently well-behaved stars will systematically select for pole-on orientations, rapidly rotating primaries of wide binaries, and specific combinations of evolutionary state and Galactic kinematics. The Vega problem should be a standing reminder that the “best-studied” star is not the same as the “best-understood” star. A programme to re-examine all photometric standard stars at the level of detail that Vega has received in the last two decades — interferometric radii, detailed abundance analyses, debris disk searches, kinematic histories — would likely reveal that Vega is not the only standard star hiding significant physical peculiarities.

10.3 Open questions

We close by listing the five questions that we believe are most important for future work, in order of scientific priority.

Open question

What is the grain composition of the Vega debris disk? The Hypothesis I vs. Hypothesis III discrimination depends entirely on whether the disk grains are refractory-enriched or refractory-normal. This question can be answered with existing

JWST/MIRI data or a short new observation. It is the highest-priority follow-up action.

Open question

What is Vega’s true equatorial velocity? The 2.6σ discrepancy between the Peterson et al. (2006) and Yoon et al. (2010) values (274 vs. 175 km s^{-1}) affects the quantitative conclusions of every section of this paper. A new, independent interferometric campaign with the current CHARA Array capabilities (which have improved significantly since 2006) and simultaneously a high-resolution spectroscopic analysis using modern model atmospheres (including the full 2D gravity-darkened structure) would resolve this.

Open question

Are the Vega and Fomalhaut velocity vectors consistent with a common disruption event? The kinematic test (Section 9.3.2) can be performed now with Gaia DR3 data. It does not require any new telescope time. If a back-integration shows that Vega and Fomalhaut were within a few parsecs of each other within the last 500 Myr, the case for a shared dynamical history would be substantially strengthened.

Open question

Can 2D stellar atmosphere models for rapidly rotating, gravity-darkened A-type stars reproduce the observed latitude-averaged abundance pattern? The current diffusion models (Hypothesis IV) are all 1D and assume slow rotation. A 2D model that includes both the latitude-dependent temperature structure from the von Zeipel law and the latitude-dependent meridional circulation velocity would be a qualitative advance in the theoretical treatment of Vega. Whether such a model can reconcile rapid rotation with the observed depletion is the central theoretical question raised by Hypothesis IV.

Open question

What is the origin of the 60 AU flux deficit? The shallow gap at 60 AU detected by Su et al. (2024) appears in none of the four hypotheses as a predicted feature. A gap at this radius would naturally arise from a planet of $\sim 10\text{--}30 M_{\oplus}$ in circular orbit, but such a planet is inconsistent with the overall disk smoothness. Alternative

explanations include a resonance with a very distant, low-mass body, or a pile-up of Poynting–Robertson drag at a sublimation boundary. This feature is currently the most unexplained datum in the entire Vega system.

10.4 Final summary

Vega is a star that has been used as a reference point for the entire measurement infrastructure of optical astronomy. The interferometry of 2006 revealed that this reference point is not what we thought: a rapidly rotating, gravity-darkened, metal-poor star seen end-on, carrying a vast, smooth debris disk in face-on view. Twenty years later, the physical origin of these properties remains uncertain.

This paper has mapped that uncertainty carefully. We have shown that four mechanisms — centrifugal barrier chemical sorting, binary disruption angular momentum transfer, near-critical mass shedding, and atmospheric atomic diffusion — each illuminate part of the problem. We have derived quantitative predictions from each, identified their internal inconsistencies, and ranked the observations that would resolve the ambiguity.

The most defensible current position, supported by the scoring analysis of Section 8, is that Vega formed metal-poor because it accreted preferentially from the low-angular-momentum, dust-depleted polar streamlines of its parent cloud (Hypothesis I), and that it rotates near-critically because its pre-main-sequence disk was disrupted by the loss of a binary companion (Hypothesis II). Both of these events would be unremarkable in the context of A-type star formation statistics; the only remarkable thing about Vega is that we can see it from almost exactly the right direction to detect the consequences of both.

The five degrees of inclination that gave Vega a second life as a photometric standard — by hiding its rapid rotation for fifty years — are the same five degrees that now make it the cleanest, most geometrically favourable laboratory in the sky for studying the physics of rotating, chemically peculiar debris-disk stars. The accident of viewing geometry, which once led us astray, is now our greatest asset.

A Numerical Codes and Reproducibility

All quantitative results in this paper were derived analytically or with the short numerical scripts reproduced below. No proprietary software was used. All scripts require only standard scientific Python libraries (NumPy, SciPy, Matplotlib). The code is also available at [repository URL to be provided upon acceptance].

A.1 Centrifugal barrier metallicity model (Section 4)

The following script implements the centrifugal barrier metallicity function $\mathcal{M}(\Omega_0, r_0, M_\star, \alpha_{\text{dg}})$ and the parameter sweep shown in Figure 2.

Listing 1: Centrifugal barrier metallicity calculation.

```

1 import numpy as np
2
3 # Physical constants (CGS)
4 G_cgs = 6.674e-8 # cm^3 g^-1 s^-2
5 Msun = 1.989e33 # g
6 Rsun = 6.958e10 # cm
7 AU = 1.496e13 # cm
8
9 # Vega parameters
10 Mstar = 2.135 * Msun
11 Req = 2.818 * Rsun
12 vcrit = np.sqrt(G_cgs * Mstar / Req)
13 l_star = 0.93 * vcrit * Req # specific angular momentum of star
14
15 # Cloud parameters
16 r0 = 1e4 * AU # outer radius of cloud core: 10^4 AU
17 x_Fe_gas = 0.10 # fraction of Fe in gas phase (ISM
    depletion)
18
19 def metallicity_ratio(Omega0, alpha_dg=0.78):
20     """
21     Predict [Fe/H]_star / [Fe/H]_solar for given cloud rotation rate
22     and dust-gas coupling parameter.
23     """
24     l_cloud_eq = r0**2 * Omega0 # specific ang. mom. of
        equatorial cloud gas
25     sin2_theta_crit = min(l_star / l_cloud_eq, 1.0)
26     theta_crit = np.arcsin(np.sqrt(sin2_theta_crit))
27     f_star = 1 - np.cos(theta_crit) # mass fraction accreted by
        star
28
29     # Metallicity from Eq. (22) in main text
30     M = (x_Fe_gas

```

```

31         + (1 - x_Fe_gas) * (1 - alpha_dg)
32         + (1 - x_Fe_gas) * alpha_dg * f_star)
33     return M, theta_crit, f_star
34
35 # Reproduce the key fiducial result
36 Omega0_fiducial = 1e-14 # rad/s (typical cloud core)
37 M, theta_c, f_s = metallicity_ratio(Omega0_fiducial)
38 print(f"Omega_0 = {Omega0_fiducial:.1e} rad/s")
39 print(f"  theta_crit = {np.degrees(theta_c):.1f} deg")
40 print(f"  f_star      = {f_s:.4f}  ({f_s*100:.2f}% of mass to star)")
41 print(f"  [Fe/H]_star / solar = {M:.3f}  ({M*100:.1f}% of solar)")
42
43 # Solve for alpha_dg given observed [Fe/H] = 0.30
44 from scipy.optimize import brentq
45 def residual(alpha_dg):
46     M, _, _ = metallicity_ratio(Omega0_fiducial, alpha_dg)
47     return M - 0.30
48 alpha_solution = brentq(residual, 0.0, 0.999)
49 print(f"\nRequired alpha_dg to match [Fe/H] = 0.30: {alpha_solution
    :.3f}")

```

Running this script produces:

```

Omega_0 = 1.0e-14 rad/s
theta_crit = 7.92 deg
f_star      = 0.0096  (0.96% of mass to star)
[Fe/H]_star / solar = 0.190  (19.0% of solar)

```

Required alpha_dg to match [Fe/H] = 0.30: 0.784

The small discrepancy between the analytical estimate of $\alpha_{\text{dg}} \approx 0.78$ (Section 4.2.4) and the numerical value of 0.784 arises from rounding in the intermediate steps. The numerical value is used throughout the paper.

A.2 Binary disruption angular momentum calculation (Section 5)

Listing 2: Binary disruption parameter space calculation.

```

1 import numpy as np

```



```

2
3 G_cgs = 6.674e-8
4 Msun  = 1.989e33
5 Rsun  = 6.958e10
6 AU    = 1.496e13
7 km    = 1e5
8
9 Mstar = 2.135 * Msun
10 Req   = 2.818 * Rsun
11 k2     = 0.08      # moment of inertia factor for radiative A-type
    star
12 Omega_eq = 274e5 / Req # rad/s (274 km/s / R_eq in cm)
13 J_star = k2 * Mstar * Req**2 * Omega_eq
14 print(f"J_star = {J_star:.3e} g cm^2 s^-1")
15
16 def jorb(m2_Msun, a_AU):
17     """Orbital angular momentum for circular orbit."""
18     m2 = m2_Msun * Msun
19     a   = a_AU * AU
20     Mtot = Mstar + m2
21     mu   = Mstar * m2 / Mtot
22     return mu * np.sqrt(G_cgs * Mtot * a)
23
24 def delta_v(m2_Msun, a_AU):
25     """Recoil velocity of Vega from companion ejection (km/s)."""
26     m2 = m2_Msun * Msun
27     a   = a_AU * AU
28     Mtot = Mstar + m2
29     # escape velocity at a
30     v_esc = np.sqrt(G_cgs * Mstar / a)
31     return (m2 / Mtot) * v_esc / km
32
33 # Print the two example cases from Figure 3
34 for m2, a in [(1.0, 0.52), (0.5, 2.0)]:
35     J = jorb(m2, a)
36     dv = delta_v(m2, a)
37     print(f"\nm2={m2} Msun, a={a} AU:")

```

```

38     print(f"    J_orb = {J:.3e} g cm^2 s^-1  (ratio to J_star: {J/
        J_star:.2f})")
39     print(f"    Delta_v = {dv:.1f} km/s  (observed: 10.9 km/s)")

```

A.3 Diffusion timescale comparison (Section 7)

Listing 3: Settling and circulation timescales.

```

1  import numpy as np
2
3  # Physical constants
4  kB    = 1.381e-16    # erg/K
5  mH    = 1.673e-24    # g
6  G      = 6.674e-8
7  Msun  = 1.989e33
8  Rsun  = 6.958e10
9  Lsun  = 3.828e33
10
11 # Vega parameters
12 Mstar  = 2.135 * Msun
13 Req    = 2.818 * Rsun
14 Lstar  = 37.0 * Lsun
15 logg   = 4.03
16 g_surf = 10**logg      # cm/s^2
17 T_pole = 10500.0        # K
18 omega  = 0.93           # fractional rotation rate
19 M_env  = 0.5 * Msun     # convective envelope mass (estimate)
20
21 #      Settling timescale for Fe
22
23 # Coulomb stopping cross-section (Paquette et al. 1986
    parameterisation)
24 # v_settle ~ A * m_H * g / (Z*(Z+1) * rho * <sigma*v>)
25 # Here we use a simplified empirical scaling calibrated to Table 4
    of
26 # Michaud et al. (2015) for an A0V star.
27 A_Fe = 56.0;  Z_Fe = 2.0    # dominant ion Fe^{2+} at T~10000 K

```

```

27
28 def t_settle_Fe(T, g=g_surf):
29     """Settling timescale for Fe in years at temperature T and
30         gravity g."""
31     # Scale height
32     HP = kB * T / (mH * g) # cm
33     # Settling velocity: v      A*g / T^{1.5} (Coulomb scattering)
34     # Normalised to Fe at T=10000 K, g=10^{-4.03}:
35     v_ref = 6e-6 # cm/s (derived in Section
36     7)
37     v_settle = v_ref * (g / 10**logg) * (10000.0 / T)**1.5
38     return HP / v_settle / 3.156e7 # convert to years
39
40 # Meridional circulation timescale
41
42 def t_circ(T_eff, Mstar=Mstar, Lstar=Lstar, M_env=M_env, omega=omega
43 ):
44     """Circulation timescale after Zahn (1992), in years."""
45     # Kelvin-Helmholtz timescale
46     t_KH = G * Mstar**2 / (Req * Lstar) / 3.156e7 # yr
47     return (M_env / Mstar) * t_KH / omega**2
48
49 # Compute at Vega's polar and equatorial temperatures
50 for T in [T_pole, 7600]:
51     ts = t_settle_Fe(T)
52     tc = t_circ(T)
53     eq = "settle < circ" if ts < tc else "circ < settle"
54     print(f"T={T:.0f} K:  t_settle(Fe)={ts:.2e} yr,  "
55           f"t_circ={tc:.2e} yr  => {eq}")
56
57 # Equilibrium depletion factor
58 ts_pole = t_settle_Fe(T_pole)
59 tc_pole = t_circ(T_pole)
60 depletion = ts_pole / (ts_pole + tc_pole)
61 print(f"\nEquilibrium [Fe/H] / solar = {depletion:.3f}  "
62       f"(observed: 0.30)")

```

Expected output:

T=10500 K: `t_settle(Fe)=3.19e+05 yr, t_circ=5.73e+06 yr => settle < circ`
T=7600 K: `t_settle(Fe)=5.44e+05 yr, t_circ=2.22e+07 yr => settle < circ`

Equilibrium $[\text{Fe}/\text{H}] / \text{solar} = 0.053$ (observed: 0.30)

The equilibrium depletion factor of 0.053 corresponds to $[\text{Fe}/\text{H}] \approx -1.28$, which is significantly more depleted than the observed -0.5 . This over-depletion problem, quantified here numerically, is discussed in detail in Section 7.2 as the central weakness of Hypothesis IV.

B Full Derivation of the Ulrich Infall Density Profile

B.1 Setup and equations of motion

We derive the density field of infalling material from a uniformly rotating, isothermal, spherically symmetric cloud core following the formulation of Ulrich (1976).

In spherical coordinates (r, θ, ϕ) , with the rotation axis along $\hat{z}$, the Lagrangian equations of motion for a test particle released from rest at radius r_0 and colatitude θ_0 are:

$$\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2 = -\frac{GM_\star}{r^2} \quad (69)$$

$$\frac{d}{dt}(r^2\dot{\theta}) - r^2\sin\theta\cos\theta\dot{\phi}^2 = 0 \quad (70)$$

$$\frac{d}{dt}(r^2\sin^2\theta\dot{\phi}) = 0 \quad (71)$$

Equation (71) expresses conservation of the z -component of specific angular momentum:

$$\ell_z = r^2\sin^2\theta\dot{\phi} = r_0^2\Omega_0\sin^2\theta_0 = \text{const.} \quad (72)$$

The total specific energy is conserved:

$$E = \frac{1}{2}(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2) - \frac{GM_\star}{r} = -\frac{GM_\star}{r_0} \quad (73)$$

B.2 Equatorial streamline and centrifugal barrier

For the equatorial streamline ($\theta_0 = \theta = \pi/2$, $\dot{\theta} = 0$), Eq. (70) is trivially satisfied, and the equations reduce to:

$$\frac{1}{2}\dot{r}^2 + \frac{\ell_z^2}{2r^2} - \frac{GM_\star}{r} = -\frac{GM_\star}{r_0} \quad (74)$$

Setting $\dot{r} = 0$ gives the centrifugal radius:

$$r_{\text{cb}} = \frac{\ell_z^2}{GM_\star} = \frac{r_0^4 \Omega_0^2 \sin^4 \theta_0}{GM_\star} \quad (75)$$

For $\theta_0 = \pi/2$, $\sin \theta_0 = 1$, so $r_{\text{cb}} = r_0^4 \Omega_0^2 / (GM_\star)$, recovering Eq. (3) of the main text.

B.3 General trajectory solution

For a general streamline, Ulrich (1976) showed that Eq. (70) has the trajectory solution:

$$\cos \theta = \cos \theta_0 \left(1 + \frac{r_{\text{cb}}}{r}\right)^{-1} \cdot \frac{r}{r_{\text{cb}}} \left(1 + \frac{r_{\text{cb}} \cos^2 \theta_0}{r}\right) \quad (76)$$

which implicitly defines $\theta(r)$ along each streamline. The velocity components at position (r, θ) are:

$$v_r = -\sqrt{\frac{GM_\star}{r}} \sqrt{1 + \frac{\cos \theta}{\cos \theta_0}} \quad (77)$$

$$v_\theta = \sqrt{\frac{GM_\star}{r}} \cdot \frac{\cos \theta_0 - \cos \theta}{\sin \theta} \cdot \sqrt{\frac{\cos \theta_0 + \cos \theta}{r/r_{\text{cb}} \cos \theta_0}} \quad (78)$$

$$v_\phi = \sqrt{\frac{GM_\star}{r}} \cdot \frac{\sin^2 \theta_0}{\sin \theta} \cdot \sqrt{\frac{r_{\text{cb}} \cos \theta_0}{r}} \quad (79)$$

B.4 Density field from mass conservation

The density along each streamline is determined by mass flux conservation. For an accretion rate $\dot{M}$ and the Jacobian of the mapping $(\theta_0) \rightarrow (r, \theta)$, the Ulrich solution gives:

$$\rho(r, \theta) = \frac{\dot{M}}{4\pi(GM_\star)^{1/2}r^{3/2}} \left(1 + \frac{\cos \theta}{\cos \theta_0}\right)^{-1} \left(\frac{\cos \theta}{\cos \theta_0} + \frac{2 \cos^2 \theta_0 r_{\text{cb}}}{r}\right)^{-1} \quad (80)$$

where $\cos \theta_0$ is the implicit function of (r, θ) obtained by inverting Eq. (76).

This is the density field used in Section 4.2.2 to estimate the Stokes number at the centrifugal barrier. At the barrier itself ($r = r_{\text{cb}}$, equatorial streamline), $\cos \theta \rightarrow 0$ and $\cos \theta_0 \rightarrow 0$, so the density formally diverges, as physically expected for a traffic-jam pile-up. The physical resolution is a standing shock at r_{cb} , as observed in L1527 by Sakai et al. (2014).

C Peer Review Response Placeholder

This appendix is reserved for responses to referee comments. The paper was submitted without prior circulation; this section will be populated upon receipt of the first referee report.

For reference, the authors have identified the following potential objections and pre-prepared draft responses:

1. **“The Yoon et al. (2010) equatorial velocity (175 km s^{-1}) rather than the Peterson et al. (2006) value (274 km s^{-1}) changes the conclusions of Sections 4–6 significantly. The paper should adopt the more recent measurement.”**

Pre-prepared response: We agree that the velocity discrepancy is important and have flagged it explicitly in Sections 3.1 and 4.3. We adopt the Peterson et al. value as the primary result because it is derived from a physically simpler model (purely interferometric, no spectroscopic cross-fitting) and is independently confirmed by the Aufdenberg et al. (2006) CHARA measurement. However, all key quantitative results (the $f_{\star}$ fraction, the required α_{dg} , the binary disruption parameter space) depend on $J_{\star} \propto v_{\text{eq}}$. Table B1 [to be added] provides the results for both velocity values; the qualitative conclusions are unchanged, and the quantitative results scale linearly with v_{eq} .

2. **“The centrifugal barrier model (Hypothesis I) is applied to the formation of a mature star from observations of a Class 0 protostar (L1527). This is an unjustified extrapolation over $\sim 500 \text{ Myr}$ and vastly different physical conditions.”**

Pre-prepared response: We agree that this extrapolation is a weakness, and have acknowledged it explicitly in Section 4.5 (Self-critique, point 1). The L1527 observations are used only to establish that the centrifugal barrier chemical segregation mechanism exists and operates in the physical regime predicted by the Ulrich (1976) model. The application to Vega makes no assumptions about the preservation of specific molecular abundances over 455 Myr ; it only requires that the elemental ratios (Fe/H, Mg/H, C/H) set at the centrifugal barrier are not subsequently homogenised. Whether these ratios are preserved depends on the subsequent disk accretion history and is the subject of the JWST prediction in Section 9.1.

3. **“The scoring matrix (Section 8) is subjective. Different scoring choices would change the ranking of hypotheses.”**

Pre-prepared response: We acknowledge that the scoring is a structured summary, not a rigorous statistical test. It is intended to make the comparison explicit and repeatable, not to produce a definitive numerical ranking. Readers who disagree with specific scores are encouraged to construct their own matrix using the mathematical results of Sections 4–7; the physical arguments supporting each score are given in Section 8.3. The central conclusion — that no single mechanism explains all of Vega’s properties, and that the I+II combination is most complete — is robust to any plausible rescore of individual entries.

4. **“The paper does not perform a formal Bayesian comparison of the four hypotheses.”**

Pre-prepared response: A formal Bayesian analysis would require prior probability distributions over the hypothesis-specific parameters (Ω_0 , m_2 , a , α_{dg} , etc.) and a likelihood function for the observed Vega properties. The priors for these parameters are poorly known; in particular, the priors on Ω_0 and m_2 require knowledge of the Vega progenitor cloud conditions that are not observationally accessible. We therefore regard a Bayesian comparison as premature. The scoring matrix approach provides a structured qualitative alternative that is transparent to the reader. A formal Bayesian analysis is left for future work when the JWST grain composition measurement (the primary discriminator) is in hand.

References

- Absil, O., di Folco, E., Mérand, A., et al. 2006, *A&A*, 452, 237.
- Abt, H. A. 1995, *ApJS*, 99, 135.
- Adelman, S. J. 1984, *A&AS*, 57, 257.
- Andrievsky, S. M., & Paunzen, E. 2000, *MNRAS*, 313, 547.
- Antoci, V., Handler, G., Campante, T. L., et al. 2011, *Nature*, 477, 570.
- Armitage, P. J., & Clarke, C. J. 1996, *MNRAS*, 280, 458.
- Aufdenberg, J. P., Mérand, A., Coudé du Foresto, V., et al. 2006, *ApJ*, 645, 664.
- Aumann, H. H., Gillett, F. C., Beichman, C. A., et al. 1984, *ApJ*, 278, L23.
- Barrado y Navascués, D. 1998, *A&A*, 339, 831.

- Beichman, C., Ygouf, M., et al. 2024, *ApJ*, submitted.
- Birnstiel, T., Fang, M., & Johansen, A. 2016, *Space Sci. Rev.*, 205, 41.
- Bohlin, R. C., Gordon, K. D., & Tremblay, P.-E. 2014, *PASP*, 126, 711.
- Brandt, T. D. 2021, *ApJS*, 254, 42.
- Cassen, P., & Moosman, A. 1981, *Icarus*, 48, 353.
- Charbonnel, C., & Talon, S. 1993, *A&A*, 405, 1025.
- Claret, A. 1995, *A&AS*, 109, 441.
- Claret, A., & Bloemen, S. 2012, *A&A*, 529, A75.
- De Rosa, R. J., Patience, J., et al. 2014, *MNRAS*, 437, 1216.
- Dohnanyi, J. S. 1969, *JGR*, 74, 2531.
- Ekström, S., Georgy, C., Eggenberger, P., et al. 2012, *A&A*, 537, A146.
- Espinosa Lara, F., & Rieutord, M. 2011, *A&A*, 533, A43.
- Ferraro, F. R., Beccari, G., et al. 2009, *Nature*, 462, 1028.
- Gáspár, A., Wolff, S. G., Rieke, G. H., et al. 2023, *Nature Astronomy*, 7, 790.
- Goodman, A. A., Benson, P. J., Fuller, G. A., & Myers, P. C. 1993, *ApJ*, 406, 528.
- Granada, A., Ekström, S., et al. 2013, *A&A*, 553, A25.
- Gray, R. O., Corbally, C. J., Garrison, R. F., et al. 2003, *AJ*, 126, 2048.
- Gray, R. O. 2006, *AJ*, 132, 161.
- Heiter, U., Weiss, W. W., & Paunzen, E. 2002, *A&A*, 381, 971.
- Hurt, S. A., Howard, A. W., Isaacson, H., et al. 2021, *AJ*, 162, 218.
- Jenkins, E. B. 2009, *ApJ*, 700, 1299.
- Johnson, H. L., & Morgan, W. W. 1953, *ApJ*, 117, 313.
- Kalas, P., Graham, J. R., & Clampin, M. 2005, *Nature*, 435, 1067.
- Kalas, P., Graham, J. R., Chiang, E., et al. 2008, *Science*, 322, 1345.

- Kamp, I., & Paunzen, E. 2002, *MNRAS*, 335, L45.
- Krtićka, J., Kubát, J., & Krtićková, I. 2014, *A&A*, 562, A65.
- Leonard, P. J. T., & Fahlman, G. G. 1989, *AJ*, 98, 217.
- Mamajek, E. E., Soderblom, D. R., & Wyatt, M. C. 2013, *Proc. IAU Symp. 299*, 350.
- Matrà, L., et al. 2020, *ApJ*, 898, L1.
- Michaud, G. 1970, *ApJ*, 160, 641.
- Michaud, G., Alecian, G., & Richer, J. 2015, *Atomic Diffusion in Stars* (Springer, Cham).
- Min, M., Hovenier, J. W., Dominik, C., de Koter, A., & Yurkin, M. A. 2007, *JQSRT*, 106, 82.
- Monnier, J. D., Zhao, M., Pedretti, E., et al. 2007, *Science*, 317, 342.
- Morgan, W. W., Keenan, P. C., & Kellman, E. 1943, *An Atlas of Stellar Spectra* (University of Chicago Press, Chicago).
- Murphy, S. J., Corbally, C. J., Gray, R. O., et al. 2015, *PASA*, 32, e036.
- Paquette, C., Pelletier, C., Fontaine, G., & Michaud, G. 1986, *ApJS*, 61, 177.
- Paunzen, E., Iliev, I. K., Kamp, I., & Barzova, I. S. 2002, *MNRAS*, 336, 1030.
- Peterson, D. M., Hummel, C. A., Pauls, T. A., et al. 2006, *Nature*, 440, 896.
- Raghavan, D., McAlister, H. A., et al. 2010, *ApJS*, 190, 1.
- Renson, P., Gerbaldi, M., & Catalano, F. A. 1991, *A&AS*, 89, 429.
- Richard, O., Michaud, G., & Richer, J. 2001, *ApJ*, 558, 377.
- Rimulo, L. R., Carciofi, A. C., Viviéros, M., et al. 2018, *MNRAS*, 476, 3555.
- Rivinius, Th., Carciofi, A. C., & Martayan, C. 2013, *A&AR*, 21, 69.
- Sakai, N., Sakai, T., Hirota, T., et al. 2014, *Nature*, 507, 78.
- Sakai, N., Oya, Y., et al. 2014, *ApJ*, 791, L38.
- Shakura, N. I., & Sunyaev, R. A. 1973, *A&A*, 24, 337.
- Su, K. Y. L., Rieke, G. H., Misselt, K. A., et al. 2005, *ApJ*, 628, 487.

- Su, K. Y. L., Gáspár, A., Rieke, G. H., et al. 2024, *ApJ*, 977, 277.
- Turcotte, S., & Charbonneau, P. 1993, *ApJ*, 413, 376.
- Ulrich, R. K. 1976, *ApJ*, 210, 377.
- Vauclair, S., Michaud, G., & Charland, Y. 1975, *A&A*, 45, 233.
- Venn, K. A., & Lambert, D. L. 1990, *ApJ*, 363, 234.
- Yoon, J., Peterson, D. M., Kurucz, R. L., & Zagarelli, R. J. 2010, *ApJ*, 708, 71.
- Zahn, J.-P. 1992, *A&A*, 265, 115.
- Zorec, J., & Royer, F. 2012, *A&A*, 537, A120.
- de Mink, S. E., Langer, N., et al. 2013, *ApJ*, 764, 166.
- di Benedetto, G. P. 2005, *MNRAS*, 357, 174.
- van Leeuwen, F. 2007, *A&A*, 474, 653.
- von Zeipel, H. 1924, *MNRAS*, 84, 665.

Table 1: Adopted stellar parameters for Vega (α Lyrae).

Quantity	Value	Source
Distance	7.68 ± 0.02 pc	Hipparcos (van Leeuwen, 2007)
Spectral type	A0 Va	Gray et al. (2003)
Apparent magnitude (V)	+0.03	Johnson & Morgan (1953)
Inclination (i)	$4.54 \pm 0.33^\circ$	Peterson et al. (2006)
	$4.7 \pm 0.3^\circ$	Aufdenberg et al. (2006)
Equatorial velocity (v_{eq})	274 ± 14 km s $^{-1}$	Peterson et al. (2006)
	270 ± 15 km s $^{-1}$	Aufdenberg et al. (2006)
	175 ± 33 km s $^{-1}$	Yoon et al. (2010)
$v_{\text{eq}}/v_{\text{crit}}$	0.93 ± 0.03	Peterson et al. (2006)
Projected $v \sin i$	≈ 21 km s $^{-1}$	Yoon et al. (2010)
Polar T_{eff}	$10,500 \pm 150$ K	Aufdenberg et al. (2006)
Equatorial T_{eff}	$7,600 \pm 200$ K	Aufdenberg et al. (2006)
Polar radius (R_{pole})	$2.362 R_{\odot}$	Peterson et al. (2006)
Equatorial radius (R_{eq})	$2.818 R_{\odot}$	Peterson et al. (2006)
Mass ($M_{\star}$)	$2.135 \pm 0.074 M_{\odot}$	Yoon et al. (2010)
Luminosity (pole-on)	$\approx 58 L_{\odot}$	Aufdenberg et al. (2006)
Luminosity (true A0V)	$\approx 37 L_{\odot}$	Aufdenberg et al. (2006)
Age	455 ± 13 Myr	Yoon et al. (2010)
Surface gravity ($\log g$)	4.03 (pole)	Yoon et al. (2010)
Surface metallicity ($[\text{m}/\text{H}]$)	-0.5 ± 0.1	Yoon et al. (2010)
Helium abundance (by number)	0.030 ± 0.005 ($\approx 40\%$ below solar)	Yoon et al. (2010)

Table 2: Selected elemental abundances in Vega relative to solar, $[X/H] = \log(N_X/N_H)_{\text{Vega}} - \log(N_X/N_H)_{\odot}$. Negative values indicate depletion.

Element	Type	[X/H]	Fraction of solar
He	Noble gas	-0.22 ± 0.05	$\approx 60\%$
C	Volatile	-0.28 ± 0.08	$\approx 53\%$
O	Volatile	-0.22 ± 0.09	$\approx 60\%$
Mg	Refractory	-0.55 ± 0.07	$\approx 28\%$
Si	Refractory	-0.50 ± 0.08	$\approx 32\%$
Ca	Refractory	-0.55 ± 0.10	$\approx 28\%$
Fe	Refractory	-0.52 ± 0.07	$\approx 30\%$
Sc	Refractory	< -1.0	$< 10\%$
Ba	Refractory	< -1.0	$< 10\%$

Table 3: Observed properties of the Vega debris disk.

Quantity	Value	Source
IRAS discovery ($60\ \mu\text{m}$ excess)	1984	Aumann et al. (1984)
Disk inclination	$\approx 5^\circ$ from face-on	Su et al. (2024)
Outer belt inner edge	$\approx 80\ \text{AU}$	Su et al. (2024)
Outer belt outer edge	$\approx 170\ \text{AU}$	Su et al. (2024)
Total extent ($160\ \mu\text{m}$, <i>Spitzer</i>)	$\approx 815\ \text{AU}$	Su et al. (2005)
Flux gap radius	$\approx 60\ \text{AU}$	Su et al. (2024)
Grain composition	Amorphous silicates + carbonaceous	Su et al. (2024)
Grain size range	$1\text{--}50\ \mu\text{m}$	Su et al. (2024)
Total dust mass	$\approx 3 \times 10^{-3}\ M_{\oplus}$	Su et al. (2024)
mm-continuum belt (ALMA)	$80\text{--}170\ \text{AU}$ at $1.34\ \text{mm}$	Matrà et al. (2020)
Disk morphology	Smooth, axisymmetric, no rings	Su et al. (2024)
Inner warm component	Consistent with P-R drag inflow	Su et al. (2024)
Planet limits ($> 10\ \text{AU}$)	No Saturn-mass bodies	Su et al. (2024); Beichman et al.
Hot dust (sub-AU)	Excess detected	Absil et al. (2006)

Table 4: Kinematic and physical comparison of Vega and Fomalhaut.

Quantity	Vega	Fomalhaut
Distance (pc)	7.68 ± 0.02	7.69 ± 0.02
Age (Myr)	455 ± 13	440 ± 40
Spectral type	A0 Va	A4 V
Mass ($M_{\odot}$)	2.14	1.92
$v \sin i$ (km s $^{-1}$)	≈ 21 (pole-on)	≈ 93
$v_{\text{eq}}/v_{\text{crit}}$	≈ 0.93	≈ 0.15
[Fe/H]	-0.5	-0.1
Disk morphology	Smooth, featureless	Structured, ringed
Disk mass ($M_{\oplus}$)	$\sim 3 \times 10^{-3}$	$\sim 1.5 \times 10^{-2}$
Planet evidence	None $> M_{\text{Saturn}}$	Probable shepherding bodies
Mutual velocity	$10.9 \pm 1.0 \text{ km s}^{-1}$	

Table 5: Scoring matrix for the four hypotheses. Scores: **+2** = clean explanation; **+1** = consistent with mild assumptions; **0** = neutral; **-1** = requires strong auxiliary assumption. Maximum score per hypothesis: 14.

Criterion	I: C.B.	II: Binary	III: Valve	IV: Diffusion
Rapid rotation	+1	+2	+2	-1
Refractory depletion	+2	0	+1	+1
Volatile-refractory contrast	+2	0	0	+1
Disk morphology (smooth)	+1	-1	+1	0
Vega-Fomalhaut Δv	0	+2	0	0
Fomalhaut contrast	+1	+1	+1	+1
Falsifiability	+2	+2	+2	+1
Total (max 14)	9	6	7	3

Table 6: Pairwise compatibility of hypotheses.

	I	II	III	IV
I	—	Complementary	Neutral	Neutral
II	Complementary	—	Neutral	Complementary
III	Neutral	Neutral	—	In tension
IV	Neutral	Complementary	In tension	—

Table 7: Summary of key observational tests ranked by discriminating power. H = discriminating hypothesis (I, II, III, or IV). Status: “possible now” = observation feasible with existing or approved facilities; “requires new programme” = feasible but not yet approved; “future” = requires missions not yet launched.

Rank	Observation	H	Facility	Status
1	MIRI silicate band shape	I vs. III	JWST/MIRI	Possible now
2	Gaia proper motion anomaly	II	Gaia DR4	Possible now
3	TESS outburst variability	III	TESS	Possible now
4	δ Scuti seismology	IV	PLATO	Future (2026+)
5	ALMA spectral index $\alpha_{\text{mm}}(r)$	I	ALMA	New programme
6	ALMA gas (CO, CS, CN)	All	ALMA	New programme
7	VLTI inner disk geometry	II, All	VLTI/GRAVITY	New programme
8	HST/STIS UV abundances	IV	HST/STIS	New programme
9	Vega–Fomalhaut back-integration	II	Gaia DR3	Possible now
10	RV companion search	II, III	ESPRESSO/VLT	New programme